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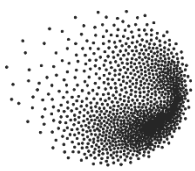
SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Characterisation of GaAs:Cr Microstrip Sensors for Single Photon Counting

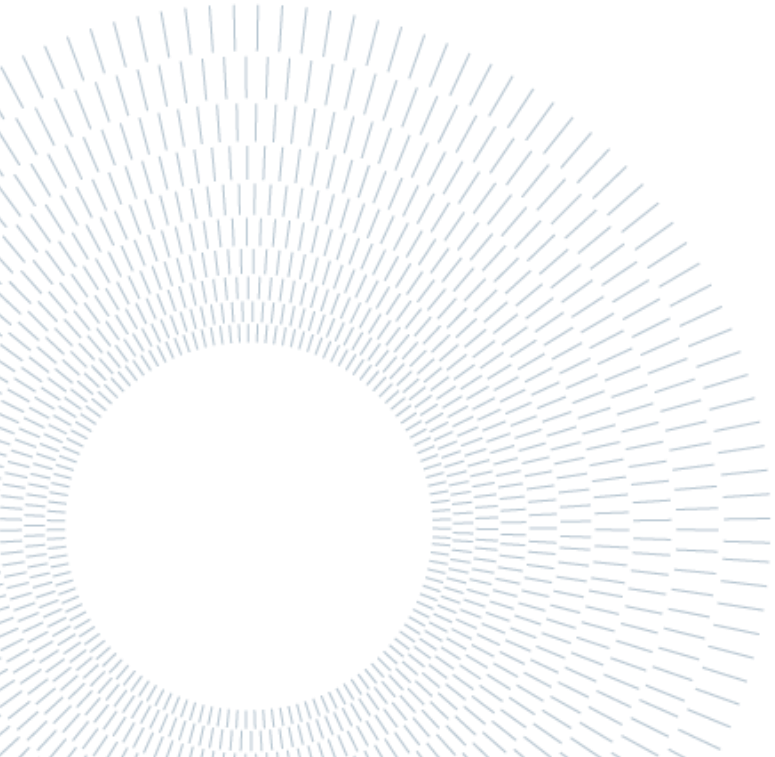
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Author: **Michele Anastasi**

Student ID: 242970
Advisor: Prof. Marco Caresana
PSI Supervisors: Dr. Anna Bergamaschi
Dr. Jonathan Franklin Mulvey
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PSI



Abstract

Single Photon Counting (SPC) microstrip detectors are widely used at synchrotron facilities where high spatial resolution and high count rate capability are required. The standard MYTHEN III uses silicon sensors, whose photon absorption efficiency drops at higher energies. Chromium compensated gallium arsenide (GaAs:Cr) represents a valid alternative for extending the energy ranges. This work therefore delves into the study of a MYTHEN III detector equipped with two prototype 500 μm thick GaAs:Cr sensors with 50 μm and 100 μm strip pitches. The mentioned detector has thus been characterized using X-ray fluorescence photons, polychromatic X-ray and an optical laser, giving particular attention to its dependence on the analogue readout settings, temperature and photon rate.

The measurements show that detector performance, in terms of gain, equivalent noise charge (ENC), high rate behaviour, is mainly controlled by the preamplifier stage, while the shaper affects only weakly its response. A dedicated trimming procedure reduced the channel to channel threshold dispersion to only a few equivalent electrons, while an unusual temperature dependence has been observed, with increasing gain and decreasing ENC over the investigated range. An effective rate on the order of a few MHz with a 90% counting efficiency has been observed, which also varies according to the readout configuration. In terms of charge transport, both sensors showed a significant charge sharing on the order of 80%, which was unexpected for the 100 μm pitch strip sensor. These values have been further verified by spatial laser measurements, which was used also to retrieve qualitative evidence of the crater effect. Eventually, spatial response investigation employing the laser showed that a large charge sharing could be exploited to the improve of spatial resolution through the MYTHEN III interstrip interpolation logic.

Overall, the results demonstrate the feasibility of employing GaAs:Cr microstrip sensors with the MYTHEN III detector, thus representing a good opportunity for extending its useful energy range by improving the detection efficiency of higher energies X-ray photons compared with conventional silicon sensors.

Key-words: GaAs:Cr; MYTHEN III; Single Photon Counting; X-ray detectors; microstrip detectors; charge sharing

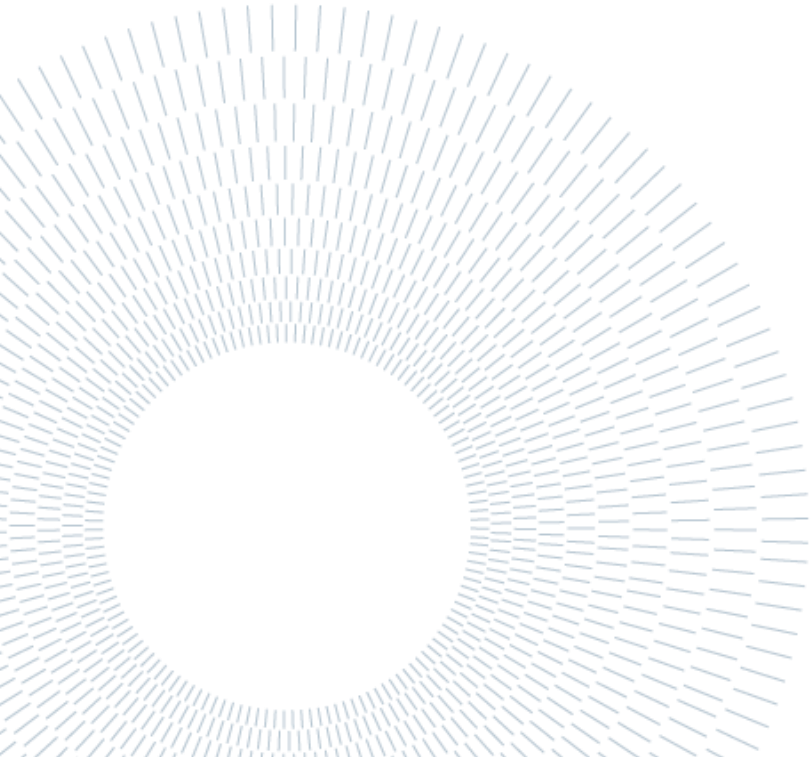
Abstract in italiano

I rilevatori *microstrip* operanti a conteggio di singolo fotone sono ampiamente usati presso i sincrotroni, dove sono richeiste eleveata risoluzione spaziale e capacità di operare ad alti rate di fotoni. La versione standard del *MYTHEN III* adopera sensori in silicio la cui efficienza di assorbimento diminiusce con l'energia. L'arsenuro di galllio compensato al cromo (GaAs:Cr) rappresenta una valida alternativa per estendere l'intervallo energetico. Questa tesi dunque approfondisce lo studio di un rilevatore *MYTHEN III* equipaggiato con due sensori prototipi in GaAs:Cr, spessi 500 μm e con larghezza delle *strip* pari a 50 μm e 100 μm. Inoltre la caratterizzazione è avvenuta mediante fotoni di fluorescenza X, radiazione policromatica e fasci laser, prestando attenzione alla dipendnza della rispotsa dalle impostazioni del readout analogico, nonchè temperatura e rate di fotoni.

Le misure mostrano che le prestazioni, in termini di guadagno, carica equivalente di rumore (*ENC*) e comportamento ad alti *rate*, sono dipendennti principalmente dallo stadio di preamplificazione, mentre lo *shaper* influenaza limitatamente. Una procedura dedicata di trimming ha ridotto la dispersione di soglia tra i canali a pochi elettroni equivalenti, mentre è stata osservata una dipendenza dalla temperatura non usuale, contraddistinta da un aumento del guadagno e diminuizione del *ENC* nell'intervallo investigato. Inoltre è stato osservato un buon comportamento ad alti flussi fotonici, con *rate* effettivi al 90% di efficienza di conteggio nell'ordine di alcuni MHz, variabili in base al *readout*. In termini di trasposto di carica, entrambi i sensori hanno mostrato una elevata condivisione di carica nell'ordine del 80%, un risultato inatteso per il sensore con pitch da 100 μm. Tali valori sono stati in seguito confermati da misure effettuate tramite laser ottico, con cui sono state pure ottenute evidenze qualitative del *crater effect*. Infine, lo studio della risposta in termini spaziale tramite laser hanno mostrato che una elevata condivisione di carica può essere sfruttata dalla logica di interpolazione *interstrip* di *MYTHEN III* per migliorare la risoluzione spaziale.

Complessivamente, i risultati dimostrano la fattibilità dell'impiego di sensori microstrip realizzato in GaAs:Cr tramite l'impiego del *MYTHEN III*, rappresentando quindi una promettente opportunità per estenderne l'intervallo energetico verso più alte energie, se comparato ai convenzionali sensori al silicio.

Parole chiave: GaAs:Cr; MYTHEN III; conteggio a singolo fotone; rilevatori fotoni; rilevatori microstrip; condivisione di carica



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1 Introduction

1.1. Scope of this work

This Master thesis inquires the response of a MYTHEN III single photon counting microstrip detector coupled to prototype GaAs:Cr sensors to extend the synchrotron X-ray application towards higher photon energies [1] [2]. The main aim is to characterize the detector performance and then evaluate the perks and drawbacks introduced by the usage of GaAs:Cr instead of standard Si sensors. Results coming from the measurements performed with the mentioned MYTHEN III GaAs:Cr module will be compared to the existing literature and to the performance of a complete silicon MYTHEN III module, including also some limited data acquired using the latter.

The first part of this work focuses on the theoretical background and description of the MYTHEN III, also covering briefly the electronics behind the readout circuitry in order to understand and predict the possible behaviour. Moreover, the theoretical framework of semiconductors is introduced and then delved into the GaAs:Cr properties, as well as some preliminary experimental results concerning electrical material properties. Eventually, the experimental setup is briefly covered along with the methods used to characterize the detector.

The second part of the work delves instead on the experimental results obtained about the MYTHEN III GaAs:Cr prototype sensors, as well as the theoretical explanation and introduction of concepts and figure of merits fundamental to evaluate the detector behaviour, which is studied also with reference to readout parameters variations, temperature and photon rate. Moreover, the characteristics of charge transport inside the semiconductor material are covered along the practical implication and effects on detector behaviour. Eventually, also the spatial response is investigated.

1.2. The MYTHEN III detector: general overview

The MYTHEN III detector, whose name stands for *Microstrip sYstem for Time-rEsolved experimeNts*, is a one-dimensional hybrid single-photon-counting (SPC) microstrip detector developed at the *Paul Scherrer Institute* (PSI) by the *Laboratory for X-ray Nanoscience and Technologies* (LXN) group. It was primarily designed to meet the demands of time resolved *X-ray Powder Diffraction* (XRPD) [3] experiments at

synchrotron radiation facilities, while its architecture makes it also compatible for selected spectroscopy and imaging applications [4].

Specifically, the detector is labelled as hybrid since it combines physically two separate parts: a semiconductor (in this case a strip sensor rather than a pixel one) which is aimed at absorbing photons, and a separate readout chip that amplifies, shapes and counts the signal. The physical connection is made through wire bonding. This is different from a monolithic detector, where detection and readouts are integrated in the same chip [4].

Existing pixel photon counting detectors such as EIGER 2 or MEDIPIX3 can also be used for powder diffraction experiments. However, thanks to the one dimensional microstrip geometry, MYTHEN III requires fewer readout channels while providing fine spatial segmentation, high frame rates and large angular coverage [3]. This makes the detector particularly suitable for powder diffraction applications. Compared with its predecessor MYTHEN II, the third generation also provides lower noise and threshold dispersion, a lower minimum detectable energy, higher frame rates and an improved count rate capability. Moreover, each electronic channel features three independent comparators and counters, which can be used for different operating modes, such as energy windowing, piling up tracking and digital inter strip interpolation [3].

1.3. X-ray powder diffraction

Synchrotron light sources provide intense and collimated photon beams usable in a wide range of different experiments in material science, chemistry, biology and so on [5]. Therefore, the evolution of X-ray detector is connected to the synchrotron facilities, due to progressively increasing photon fluxes and energies [1] [3]. Then a suitable detector would be capable of providing, ideally, fast readout with low noise alongside high rates capability and high spatial resolution and energy range [3].

One of the most noteworthy applications is the *X-ray powder diffractions* (XRDP), a non-destructive technique fundamental for the crystalline structures analysis [5]. A powder consists of many crystals with different orientations. Therefore, several crystallographic planes can contemporarily meet the diffraction condition and then generate diffraction cones around the incident beam. The resulting diffraction pattern can therefore be described mainly as the intensity distribution as a function of the scattering angle. Therefore, XRPD experiments require high spatial and angular resolution in order to discern close diffraction peaks [3]. However, only one-dimensional data is needed, eventually simplifying the electronics and reducing the data throughput and then allowing higher fidelity time resolved measurements (also made possible thanks to the faster frame rate of the detector) [3].

1.4. High Z sensors and GaAs:Cr

The standard MYTHEN III detector employs silicon strip sensors [1]. However, although silicon provides noteworthy performance, due to its high crystalline quality and nowadays consolidated productive processes, its absorption efficiency decreases rapidly at higher X-ray energies. Therefore, high Z semiconductor sensors can be used to extend the efficient detection range of the system, with chromium compensated GaAs representing one possible alternative [2] [6].

Indeed, a 500 μm thick GaAs sensor exhibits a photon absorption efficiency of approximately 98% at 30 keV, compared with approximately 15%[1] for a silicon sensor of the same thickness [2].

However, the use of high Z material is challenging, due to larger leakage currents, reduced material uniformity and consequent different or even worse charge transport properties. These mentioned effects translate to detector performance and characteristics in term of noise, spatial response [6].

[1] Calculated using the Beer–Lambert law, $\eta = 1 - e^{-\mu x}$, with $\frac{\mu}{\rho} = 1.436 \frac{cm^2}{g}$ and $\rho = 2.33 \frac{g}{cm^3}$ [48], for a 500 μm thick Si sensor.

2 MYTHEN III framework

The MYTHEN III ASIC developed at PSI features 128 independent readout channels, while a complete detector module combines ten ASICs to read out 1280 strips in parallel [4] [3]. In this thesis, measurements and tests have instead been performed using a reduced prototype made up of two readout chips connected to sensors with 50 μm and 100 μm strip pitch, respectively. Nevertheless, the architecture of the complete MYTHEN III module and detector system is introduced to provide the general framework in which the ASIC operates.

The following sections cover the general sensors characteristics, the ASIC architecture and its configuration, the analogue front-end behaviour, the available operation modes, and finally the module and complete detector system. Moreover, a theoretical overview of the main detector performance parameters relevant to the present work are subsequently introduced, while the associated experimental data is presented in the next chapters.

2.1. Sensors

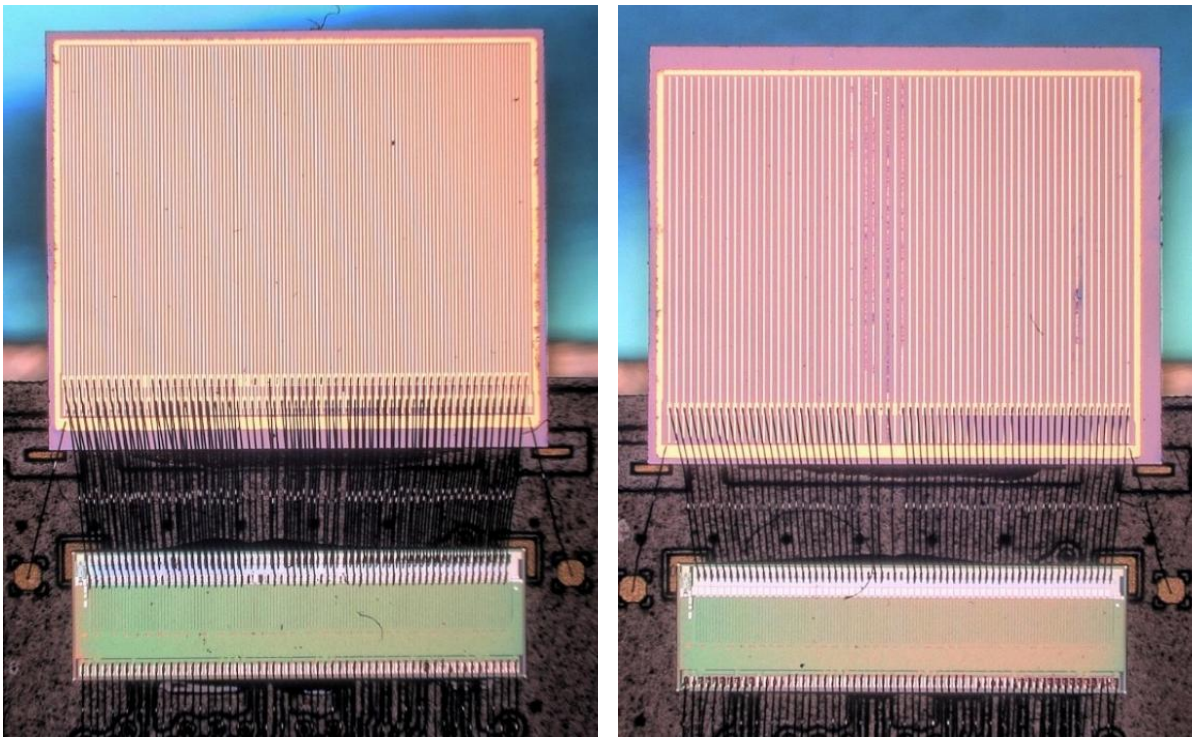
Throughout the years, the main option has been the employment of n-doped Si segmented on one side into 50 μm-pitch p+-doped strips, to have the behaving in a reserve biased diode to get a reasonable wide depleted region. At the same time, the backplane presents an n+ doped layer along with a thin Al layer so that a good electric contact for the bias voltage is provided [5].

For the MYTHEN sensors considered here, the layout is designed so that the radiation enters from the side opposite to the strip implants (backside illumination). This configuration provides a more uniform absorption efficiency across the sensor, since photons enter through the continuous backplane rather than the segmented strip side [5] [7].

Moreover, backside illumination can lead to signal attenuation for low energy photons, owing to the recombination of pairs generated near the rear surface, where the electric field may not be enough intense to ensure an efficient charge collection [5]. This would be a matter of concern for Si based detector but, in contrast, can be neglected for chromium compensated GaAs sensors which can be operated with ohmic contacts and an active volume extending through the sensor thickness, while providing a more uniform electric field (due to Cr compensation as detailed in Section

3.3.4) [8]. Moreover, their higher atomic number makes them attractive for X-ray detection at higher photon energies, where the efficiency of silicon rapidly decreases [2] [1].

In this work, a MYTHEN III prototype equipped with two chromium compensated GaAs microstrip sensors was studied, which have been produced and provided by DECTRIS AG, a spin-off company of the Paul Scherrer Institute [9]. Only two sensor positions of the module were instrumented, with strip pitches of 50 μm and 100 μm[2], respectively, while for both the thickness is 500 μm. Photographs of the two prototype sensors are shown in Figure 2.1, altogether with the corresponding MYTHEN III readout ASICs and wire bond connections: some missing wire bonds are visible, corresponding to inactive channels in the subsequent measurements [5]. For the 100 μm even some electrode strips are missing.



(a) 50 μm pitch (b) 100 μm pitch

Figure 2.1 Photographs of the two chromium-compensated GaAs microstrip sensors together with ASIC chips, respectively 50 μm (a) and 100 μm (b) strip pitch.

[2] The two sensors, from this time onwards, will be referred simply as “50 μm” and “100 μm” ones.

2.2. ASIC readout

The Mythen III "Application-specific integrated circuit", also known as ASIC (which must be discerned from an entire module itself, made up of 10 single ASIC), is produced with the newer 110 nm UMC technology, which shows better performance in terms of stability, variance among transistors and higher transistor per area density in comparison to the former MYTHEN II's 250 nm UMC production technology [7].

The ASIC is responsible for the amplification and shaping of the analogue pulse and then its conversion in digital counts for each strip (or more exactly for every counter of the three which every strip is provided of). The readout can read the 128 strips over four parallel data output lines [3].

2.2.1. Overview

The readout chip, wire bonded both to the sensor and to the HDI module itself, features 128 independent channels comprehensive, for each of them, of an independent preamplifier and shaper as well as a set of three comparators (whose threshold is global for the whole chip but can be adjusted per channel though a 6-bit trim bit) coupled with three independent 24-bit counters, whose logic varies according to the operation mode (which will be accurately covered in the next chapters). In this way the intrinsic noise, along the low-energy fluorescence photons, would be rejected. Furthermore, every counter is gateable, therefore a digital signal acts as a shutter for the acquisition time in which photons are counted [3]. The simplified architecture is illustrated below:

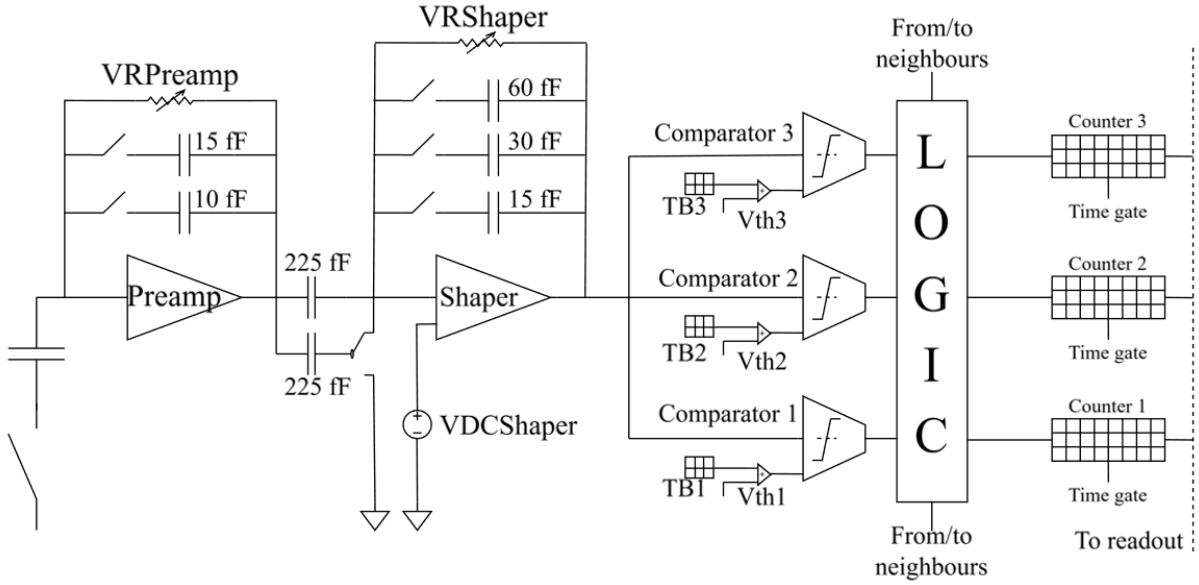


Figure 2.2 Simplified architecture of electronics of one ASIC chain. Based on schematics from Andrä et al. [3] [7].

Every amplifying stage exhibits an adjustable feedback resistance, which for both preamplifier and shaper come with voltage controlled MOSFET transistors,

specifically a p-type for the former, whereas for the latter there is the possibility of selecting between the usage of p-type or an n-type MOSFET accordingly to the pulse polarity which in turn depends on which charge (electrons or holes) is collected on the strips electrodes: indeed, the readout chip can manage flexibly both polarities by configuring properly the chip status register (see Section 2.2.5). Analogously, the mentioned MOSFETs are adjusted by setting the proper voltage value of *VR,preamp* and *VR,shaper*[3] for the preamplifier and the shaper, respectively.

Additionally, a set of capacitors in the readout chains is provided and adjusted for the entire chip to modify the gain for both the preamplifier and the shaper. The cited capacitances are tuned only in discrete steps by the opening or closing of some switches (transistor operated in a region so that they act as). These are implemented by MOSFETs operated between their ON and OFF states, differently from the voltage-controlled MOSFETs introduced above, which are operated in the linear region to behave as equivalent resistances.

The available combinable sets are in the following [7]:

- *Preamplifier* (C_{preamp}): 10, 15 fF
- *Shaper* (C_{shaper}): 15, 30, 60 fF
- *AC coupling* (C_{AC}): 225, 450 fF (not combinable)[4]

The feedback capacitances are practically activated or turned off by sending to the chips the right combination of active bits, except for the AC coupling that are complementary selectable. The choice happens at a chip status register level that also allows for the mode selection (such as standard, interpolation mode, and so on) and for the polarity set-up of the comparator output, which must be tuned accordingly to the sensor one.

Furthermore, the outward output feeds the set of three comparators which every readout channel is provided with. Each comparator answers positively only the incoming pulse crosses over a certain threshold defined by a global reference voltage [3] [7].

Since unfortunately each comparator behaves slightly differently, the effective threshold hence is not uniform throughout the ASIC. For this reason, every comparator is provided with six tuneable *trimbits* (called TB_N) able to fine-adjust the comparator, with an intensity defined by the reference voltage *Vtrim*. Thereby it is possible to make up for the discrepancies in terms of response in the counting logic process (which is provided with 24 bit counters) due to the inevitable variance among the transistors [3]. This aspect is addressed in Section 6.6.

[3] From here onwards, the VR,shaperNpol will be referred simply as VR,shaper when dealing with GaAs
[4] When 225 fF as coupling capacitance is selected, then one capacitor is grounded

2.2.2. Schematic description

In MYTHEN III's ASIC readout chips, every channel includes a charge sensitive preamplifier (CSP) followed by a shaper: both stages possess adjustable feedback networks that allow for adjusting parameters like the gain and shaping time, leading to different electronic speed as well as noise [3].

2.2.2.1. Preamplifier

The sensors' signal can be represented by a current $i_{det}(t)$, which comes from the charge drifting and inducting on the electrode (see Shockley-Ramo's theorem, Section 3.2.2). The preamplifier must therefore convert the current into a voltage [10].

In a simplified model, the feedback network can be represented by an effective resistance $R_{f,pre}$ in parallel with the feedback capacitance $C_{f,pre}$, where the equivalent resistor is operated by a p-FET [7]. The associated transfer function (that coincides with the feedback transimpedance) can be written as follows:

$$Z_{f,pre}(s) = H_{pre}(s) = \frac{R_{f,pre}}{1 + sR_{f,pre}C_{f,pre}} \tag{2.1}$$

Therefore:

$$V_{out}(s) \cong -I_{det}(s) \cdot H_{pre}(s) \tag{2.2}$$

Assuming $t_{collection} \ll R_f C_f$, the charge is mainly stored in $C_{f,pre}$, with $R_{f,pre}$ slowly discharging C_f, hence obtaining the classical CSP gain behaviour:

$$G_{CSP} = \frac{\Delta V_{pre}}{Q} \cong -\frac{1}{C_{in} + C_{f,pre}} \tag{2.3}$$

An ideal CSP would show only a dependence on the feedback capacitance $C_{f,pre}$, but for semiconductor detectors it should also be considered the detector capacitance C_{in}, which also affects negatively the noise and the rise time [10].

The gain condition expressed in (2.3) is mainly verified for higher $R_{f,pre}$, since it reduces the capacitor discharge during the pulse formation, thus making easier for the amplifier to reach the ideal limit $V_{max} = Q/C$. Indeed, when R_f is sharply decreased, the charge is partially depleated even during the collection, so a dependance from $R_{f,pre}$, as well as from the current injection period, is introduced: their effects can be seen as a *ballistic deficit* on the pulse itself, thus affecting the effective gain and the return to baseline speed. This is due to some of the charge discharging on the resistance even during the pulse formation, then reducing the final amplitude [7]. On the other hand, an excessively high $R_{f,pre}$ would constitute a problem for the electronics, due to

the leakage current passing though the feedback resistance that brings about an excessive voltage drop over the latter, therefore ending up saturating the CPS:

$$\Delta V_{DC} \sim I_{leak} R_{f,pre} \tag{2.4}$$

Moreover, for a simplified rectangular current pulse of duration T_c, and considering an ideal CSP, the ideal peak voltage can be therefore derived as:

$$V_{peak} = \frac{Q}{C_f} \frac{\tau_{f,pre}}{T_c} \left(1 - e^{-T_c/\tau_{f,pre}}\right) \tag{2.5}$$

where $\tau_{f,pre} = R_{f,pre} C_{f,pre}$ is the CSP time constant (which is different from the one describing the whole electronic chain dynamics).

In conclusion, an ideal amplifier "amplifies" at the same way both signal and noise, thus keeping a constant SNR. But every further stage introduces its own noise. Hence if the first one (the preamplifier) provides a large enough signal, the noise introduced from the shaper, comparators and so on, relatively weights less. So, in terms of gain, the preamplifier is on purpose the dominant stage [10].

2.2.2.2. Shaper

The shaper receives the pulse from the preamplifier through a coupling capacitance C_{AC}. Like the CSP, the charge pulse is integrated on the feedback capacitance $C_{f,sh}$. Its gain can be described (neglecting parasitic capacitance, for instance) as follows:

$$G_{sh} = \frac{C_{AC}}{C_{f,sh}} \tag{2.6}$$

The shaper works around a well defined DC point, set up by the *VDC,shaper* voltage which eventually determines the baseline level. Given this continue voltage component, if the $R_{f,sh}$ grows excessively then the required voltage to close the feedback could lead the amplifier (specifically, the feedback transistor) to saturation or out of MOSFETs' linear working range.

The feedback and coupling impedances can be respectively written as follows:

$$Z_f(s) = \frac{R_{sh}}{1 + sR_{f,sh}C_{f,sh}} \tag{2.7}$$

$$Z_{in}(s) = \frac{1}{sC_{AC}} \tag{2.8}$$

Then, for this ideal inverting amplifier the transfer function is:

$$H_{sh}(s) = -\frac{Z_f}{Z_{in}} = -\frac{sR_{f,sh}C_{AC}}{1+sR_{f,sh}C_{f,sh}} = -\frac{C_{AC}}{C_{f,sh}}\frac{sR_{f,sh}C_{f,sh}}{1+sR_{f,sh}C_{f_{sh}}} \quad (2.9)$$

which for $s \to 0$ it tends to zero as well, while for $\omega R_{f,sh}C_{f,sh} \gg 1$ the function saturates to $H_{sh} \cong -\frac{C_{AC}}{C_{f,sh}}$, then leaving out the resistance and depending only on the capacitance ratio. Instead, the resistance still determines the corner frequency $\omega_C = \frac{1}{R_{f,sh}C_{f,sh}}$ which modifies the pulse shape and the band width (and consequently the noise cut).

No deleterious effects, like the ones affecting the preamplifier due to excessive leakage currents, arise when $R_{f,sh}$ increases too much, since the coupling capacitor gets rid of the eventual DC components incoming. Nevertheless, the shaper has its own biasing and feedback network, which must remain within its valid operating range, otherwise transistors making up the shaper would be push outside of their ideal working points. As already mentioned, the coupling capacitance serves as decoupler between the two stages, to allow them for working in different DC point by getting rid of the DC components (for instance, the *VDC,shaper* voltage does not interfere with the preamplifier).

2.2.2.3. Overall gain

Ideally (neglecting also the input capacitance), the overall transfer function is as follows, where the time constant of preamplifier and shaper are respectively $\tau_{f,pre} = R_{f,pre}C_{f,pre}$ and $\tau_{f,sh} = R_{f,sh}C_{f,sh}$:

$$H(s) = H_{pre}(s)H_{sh}(s) = R_{f,pre}\frac{C_{AC}}{C_{f,sh}}\frac{s\,\tau_{sh}}{(1+s\,\tau_{pre})(1+s\tau_{sh})} \quad (2.10)$$

And, assuming as before $t_{collection} \ll R_fC_f$, an ideal gain as follows:

$$G \propto \frac{1}{C_{f,pre}}\frac{C_{AC}}{C_{f,sh}} \quad (2.11)$$

The band pass transfer function is depicted below in Figure 2.3, normalised to the factor $R_{f,pre}\frac{C_{AC}}{C_{f,sh}}$. For illustrative purposes, the overall time shaping has been fixed approximately at 100 ns (from which the peak frequency $f_{peak} \cong 1.59\,MHz$ has been derived), as found for standard settings in Bergamaschi et al. [4]. The single contributes, since not specified, have been graphically separated (the ratio of the two is 10) while keeping $\tau_{eqv} = \sqrt{\tau_{\mathrm{pre}}\tau_{\mathrm{sh}}} = 100$ ns [5] for graphical purposes.

[5] This "equivalent" τ_{eqv} is the inverse of the pulsation ω_0 which solves $\frac{d|H(j\omega)^2|}{d\omega} = 0$, thus the ω_0 that maximises the transfer function module. The solution is eventually $\omega_0 = {}^{1}\!\Big/\!_{\sqrt{\tau_{pre}\tau_{sh}}} = {}^{1}\!/\!_{\tau_{eqv}}$

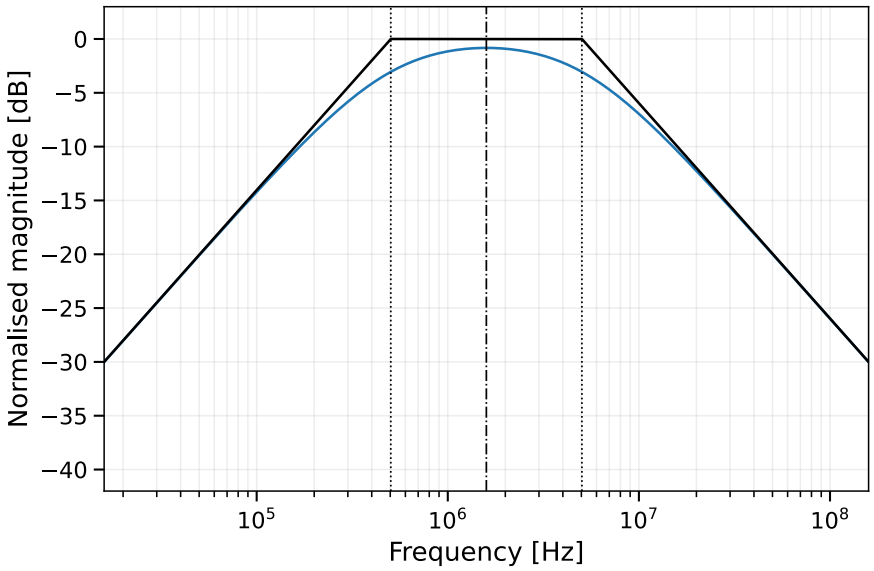


Figure 2.3 Normalized band pass transfer function, with its Bode asymptotic approximation, centred on frequency $f_{peak} = {}^{1}/_{2\pi\tau_{eqv}} \cong 1.59\ MHz$ [6]

In conclusion, the overall function behaves as a band pass filter, with C_{AC} suppressing lower frequency components of noise (which is assumed as *white*) and the preamplifier attenuating the higher ones, eventually cutting out most of the noise and producing a semi gaussian shaped pulse.

In terms of gain, the feedback capacitances strongly influence it, as seen in Eq. (2.11), while the feedback resistances mainly control the temporal response, such as the corner frequencies and therefore the pulse duration and return to baseline. They can also affect the gain by deviating the readout from the ideal conditions, for example due to *ballistic deficit* effects [7].

[6] The two assumed shaping time constants, from which pole frequencies are derived, are respectively 316.23 ns and 31.62 ns, whose corresponding frequencies are represented in Figure 2.3 by the dotted lines, while the dash-dot stands for the peak one.

2.2.3. DAC units

The MOSFETs mentioned in Section 2.2.1, which act as equivalent resistances when operated in their linear region, are controlled by voltages. These voltages, together with the other analogue reference voltages of the ASIC, are set digitally by the user. Therefore, *a Digital to Analog Converter* (DAC) is required, consisting of an electronic circuit that converts a digital input value into the corresponding analogue voltage.

To mention one of the possible applications, for the ASIC operation the comparator threshold must be set by the outside on the readout board: hence an integer (such as 500, 1000, 2000 and so on) is sent to the chips and therefore it is translated in millivolt.

However, the *DAC unit* conversion into physical voltage depends on the readout hardware [7]. For the *Strip Readout Board* (SRB), the conversion is as follows:

$$1\ DAC\ unit = 0.5\ mV$$

while it is slightly larger for the *Chip Test Board* (CTB), used for single-chip measurements:

$$1\ DAC\ unit = 0.61\ mV$$

In the measurements in this work, the CTB conversion is relevant for the analogue reference voltages used to configure the readout, such as the preamplifier and shaper feedback settings. For the complete module instead, the SRB conversion applies to the voltages set by the board, such as the comparator thresholds which define the voltage levels that the analogue pulse must cross to generate a count.

2.2.4. Readout speed and dynamic range

The digital readout architecture also determines the acquisition speed and the amount of data that can be stored for each frame. Each MYTHEN III channel provides three independents 24 bit counters, whose readout depth can be reduced also to 16 or 8 bit: a lower bit depth decreases the amount of data transferred and therefore allows a higher maximum frame rate. As reported in Andrä et al. [3], the maximum frame rate ranges from 40 kHz, when all three 24 bit counters are read, up to 360 kHz when only one 8 bit counter is used. Moreover, MYTHEN III supports dead time free readout: the content of a counter can be transferred to an internal buffer while a new acquisition is already started [7].

Additionally, the counter depth also defines the dynamic range of a single frame, which defines the maximum number of counts that can be stored per frame and per channel: when the full 24 bit depth is employed, the maximum is $2^{24} - 1$ counts per counter. For SPC, the total number of recorded photons can in principle be extended without a fixed upper limit if the consecutive frames are summed up. However, a trade off exists between dynamic range and frame rate, since reducing the counter depth from 24 to 16 or 8 bits allows a higher maximum frame rate.

2.2.5. Chip status register

The operation of the MYTHEN III ASIC is configured through an 18 bit *Chip Status Register* (CSR), which defines the operating modes of the chip (standard, interpolation, pump-probe and so on) and controls additional feedback equivalent resistors and capacitances in the preamplifier and shaper, thus managing parameters like the gain.

The CSR includes configuration bits for several functions of the ASIC, some of which have been extensively employed and then are relevant for the operation modes considered in this work [7]. The main settings are summarized in Table 2.1.

Table 2.1 Main Chip Status Register (CSR) settings used in this work.

CRS setting	Function
INVPOL	Enables negative polarity operation
INTERP	Enables on-chip interpolation
C10pre	Adds 10 fF to the preamplifier feedback capacitance
C15pre	Adds 15 fF to the preamplifier feedback capacitance
C15	Adds 15 fF to the shaper feedback capacitance
C30	Adds 30 fF to the shaper feedback capacitance
C60	Adds 60 fF to the shaper feedback capacitance
C225	Controls the shaper AC-coupling capacitance

While the CSR defines the global configuration of the ASIC, the settings of the individual channels are controlled through the *Strip Status Register* (SSR). The SSR contains the trim and mask bits associated with the three comparators of each channel, allowing their thresholds to be individually adjusted and selected comparators to be disabled. In practice, the ASIC configuration is loaded through a configuration file using the detector control software.

2.3. The MYTHEN III complete module

Each module presents ten ASICs glued to a custom HDI, which stands for *High-Density Interconnect*. The HDI provides the electrical connection between the ASICs and the readout board. The latter instead hosts the FPGA module with an embedded Linux system, reachable by using a 1 Gbps Ethernet connection. There is also the possibility of streaming data (using a UDP protocol, while the TCP/IP one manages the control purposes) either over the 1 Gbps control or a 10 Gbps fiber-connection is case of larger amount of streamed data [3].

For diffraction experimenters or more generally whenever a larger active area is required, the parallel design of the MYTHEN III allow for the combination of more modules. To synchronize and manage the control and data acquisition of more different modules, one of them must therefore be designated as master, which then distributes the trigger to the remaining modules through the synchronization connection.

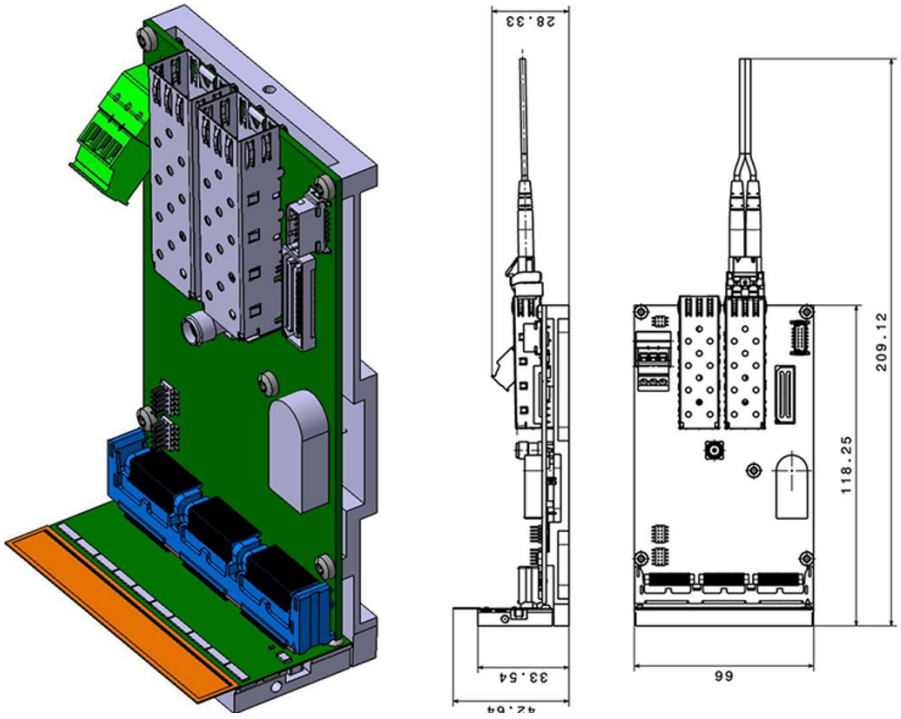


Figure 2.4 Sketch of a complete MYTHEN III module (a) and readout board (b). Reproduced and adapted from Andrä et al. [3]

Indeed, what must be considered is one of the main purposes of the MYTHEN III, namely the powder diffraction experiments, which require wide angular coverage. The complete MYTHEN III detector installed at the MS beamline of the SLS (Figure 2.5) consists of 48 modules, corresponding to 61 440 readout channels, arranged in two parallel rows of 24 modules each. The two rows are shifted by approximately 1° allowing the gaps between neighbouring modules to be covered and providing a continuous angular range of 120° in 2θ [3].

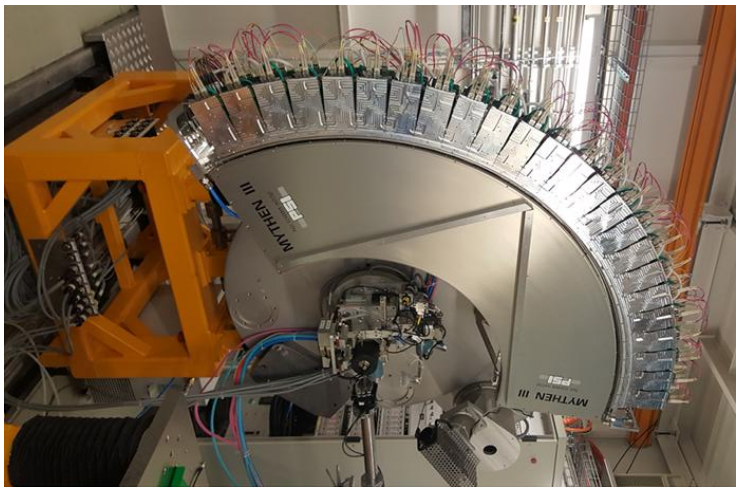
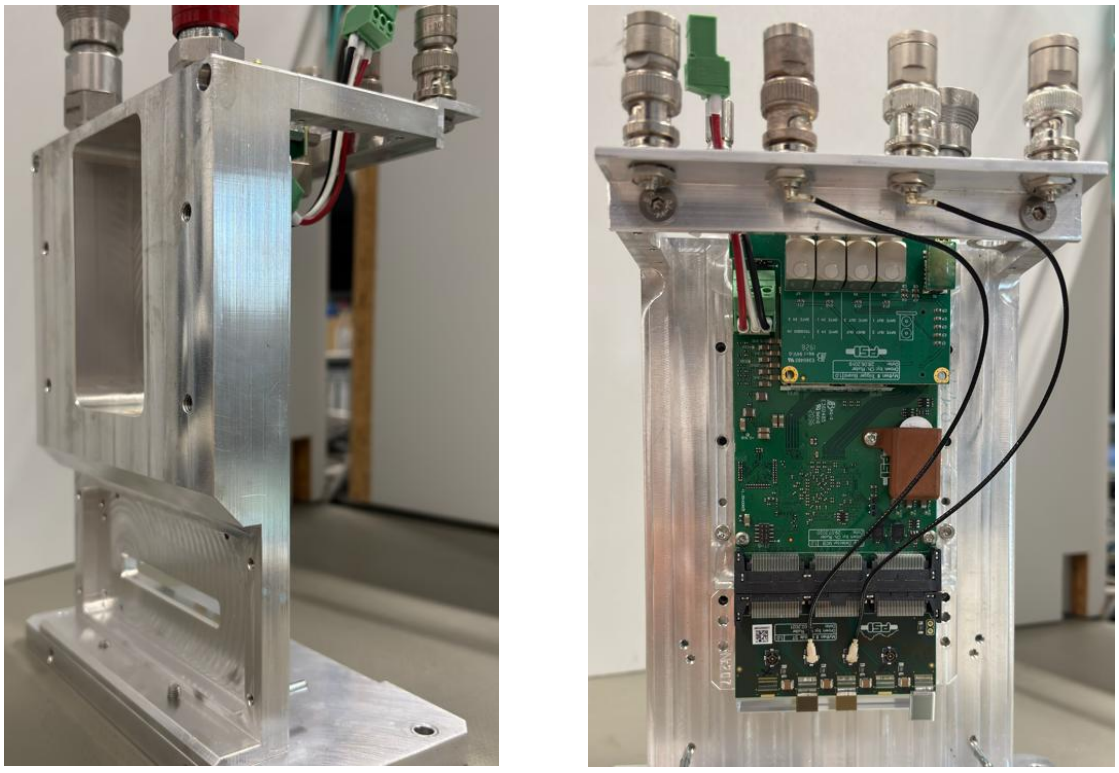


Figure 2.5 Row of modules installed at the MS beamline of the SLS. Reproduced and adapted from Andrä et al. [3]

The complete MYTHEN III prototype employed in this work is shown in Figure 2.6 from two different sides. The module is mounted on a support used for its mechanical installation and cooling. The electrical connections and the wiring used for the sensor bias are also visible. Other connections are the power supply, the 1 Gbps Ethernet and 10 Gbps fiber connection, the inlet and outlet of the cooling system.



(a) Side view with entrance window (b) Readout side

Figure 2.6 Prototype MYTHEN III employed in this work: (a) side view with the entrance window visible; (b) readout side showing, from left to right, the GaAs sensors of 50 µm and 100 µm strips pitch, while the last silicon sensor was not operational.

2.4. Operative modes

The MYTHEN III chip can be configured with the purpose of making the most out of the incoming signal in different modes, hence different logic can be tweaked at a chip-level so that the hardware can work as a simple counter or in a way that the spatial resolution is hugely improved, as explained below [3] [7]. Note that some of the offered modes, such as pump probe and the testing ones, will not be covered here since they have not been used in this work.

2.4.1. Standard mode

In standard counting mode, the comparator simply records when the pulse gets over the comparator thresholds for a certain set exposure time. Moreover, each counter can be gated independently, allowing different acquisition time windows, while the comparator thresholds (expressed in DAC units) can also be set independently: then it is possible to take advantage of different threshold for the comparators in order to set up an energy window (similarly to an MCA, despite being a single photon counter) or to get rid of piling up negative effects.

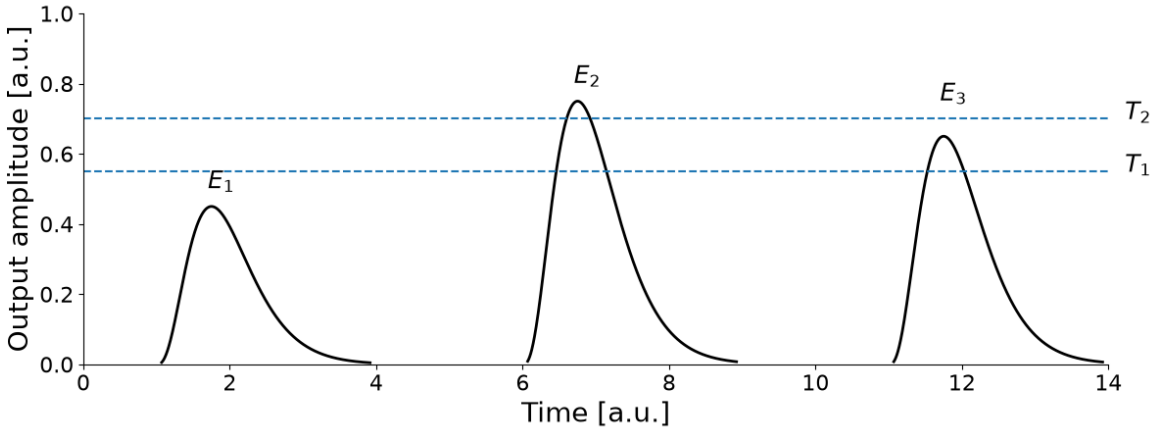


Figure 2.7 Schematic representation of energy discrimination in standard mode. The setup of different comparator thresholds would make possible to discern different energy ranges.

2.4.2. Inter-strip interpolation mode

The interpolation mode makes the most out of the charge sharing phenomenon, that consist in the partial collection of the charge cloud by multiple neighbour strips rather than one (this topic will be extensively addressed in Section 8.1). This will lead to a spread of the signal such that the pulse could be rejected by the counters due to the signals' resulting low amplitudes, or counted twice, according to the comparator threshold.

For this purpose, two comparators per channel are active, with a high (for the central) and a low threshold (for the side ones) set up: the *High Threshold Comparator* (HTC) identifies the photon event, while the *Low Threshold Comparator* (LTC) is used to detect

the presence of shared charge in the neighbouring strips [4]. In the MYTHEN III configuration, *Vth1* is used as the low threshold for the LTC, while *Vth2* is used as the high threshold for the HTC [7] (in Figure 2.8 these are respectively referred to as *VthHigh* and *VthLow*).

When the high threshold is exceeded, the interpolation logic checks out if the low one is also overcome in the neighbouring strips. So, depending on the coincidence, the event is assigned to the left counter (LC), central one (CC) or right one (RC) of the strips: if neither neighbouring strip exceeds the low threshold, the photon is assigned to the CC. If the low threshold is exceeded by the left or right neighbour, the event is instead assigned to the corresponding LC or RC, indicating that the photon was absorbed close to that strip boundary.

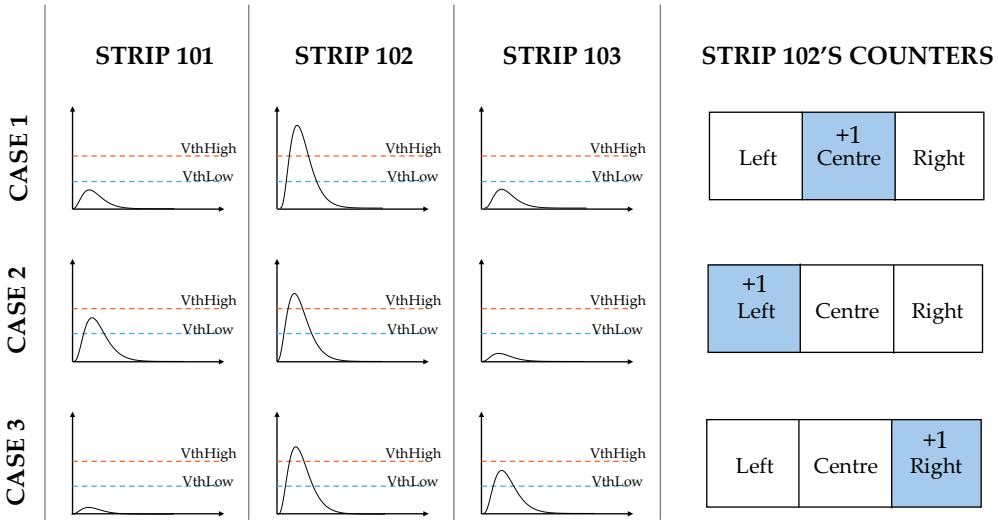


Figure 2.8 Schematic of the MYTHEN III interpolation logic. Based on Andrä [7], Fig. 3.2.2.

Eventually, the low threshold therefore allows the small fraction of charge shared with a neighbouring strip to be detected even when it would remain below the normal photon counting threshold. Furthermore, the interpolation logic therefore associates the events with three virtual regions within each physical strip providing positional information below the physical strip pitch without the need of increasing the number of sensor electrodes or readout channels.

An additional advantage of this approach is that the central counters come out as (almost) free of charge sharing effects, since the event affected by charge sharing will be redirected to side counters, if the low threshold is set as low as possible to work efficiently (but high enough to avoid false coincidences due to noise). In this way the central counter behaves as if a physical collimating mask were placed in front of the sensor [7].

The interpolation mode cannot be used with the 100 µm pitch sensor, since the ASIC is only designed for a 50 µm, hence only every second ASIC channel is connected. Therefore, two physically neighbouring channels are not connected, and the interpolation logic cannot work.

2.4.3. Negative polarity mode

For this operative mode, the standard shaper feedback transistor is disabled and substituted by the dedicated negative-polarity one. Moreover, the shaper baseline and threshold levels must be readjusted accordingly (see Section 6.1).

This configuration is particularly relevant for GaAs:Cr sensors, where the detector is operated in electron collection mode and thus the ASIC must process incoming negative signals.

2.5. Comparison with charge integrating detectors

For charge integrating (CI) detectors, the rate capability is mainly given by the product of the frame rate and the dynamic range, since each frame can integrate only a finite amount of charge. For SPC detector instead, the maximum rate does is not directly determined by the frame rate, rather by the minimum time interval required to discern consecutive photon pulses. Therefore, a SPC detector may provide a very high frame rate while still suffering from counting losses at high flux due to pile up [7]. This difference becomes particularly relevant at very high fluxes, where SPC detectors suffer from saturation due to overlapping analogue pulses, while CI detectors could better integrate the charge produced by multiple photons within the same frame: indeed, the CI detector development at PSI has been strongly motivated by XFEL, photon sources where the fluxes are very high [11].

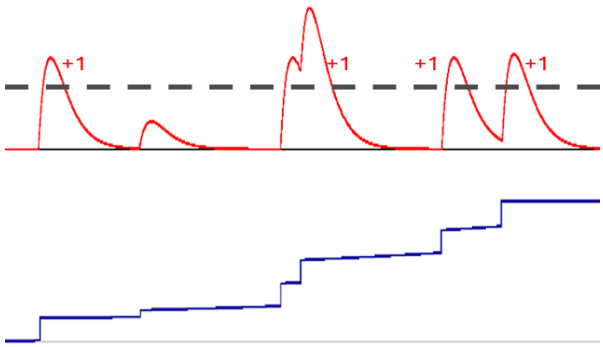


Figure 2.9 Photon counting comparison between SPC (upper) and CI (lower) detectors. The deleterious effects for SPC (pile up) and CI (noise/fluorescence/low energy background) are also depicted. Reproduced and adapted from [12].

Moreover, unlike CI detectors, where the measured signal is proportional to the total charge collected, in a SPC detector each pulse crossing the threshold normally contributes one count independently of its amplitude, Therefore, higher energies photons are not given a proportionally larger weight. Moreover, thanks to the threshold, the electronic noise (for instance, shot noise due to leakage current), as well as the fluorescence photons (if their energies lie below the threshold), can be discarded and suppressed [7].

3 Semiconductor Sensors for X-ray Detection

The employment of semiconductor for charged particles and X-ray and gamma radiation is common due to their outstanding qualities in terms of energetic and spatial resolution, given the larger number of information carriers (electron-hole pairs) generated for each radiation event, especially when compared to gas and scintillator: for these reasons, strip or pixel semiconductor sensors have been devised. Another outstanding advantage are their compact dimensions and faster timing resolution given by their charge carrier mobility. The most studied and used material for semiconductor sensors are Silicon (Si), Germanium (Ge), Gallium Arsenide (GaAs) and Cadmium Tellurium (CdTe), with the two latter able to cover also the higher energies spectrum.

Table 3.1 Main material properties of Si, GaAs and CdTe relevant for semiconductor radiation detectors. Data adapted from Fiederle et al. [13] and Chen et al. [14].

Material properties	Si	GaAs	CdTe
Crystal structure	Diamond	Zinc blende	Zinc blende
Atomic number Z	14	31, 33	48, 52
Density [g cm^{-3}]	2.33	5.32	5.85
Band gap [eV]	1.12	1.43	1.44
Pair creation energy [eV]	3.62	4.2	4.43
Resistivity [Ω cm]	10^4	10^9	10^9
Electron mobility [cm^2 V^{-1} s^{-1}]	1350	8500	1100
Hole mobility [cm^2 V^{-1} s^{-1}]	480	400	100
Electron μτ-product [cm^2 V^{-1}]	>1	10^{-4}	10^{-3}
Hole μτ-product [cm^2 V^{-1}]	>1	10^{-6}	10^{-4}

In this work, both Si and chromium compensated GaAs (GaAs:Cr) are considered, since the former provides the reference of the MYTHEN detector, while the latter is investigated as an alternative sensor material for extending its operation towards higher X-ray energies. Therefore, the following sections first introduce the main

physical principles of semiconductor sensors and microstrip detection, before discussing the specific ones of GaAs:Cr.

3.1. Fundamentals of Semiconductor Detectors

3.1.1. Band Structure and Charge Transport

The properties of a semiconductor material are defined by the allowed and prohibit energy bands and gaps which eventually depends on their periodic arrangement of atoms in the lattice. Hence the resulting band structure governs the electrical and thermal behaviour.

In a crystalline solid, electrons can only occupy specific energy ranges: the most relevant are the valence and the conduction band. The valence band contains covalent binding electrons (outer shells ones) and, for semiconductors and insulators, it is typically filled at lower temperatures. Instead, the conduction band lies at higher energy and contains electrons free to move through the periodic lattice and so they actively contribute to electrical and thermal conduction.

Indeed, the mentioned bands are separated by the forbidden gap, whose width determines if the material behaves as a conductor, semiconductor, or insulator. In the absence of thermal excitation (hence theoretical zero temperature), both semiconductors and insulators would have a filled valence band and an empty conduction band. For the metal instead the highest occupied band is partially filled and so the electrons can easily move without significant earlier excitation: this translates into much higher conductivity with respect to the one pertaining to the other class of materials where first the electrons must overcome the bandgap [10] [15] [16].

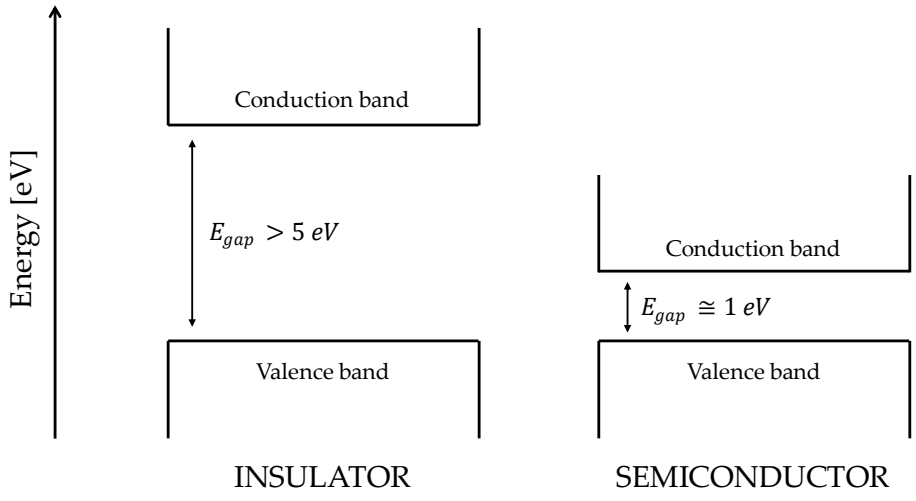


Figure 3.1 Representation of insulator and semiconductor forbidden gaps

The mentioned excitation process leads to the creation of an electron-hole pair [10] [15], which represents the fundamental unit of the generated charge (analogous to the ion pairs for gas detectors) and both contribute to the material conductivity. The probability of thermal generation follows the following probability per unit time, where T is the absolute temperature, E_g the bandgap energy, k the Boltzmann constant and C a proportional constant characteristic of the material [16]:

$$p(T) = CT^{\frac{3}{2}} \mathrm{e}^{-\frac{E_g}{2kT}} \tag{3.1}$$

Unless separated by an external field, these thermally generated pairs eventually recombine, then achieving a temperature-dependent equilibrium concentration. In practice, crystal impurities create recombination centres that disturb the band structure by introducing energy levels in the bandgap, with the result of enhancing recombination, thereby drastically reducing the lifetime of carriers. Eventually, this results in charge loss in case of the collection time exceeding the lifetime and consequently degrades the energy resolution [10] [16]..

Instead, when an external electric field is applied, charge carriers experience a net migration, superimposed on their random thermal motion, which spreads as a function of time. At low to moderate electric field intensities, the drift velocity of the carriers grows linearly with the electrical field, as depicted below [10]:

$$v = \mu E \tag{3.2}$$

However, at higher electric field values typically used in semiconductor detectors, the drift velocity increases sublinearly and eventually saturates: however the latter, on the order of $10^7 cm/s$, allows for highly rapid charge collection, often under 10 ns for typical detector dimensions which leads to fast timing resolution. For this reason, most detectors operate with electric fields high enough to saturate drift velocity.

The broadening disruption of charge follows approximately a gaussian function with the following standard deviation, where $D = \mu kT/e$ is the diffusion coefficient and t the time. Assuming also the linear dependence in Eq. (3.2):

$$\sigma = \sqrt{2Dt} = \sqrt{\frac{2kTx}{eE}} \tag{3.3}$$

Which puts in clear view the dependence on the drift distance: indeed, for thicker sensors where charge carriers must travel longer, this increased spatial broadening puts a cap to the achievable position, as well as worse timing resolution. Since the diffusion constant depends on the mobility and at the same time the collection time is inversely proportional, so the diffusion has the same value for both the type of charge

carriers [10] [17]. Moreover, the spread out of charge carries relies not only on the diffusion process but also on the Coulombian repulsion, although being relevant only for higher X-ray energies or charged particles interactions [17].

3.1.2. Intrinsic and Doped Semiconductor

Due to thermal generation, in an ideal pure semiconductor the number of electrons equals the number of holes. This class of materials is called *intrinsic semiconductors*. However, getting totally intrinsic materials is almost impossible in practice, due to domination by residual impurities that affect the electrical properties even if in small quantities. The concentration of free career in *i-type semiconductors* is defined as follows, where N_c and N_v are the effective densities of states in conduction and valence band, respectively [10] [16]:

$$n_i^2 = N_C N_V \exp\left(-\frac{E_C - E_V}{k_B T}\right) = N_C N_V \exp\left(-\frac{E_g}{k_B T}\right) \tag{3.4}$$

For doped materials, the latter is affected by the concentration of injected dopants, although also the impurities have a role in. This leads to the alteration of fundamental figures of merit, both negatively and positively, such as the resistance and then the leakage current [10].

3.1.3. Silicon p-n Junctions

For an X-ray detection sensor based on silicon, it is necessary to form a p-n junction to make the device usable. Without such a junction, the dark current would be excessively large compared with the small current generated by the photon induced charge carriers, making the detector signal difficult to measure [10] [15].

The p-n junction is the interface between p-type and n-type regions, with the two latter formed by intentionally injecting impurity atoms into the crystal lattice, a process known as *doping*. Thanks to the latter, it is possible to create shallow energy levels close to the conduction or valence band, so that only a small amount of thermal energy is needed to ionize the dopants and release free carriers. In n-type material, donor impurities provide electrons as majority carriers, whereas in p-type material acceptor impurities create holes as majority carriers [10] [16].

At the junction, the concentration gradient pushes electrons and holes to diffuse in opposite directions, where they recombine. This leads to the creation of ionized regions: positive on the n-side and negative on the p-side. As a result, a built-in electric field and contact potential are generated, which oppose further diffusion. This region, depleted of mobile carriers, is called the depletion zone and it is characterized by a substantial resistivity value [10] [16].

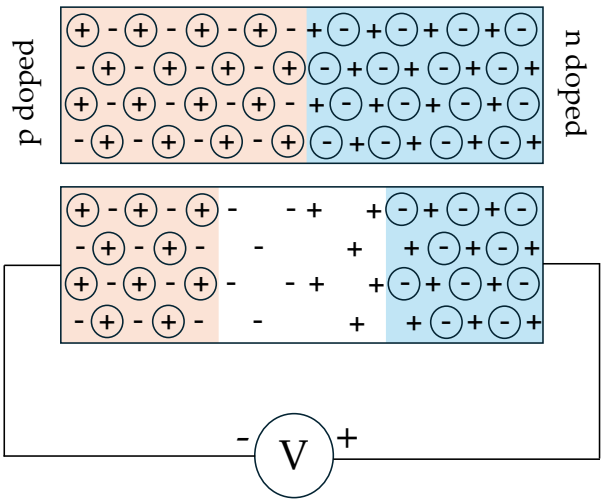


Figure 3.2 Representation of p-, n-type and depleted regions

Consequently, this zone becomes the active detection volume since any electron-hole pair created there by ionizing radiation is rapidly swept away by the electric field and collected at the electrodes as a measurable current signal. The size of the depleted region widens along the reverse bias voltage rise, as shown in the following relationships (the second considers the built-in voltage V_{bi} and the minority approximation), where ε is relative electrical permeability and N_{min} the minority dopant concentration [10]:

$$w_d = x_p + x_n = \sqrt{\frac{2\epsilon V}{e}\frac{N_A + N_D}{N_A N_D}} \cong \sqrt{\frac{2\varepsilon}{e}\frac{V_b + V_{bi}}{N_{min}}} \tag{3.5}$$

Moreover, the junction acts as a capacitance since it has been depleted of free charges, which can be expressed as [10]:

$$C = \epsilon\frac{A}{w_d} = A\sqrt{\frac{\epsilon e N}{2(V + V_{bi})}} \tag{3.6}$$

Or, expressing it as a specific capacitance (technical unit), which is expected to be in the order of less than 1 $\frac{pF}{mm^2}$ [10]

$$\frac{C}{A} = \frac{\epsilon}{w_d} \approx 1\left[\frac{pF}{cm}\right]\frac{1}{w_d} \tag{3.7}$$

3.1.4. Compensated Semiconductors

As already mentioned, in practical semiconductor manufacturing it is impossible to achieve perfectly pure intrinsic materials, as residual impurities always manage to dictate the electrical properties. However, if donor and acceptor impurities are present in exactly equal concentrations, the material is then called *compensated*. In this state, the extra electrons provided by donor impurities are captured by the available sites of acceptor impurities, or vice versa. From a solid state physics point of view, the compensator manages to shift the Fermi level (that define the upper limit of occupied states if at zero temperature) into the gap band and ultimately achieving a semi-conductive material. So, the mutual cancellation gives the material close-intrinsic properties despite the presence of dopants. In this way, compensated semiconductors are commonly designated as *i-type* material and are characterized by higher resistivity even without the presence of a depleted zone [8] [15].

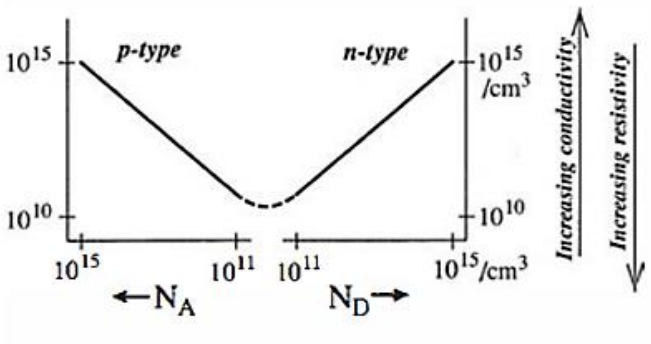


Figure 3.3 Plot depicting the conductivity of a semiconductor according to the acceptors and donors concentration, with a minimum where the latter compensate each other. Figure extracted from Knoll [15]

In practice, exact compensation during production stage is extremely difficult [8] for standard materials as even a microscopic imbalance in dopant concentration quickly causes the material to revert to either n-type or p-type behaviour. Hence, the only feasible way to get compensation is through dopant injection (see Section 3.3.4), rather than fabrication stages.

For the capacitance of the free-mobile careers bulk, the same equation of the p-n junction (3.6) can be used, provided that the depleted region width is taken equal to the sensor thickness itself.

3.1.5. Trapping and Recombination

Referring to the electron-hole drift, theoretical models predict that carrier lifetimes in perfectly pure crystals could reach up to a second. Nonetheless, experimentally observed lifetimes are typically orders of magnitude shorter (even four orders of magnitude): this difference is explained by the negative effects of residual impurities and structural defects presence within the material [18].

Deep level energies coming from impurities create energy states near the middle of the bandgap that can behave as either traps or recombination centres, since they occupy substitutional lattice positions and they are consequently called *deep impurities*: this is different from donors or acceptors established level, whose energy is closer, respectively, to conduction or valence band and for this reason are called *shallow levels* [19].

So, impurities presence leads to having more free charge concentration (and so more leakage current as explained in the next chapter) as well as traps, which capture and temporarily immobilize carriers and so affect both the mobility and the lifetime. If the pairs are captured by traps and the release time exceeds the collection time, the carrier is effectively lost from the measured signal, but it is released at the end. Instead, the recombination centres can capture sequentially both an electron and a hole, so leading to their mutual cancellation in time.

For instance, in GaAs:Cr this mechanism is particularly relevant for holes. Chromium acts as a deep acceptor and introduces deep levels responsible for the compensation of the residual donor concentration of the original n-type GaAs. Therefore, Cr shows an electron capture cross section relatively small, resulting in considerably better electron transport, whereas holes are trapped much more efficiently due to the higher cross section offered [20] [21]. Consequently, the hole lifetime is significantly shorter than the electron lifetime.

Both the mentioned mechanisms therefore reduce the average carrier lifetime, causing charge loss and directly degrading the detector's energy resolution. The magnitude of this loss is characterized by the trapping length, which in turn depends on the so-called *mobility-lifetime* or *mu-tau* product [8]: the latter's unit is $cm^2\ V^{-1}$ while for the electric field is V/cm, so the results is a length: $\lambda = \mu\tau E$

The trapping length thus must be significantly larger than the physical thickness of the sensor to ensure complete average charge collection during operation, so a good trade off with the absorption efficiency must be looked for. Indeed, low Z materials can make up for the low stopping power by increasing the thickness at the expense of more trapping and charge sharing [8].

For instance, considering the sensors studied in this work and the results in Greiffenberg et al. [6], where for a 500 μm GaAs:Cr operated at -300 V, the electron μτ product had been estimated as $\mu_e\tau_e = (4.730 \pm 0.003)\ 10^{-4}\ cm^2V^{-1}$. Considering also an electric field $E \cong \frac{300}{0.05} = 6000\ V\ cm^{-1}$, eventually:

$$\lambda_e = \mu_e\tau_e E \cong 2.84\ cm \tag{3.8}$$

which is about 57 times the sensor thickness, consistently with the charge collection efficiency (CCE) estimated at 98.2% [6].

3.1.6. Leakage current and Resistivity

To achieve efficient charge collection, external voltage (usually hundreds of volts and depending on the sensor material and thickness) must be applied across the detector to have a strong electric field. Even in the absence of ionizing radiation and despite the high resistivity achieved by the presence of the depleted zone or the compensation-induced effects, this applied voltage induces a steady-state leakage current. The statistical fluctuation of this current is visible as parallel noise superimposed on the output signal, fundamentally hindering the smaller signals and introducing uncertainty on the pulse amplitude by deforming the optimal Gaussian shape [10].

Although the SPC technique manages well to cut off the former, there is significant resolution degradation coming from amplitude fluctuations due to the latter; so, the leakage current in general must be kept as low as achievable to avoid significant resolution degradation. Three general leakage current contributions can be identified as three main origins [7]:

- *minority-carrier drift*: the drift of minority carriers across the junction, which typically contributes only a few nA/cm²;
- *thermal generation*: the creation of electron-hole pairs caused by deep recombination or trapping centres within the depletion zone, which can incentive additional carrier creation and contribute up to several μA/cm² even at room temperature. In practice, bulk thermal generation is usually reduced by cooling the sensor.
- *surface leakage*: surface currents generated at the interface between the semiconductor bulk and the passivation layer, such as SiO_2; these surface effects can often dominate the bulk leakage and are therefore mitigated by appropriate surface processing, guard rings, etc.

While reverse biased p-n junctions are mainly affected by junction-related leakage mechanisms, compensated semiconductors are more strongly influenced by defect related bulk generation and surface effects due to the intentional presence of deep centres that reduce the free carrier concentration and increase resistivity but also degrade the quality of the semiconductor in terms of dark current. The main contributions to the leakage current can be summarized as follows [10]:

$$I_{dark} \propto T^2 e^{-\frac{E_g}{2kT}} \tag{3.9}$$

Since the leakage current must be kept as low as achievable, being one of the major sources of noise, despite applying at the same time high bias voltage, hence the resistivity showed by the material must be high enough. The latter is described as follows [8]:

$$\rho = \frac{1}{en_i(\mu_e + \mu_h)} \tag{3.10}$$

A practical and trivial way to measure the resistivity of a whole sensor [8] is by applying the Ohm's law and considering the bias voltage and the measured (leakage) current:

$$U = RI = \rho\frac{d}{A}I \leftrightarrow \rho = \frac{U}{I}\frac{A}{d} \tag{3.11}$$

with A and d being the sensor's area and thickness, respectively.

Considering the semiconductor studied in this work, the chromium compensation increases the bulk resistivity to values sufficiently high for the material to be operated essentially as a photoconductor rather than as a depleted p–n diode [1]. Consequently, nearly ohmic metallic contacts can be employed (such as Nickel [2]), and most of the applied bias voltage drops across the sensor bulk. The electric field can therefore extend almost uniformly throughout the sensor, allowing the whole volume to contribute to charge collection [8].

Moreover, the leakage current is also a good indicator of the degree of damage caused by radiation, which is one of the key factors limiting the semiconductor's lifetime. However, low-energy photons generally produce negligible permanent damage in semiconductor detectors, since most of their interaction leads to ionization rather than displacement damage. On the other hand, charged particles and neutrons are much more perilous at creating lattice defects that may introduce newer deep levels that enhance the levels of dark current [22].

3.1.7. Energy Resolution

When radiation deposits an energy Ein a sensor, this energy is converted into a finite number $N$of signal quanta, such as scintillation photons, ion pairs in gases, or electron–hole pairs in semiconductors. The average number of generated quanta is:

$$N = \frac{E}{E_i} \tag{3.12}$$

where E_i is the average energy required to produce one signal quantum.

Because the number of generated quanta fluctuates statistically, the measured signal amplitude also fluctuates. The relative energy resolution can be written as follows:

$$\frac{\Delta E}{E} = \frac{\Delta N}{N} = \frac{\sqrt{FN}}{N} = \sqrt{\frac{FE_i}{E}} \tag{3.13}$$

where F is the Fano factor [10]. This factor accounts for the fact that the generation of signal quanta is not purely Poissonian: different excitation processes can share the deposited energy, reducing the statistical fluctuations.

In semiconductor detectors, part of the deposited energy produces electron–hole pairs, while another part excites lattice vibrations, or phonons. Since phonons have much lower energies than electron–hole pairs, their production reduces the fluctuations in the number of charge carriers. As a result, semiconductors have a small Fano factor: generally, in semiconductors, F is approximately 0.1 [8].

In conclusion, for many semiconductor detectors, this intrinsic contribution is sufficiently small so that the total energy resolution is instead dominated by external contributions (such as electronic noise and leakage current fluctuation). Indeed, in the case of MYTHEN III, the *equivalent noise charge* (ENC) and the corresponding minimum threshold determine the minimum detectable signal, with the minimum detectable energy being roughly 10 ENC (see Section 6.4).

Instead, the mechanism by which the moving charge carries induce a signal on the electrodes will be addressed in Section 3.2.2.

3.2. Semiconductor Microstrip Sensors

Microstrip semiconductor sensors are fundamentally characterized by the presence of segmented electrodes, which provide a direct one-dimensional reference for position according to the strips collecting charge: indeed, the magnitude of the measured signal can depend on the relative position of charge formation in the sensor.

3.2.1. Electrode Structure

In silicon strip detectors, highly doped p+ strips are typically implanted on a lightly doped n-type substrate [5] [2], so that each strip acts as a collecting electrode (for vacancies) separated from the neighbouring ones by the surface passivation and the local electric field structure.

The situation is different in compensated GaAs, since detectors are based on a compensated bulk material rather than on a conventional p-n junction, hence segmentation does not come from the same doping junction structure used in silicon. As a result, the whole sensor volume can be made active [8] [2], while the strip geometry mainly serves the purpose of charge collection and thus position readout. Eventually, this makes the GaAs approach conceptually simpler on the detector side.

In both cases, the biasing polarization happens on the backside though an ohmic contact: for the sensors examined in this thesis, the latter consisted in a few microns thick strip electrodes made of Nickel [2].

3.2.2. Signal Induction and Weighting Potential

When an X-ray is absorbed in the semiconductor, the generated pairs drift towards the electrodes under the action of the electric filed: their motion though the sensor volume induces a current on the readout electrodes. This process is well described by the Shockley-Ramo theorem [23] [24], which states that for a charge q moving from a position $\overrightarrow{x_i}$ to a position $\overrightarrow{x_f}$, the inducted one on the electrode L can be expressed as follows:

$$\Delta Q_L = \int_{\overrightarrow{x_i}}^{\overrightarrow{x_f}} q\overrightarrow{E_0} \cdot d\vec{x} = -q[\varphi_0(\overrightarrow{x_f}) - \varphi_0(\overrightarrow{x_i})] = -q\Delta\varphi_0 \qquad (3.14)$$

The weighting field must not be confused with the actual electric field inside the sensor: the latter determines the motion of the charge carriers, whereas the former determines how their motion contributes to the signal measured by the electrode. Furthermore, the weighting potential can be computed by putting the electrode of interest to unit potential, the other ones to zero and then solving the Laplace equation $\nabla^2\varphi_0 = 0$ [8] [25]. This will be carried out properly in Appendix B. Instead, the induced current on electrode can be derived from (3.14):

$$i_L(t) = \frac{dQ_L(t)}{dt} = q\overrightarrow{E_0}(\vec{x}) \cdot \frac{d\vec{x}(t)}{dt} = q\vec{v}(t) \cdot \overrightarrow{E_0}(\vec{x}) = q\mu\vec{E}(\vec{x}) \cdot \overrightarrow{E_0}(\vec{x}) \qquad (3.15)$$

where $\overrightarrow{E_0}(\vec{x})$ is the weighting field.

Moreover, the contribution from electrons and holes can be treated separately and their signal can be summed up despite having opposite charge, since they travel opposite directions [8].

In a segmented detector, the weighting potential strongly depends on the position relative to the readout electrodes. Therefore, moving charge carriers can induce transient signals also on neighbouring strips even when they are eventually collected by a different electrode [6] [25]. Moreover, the loss of moving charges along the path can lead to incomplete signal collection and making the response dependent on the interaction depth. This aspect will be discussed in Section 8.2 in relation to the so called *crater effects* observed in GaAs:Cr sensors [6].

3.2.3. Spatial Resolution

The spatial resolution is defined as the minimum spatial distance to have the detector still capable of discerning in space two different photon interaction positions.

Basically, the major constrains in spatial resolution is the electrode width itself which may also depends on the space required from the readout electronics, so it is possible to estimate the lower bound of strip pitch practically achievable in the order of 20-50 μm, while it can be about 10 μm for the electrodes [10].

As it will be deepened in Section 9.1.1, the response function of the single strip is approximate as a box-like function with the pixel size as superior limit: this function is called *Point Spread Function* (PSF) and its frequency-space *Fourier Transform* is called *Modulation Transfer Function* (MTF) [4].

The standard deviation of the PSF can be estimated as follows, by assuming an ideal response i.e. no charge sharing and that a photon can imping the strip with uniform probability along the pitch (expressed in the integral by the uniform probability density $1/p$ over the strip pitch, corresponding to a box shaped PSF):

$$\sigma^2 = \int_{-p/2}^{p/2} x^2 \frac{1}{p} dx = \frac{1}{p}\frac{p^3}{12} = \frac{p^2}{12} \tag{3.16}$$

hence the root mean square (RMS) is simply $\sigma = \frac{p}{\sqrt{12}}$.

Some physical phenomena contribute to worsen the spatial resolution. As already mentioned, the thermal diffusion (and, for noticeably higher local careers density, also the Coulombian repulsion) leads to the worsening of spatial resolution due to charge sharing. However, the latter could be exploited to improve the resolution via the specific interpolation MYTHEN III feature (see Section 9.4).

Another issue to be factored in is the autofluorescence phenomenon [8], which consists of the emission of characteristic secondary X-ray photons following photoelectric absorption within the sensor material, with energy:

$$E_{K\alpha} = E_K - E_L \tag{3.17}$$

Since photoelectric absorption is more likely to involve inner-shell electrons, the emitted fluorescence photons are often K-shell lines. For instance, the binding energy of K shell electrons for Si, Ga and As are respectively 1.84, 10.37 and 11.87 keV [26], which corresponds to 1.74, 9.25 and 10.54 keV Kα fluorescence photons [27].

These photons may escape the initial interaction site and be lost or at least reabsorbed in the neighbour strips if emitted already close to the border, or even in the same channel at a different position along its length for some angles. Eventually, this can lead to the production of charge sharing-like events, as well as spectral tails or escape

peaks, thereby degrading both spatial and energy resolution. The effect is particularly relevant in high Z compound semiconductors, such as GaAs and CdTe, where autofluorescence photons have higher energy and therefore higher probability of escaping and degrading energy and spatial resolution [1].

A further contribution to the spatial spread out of charge careers of the signal is related to the finite range of the photoelectron generated after X-ray absorption. For the energy range and pixel dimensions considered for Si, the photoelectron range can be small compared with the detector pitch and may therefore be neglected, but this conclusion cannot be automatically extended up to about 50-60 keV (upper GaAs energy cap) [11].

3.2.4. Interstrip Capacitance

Due to the electrode geometry, described in Section 3.2.1, a capacitance between the readout electrode and the backplane. This can be simply described approximately as:

$$C_{back} \cong \varepsilon \frac{A_{strip}}{d_{thickness}} \tag{3.18}$$

In segmented strip detectors, however, an additional contribution arises from the capacitive coupling between neighbouring strips that can represent an important contribution overall.

The interstrip capacitance depends on the sensor geometry, the strip width w, the strip pitch p, and the interstrip gap. For silicon microstrip detectors, its capacitance per unit strip length can be described by the empirical relation proposed by Demaria et al. [28], where C_s is the interstrip capacitance and l the strip length:

$$\frac{C_s}{l} = \left(0.03 + 1.62\frac{w + 20[\mu m]}{p}\right)\left[\frac{pF}{cm}\right] \tag{3.19}$$

For typical microstrip geometries, the resulting specific interstrip capacitance is therefore of the order of 1 pF/cm, although the actual value depends on the electrode geometry.

Therefore, the overall sensor capacitance includes both the interstrip and the backplane, which contribute eventually to the total input capacitance of the preamplifier and thus influencing the gain and noise.

3.3. GaAs:Cr as a Sensor Material

The study of GaAs semiconductors comes into interest when there is the need of dealing with the detection of higher energy photon with an adequate efficiency. Despite Si semiconductors offer noteworthy performance in terms of uniform response among the strips (or pixel, in two-dimensional detector systems) and charge transport efficiency, its absorption efficiency goes down steeply even at 20 keV due to its relatively low atomic number (Z=14).

For this reason, high-Z materials have come under study to extend the detector performance to higher energies thanks to their higher atomic number. Materials like CdTe (48/52) or GaAs (31/33) thus provide better absorption efficiency in the harder photon energy ranges where silicon becomes less effective and so constitute a valid alternative for higher-energies setups like synchrotrons [1].

The GaAs semiconductor detector-class represents a valuable alternative also for outstanding properties in terms of high electron mobility, such as faster response and charge collection, satisfactory resistance achieved through Cr compensation, plus a wide enough bandgap which allows for room-temperature measurements along with the employment of GaAs even in high-frequency electronic devices [8].

At the same time, compound-class semiconductors still do not grant a proper uniform response like silicon and are more propense to show auto-fluorescence effects, as well as polarization and afterglow due to charge trapping [6].

3.3.1. Crystal and Band Structure of GaAs

Following the general behaviour of II-VI and III-V semiconductor compounds, the GaAs crystallize in a zinc blende structure. In this way, both the Ga and As atoms are tetrahedrally coordinated and thus forming two distinct interpenetrating FCC lattices with an (1/4, ¼, ¼) offset along the unit cell diagonal. The lattice constant is 5.653 Å.

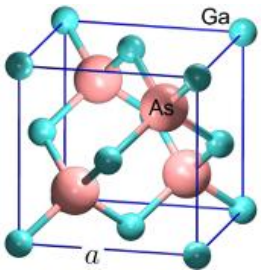
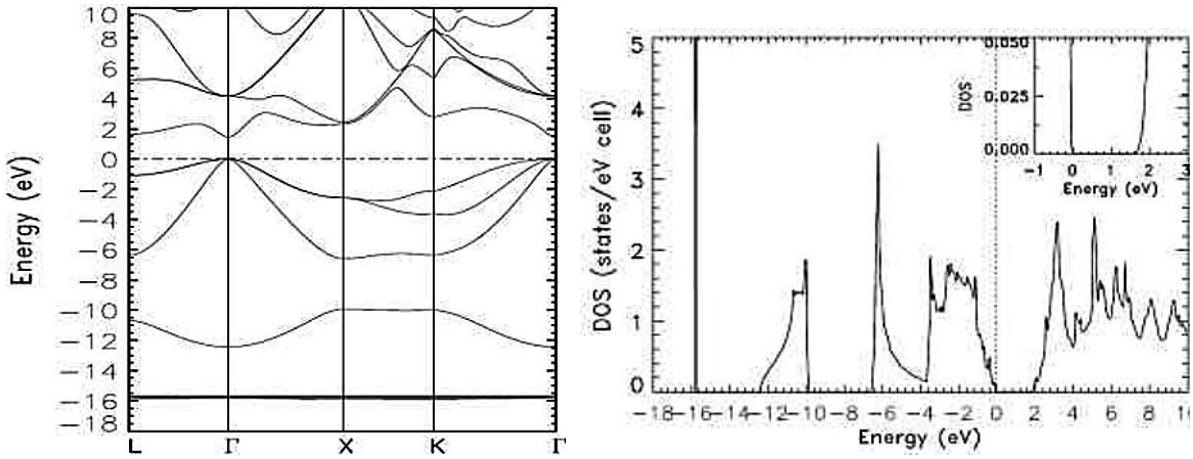


Figure 3.4 Scheme of the GaAs zinc blende structure. Reproduced from [29]

GaAs is a direct bandgap semiconductor, since the conduction band minimum and the valence band maximum are both located at the Γ point (I.e. null wave vector) of the Brillouin zone. This discerns it from other semiconductors like silicon and germanium, which present a peak between the maximum of valence and minimum of conductive bands.



(a) Band structure of GaAs (b) Density of states of GaAs

Figure 3.5 Electronic properties of zinc-blende GaAs: (a) calculated band structure and (b) calculated total density of states. Adapted and reproduced from Diakite et al. [30].

The above plots depict respectively the band structure of GaAs alongside its own density of states (DOS) which represents the number of states per unit energy, which is defined as follows, where n is the band index and the integration is performed over the Brillouin zone:

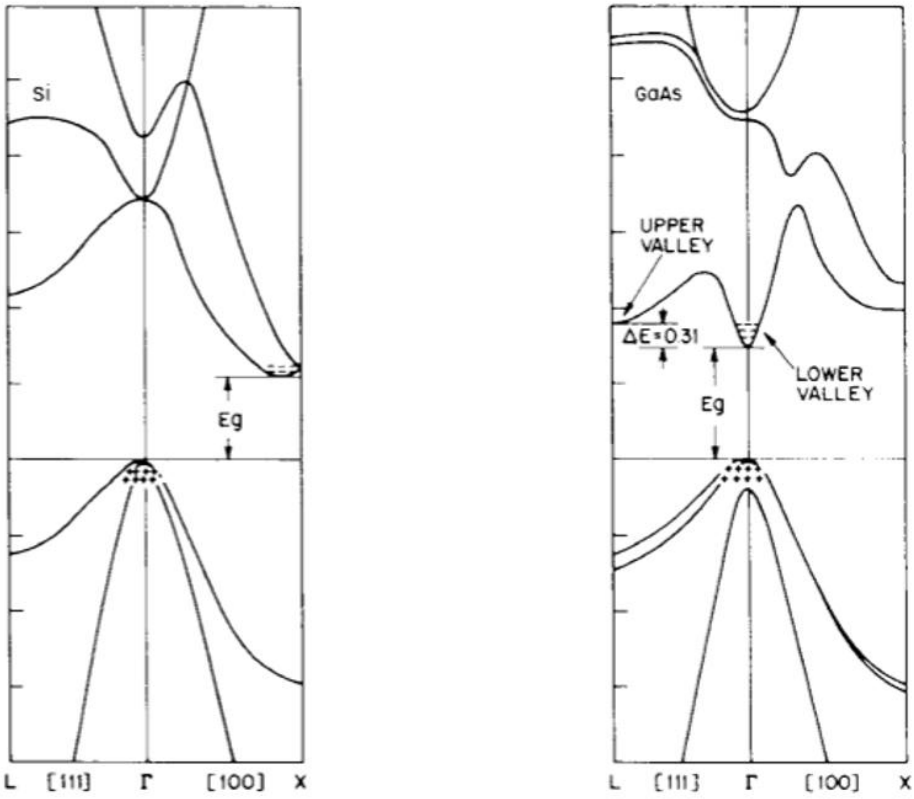
$$g(E) = \frac{dN}{dE} = \sum_n \int \delta\big(E - E_n(\boldsymbol{k})\big) \frac{d^3\boldsymbol{k}}{(2\pi)^3} \tag{3.20}$$

Moreover, for optical transitions, it is useful to introduce the joint density of states (JDOS):

$$J(E) = \frac{1}{(2\pi)^3} \int \delta(E_C(\boldsymbol{k}) - E_V(\boldsymbol{k}) - E) d^3\boldsymbol{k} \tag{3.21}$$

Where $E_C(\boldsymbol{k})$ is the conduction band energy, $E_V(\boldsymbol{k})$ the valence band energy and E the photon energy. Consequently, while the density of states describes the number of electronic states available at a given energy within a single band, the joint density of states describes the number of pairs of initial and final states that satisfy, at the same time, the energy and momentum conservation required for an optical transition.

For direct gap semiconductors such as GaAs, the JDOS becomes remarkably larger than zero immediately above the bandgap energy, resulting in a steep increase in the absorption coefficient. In indirect semiconductors like Si and Ge, by contrast, the additional phonon reduces the number of allowed transitions close to the absorption edge, thus leading to a more gradual increase in absorption and so to a decrease of the transition probability itself. Besides, the availability of phonons with enough momentum depends on the temperature, so are the latter's absorption capabilities.



(a) Silicon band structure. (b) GaAs band structure

Figure 3.6 The band structure for some wavevector, showing several peaks and valleys. Reproduced and adapted from Spieler [10]

In contrast to optical absorption for energies near the bandgap values, higher energy quanta carry a significant amount of momentum. Therefore, the absorption process must satisfy both energy and momentum conservation even for a direct bandgap. Consequently, the excess momentum must be transferred to other excitations of the crystal, such as phonons and lattice vibrations. A significant fraction of the absorbed energy is converted into non-ionizing excitations, and this explains the experimental observation for which the energy required to generate a pair is larger than the bandgap: this is approximately expressed by the following relationship [31]:

$$E_i \approx 2.8E_g + 0.6\ eV \tag{3.22}$$

When dealing with high-energy photons, the direct or indirect character of the bandgap plays a secondary role in the initial absorption process, as the photoexcited carriers are generated far above the band edges. Nevertheless, the bandgap magnitude is a main factor in determining the mean energy required to create an electron–hole pair.

3.3.2. Lattice Defects in GaAs

As already mentioned in Section 3.1.5, the deviation from a perfect lattice due to defects that end up altering properties such as leakage current, resistivity, mobility, lifetime and so on.

The defects can generally be discerned according to their origin (intrinsic and extrinsic) together with their dimensional extension in the atomic structure:

- Intrinsic defects are defects that are present in the material from the beginning, as part of its own crystal structure.
- Extrinsic defects are defects caused by foreign atoms, either unintentionally introduced as impurities or intentionally added as dopants or compensators (like chromium).
- Point defects include vacancies, interstitials, and, in compound semiconductors, antisite defects, where atoms of different elements exchange places.
- Line defects are called dislocations and cause mechanical strain in the crystal lattice.
- Planar defects occur at interfaces between differently oriented crystal grains or between different phases.
- Bulk defects are three-dimensional defect clusters, such as precipitates or inclusions.

For GaAs, an important point defect is the arsenic antisite AsGa, in which an As atom occupies a Ga lattice site. This antisite behaves as a deep double donor and introduces two levels in the mid gap: the deeper (As_{Ga}^{0}), located about 0.75 eV below the CB, while the second (As_{Ga}^{+}) lies about 0.5 eV above the VB. Therefore, they represent different charge states associated with the same arsenic antisite defect. Moreover, their influence on dark current is not simply proportional to its own concentration, as it also plays a role in the compensation mechanism of GaAs [8].

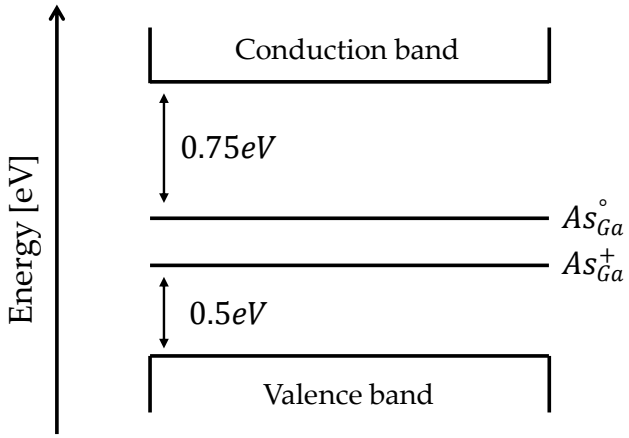


Figure 3.7 Defects states within GaAs forbidden gap. Based on Kaminska et al. [32]

3.3.3. Impact of Lattice Defects on Charge Transport

Accordingly, while silicon stands out for being easier to use and characterize and for its well-established fabrication processes, the presence of multiple elements makes the use of this kind of semiconductor more challenging for the following reasons:

- *Crystal defects*: it is difficult to produce a two-element crystal without compromising the exact stoichiometry or creating structural defects, so it is common the formation of vacancies and interstitials. The problem is then upscaled due to the presence of injected Chromium which makes up for the low resistivity of the initial compound.
- *Charge trapping*: due to the presence of abovementioned defects or impurities (even the Chromium), new energetic levels appear in the prohibited gap band which can trap the charges, particularly the holes.
- *Polarization and afterglow effects*: since some charges are trapped, thus a static special charge is accumulated in the crystal ending up with a distorted electric field. Indeed, this is one of the major drawbacks of CdTe detectors, where polarization (effective field decreased) and afterglow effects (traps slowly released with delay) take down the material quality.
- *Crater effect*: the shorter lifetime of holes with respect to the electrons causes a reduced signal in the collecting strip and a negative in the adjacent ones. This effects present dependences on the depth of absorption. This aspect will be discussed in Section 8.2.

3.3.4. Production Techniques of Cr-Compensated GaAs

The way of obtaining a chromium compensated GaAs detector has been presented by researchers from the *Tomsk State University* (TSU) that allow for getting a high resistivity GaAs by diffusing Cr in already commercially available n-type GaAs sensors, thus improving up to the order of $10^9 \Omega\, cm$ and eventually yielding adequate detector-grade semiconductor [6].

The compensation process works by introducing deep energy levels into the band gap. So, the Chromium captures the excess free electrons naturally present in the n-type material, effectively turning the bulk into a semi-insulating crystal. From a band diagram perspective, the Cr atom substitutes a Ga atom (thus forming a Cr Ga-antisite), eventually behaving as a mid-gap deep acceptor [8]:

$$E_{Cr} \cong E_V + 0.78 \tag{3.23}$$

The first method was devised in 1964 by *Cronin* and *Haisty* [33] which consist in the addition of elemental Cr during the production process.

Adapting a compensation model already proposed by *Martin*, the concentration of added Cr must abide by the following condition [8]:

$$N_{Cr_{Ga}} > N_{SD} - N_{SA} > 0 \tag{3.24}$$

where the N_{SD} and N_{SA} stand for, respectively, concentration of shallow donors and acceptors. Under these conditions, the shallow donor electrons first compensate the shallow impurities, and the remaining excess electrons are trapped by the deep Cr acceptor centres, resulting in a slightly overcompensated, high-resistivity p-type material.

The second method was introduced by the above-mentioned TSU's Ayzenshtat et al. [34] and the compensation is achieved through post growth high temperature diffusion: Cr is evaporated onto both sides of a n-type GaAs wafer and then annealed at high temperatures to ensure an optimal bulk diffusion. At the end the wafer is polished to remove residuals on surface layers, then leaving a uniform Cr concentration in the bulk.

In the same manner as before, the below conditions must hold [8]:

$$N_{Cr} > N_D > N_{BG,i} > 0 \tag{3.25}$$

where N_D and $N_{BG,i}$ stand for electrons coming from n-type doping and the ones from background plus intrinsic defect. This post growth diffusion method is advantageous as it allows for the use of standard and already commercially available n-type GaAs wafers without requiring further doping procedures during the production phase (crystal growth stage).

As proved by Tyazhev et al. [20], the resulting GaAs:Cr exhibits excellent electric field uniformity and reduced current oscillations in time of dark current (since it could show sharp variation rather than a steadier behaviour).

3.3.5. Absorption efficiency

The total linear attenuation coefficient of GaAs, together with the fraction of absorbed photons for a 500 μm-thick sensor, is shown below up to 60 keV, including the Ga and As K-absorption edges.

In this energy range, the predominant interaction mechanism is only the photoelectric absorption. According to the Beer Lambert law, the fraction of absorbed photons can be expressed as follows, where $\mu(E)$ is the energy-dependent linear attenuation coefficient and $d$the sensor thickness [15]:

$$\eta_{ABS}(E) = 1 - e^{-\mu(E)d} \tag{3.26}$$

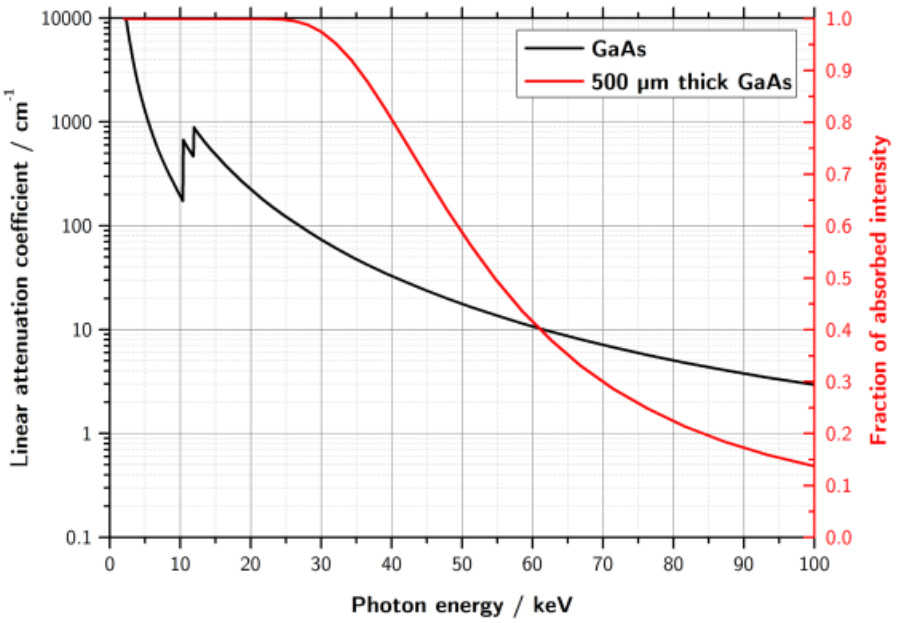


Figure 3.8 Linear attenuation coefficient of GaAs (black) as well as the fraction of absorbed photons for a 500 µm thick sensor (red) versus energy. Reproduced from [8]

Silicon is highly uniform and technologically mature, but its efficiency decreases abruptly already at 20 keV. The higher atomic numbers of Ga and As with respect to Si considerably improve the absorption efficiency of GaAs at higher X-ray energies. In particular, 500 μm-thick GaAs sensors provide nearly full absorption at 30 keV and approximately 50% absorption efficiency at 50 keV [7]. Figure 3.9 below shows the absorption efficiency for some common sensor materials and different thicknesses.

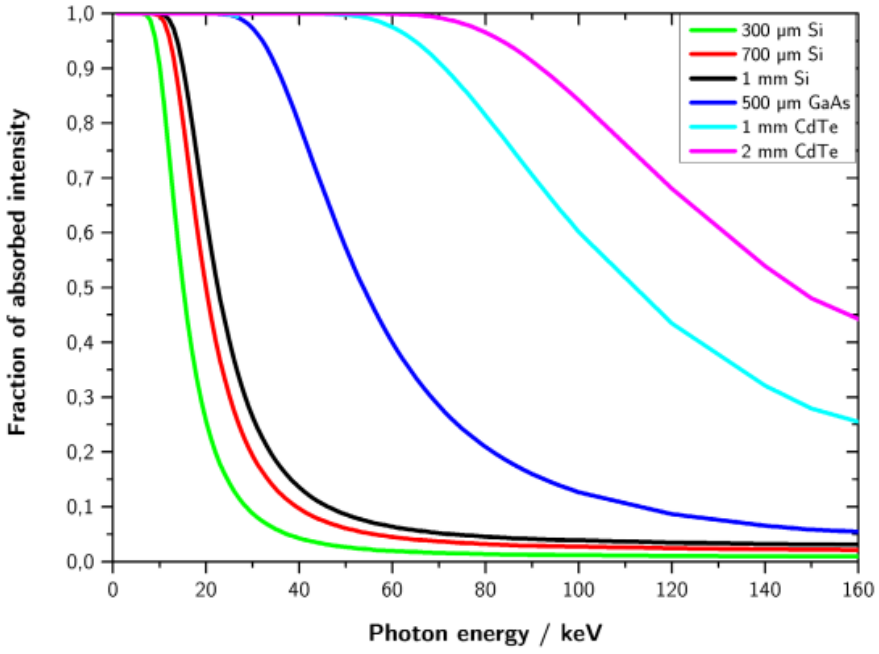
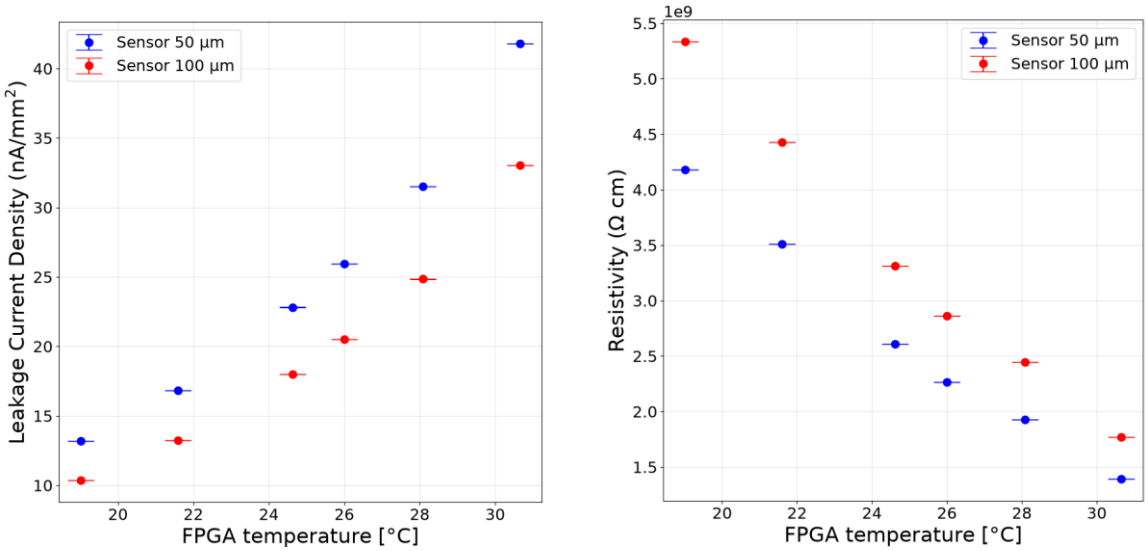


Figure 3.9 Fraction of absorbed photons versus energy for the most common semiconductor material and some common thickness. Reproduced from [8]

3.3.6. Basic electrical characterization of the GaAs:Cr sensor

Based on the theoretical discussion of leakage current and resistivity introduced in Section 3.1.6, a basic electrical characterization of both GaAs:Cr sensors has been carried out by measuring the leakage current under different bias voltages and temperatures. From the resulting current-voltage curve, the sensor resistance and resistivity can be estimated. However, the only available temperature sensor was the one in the FPGA module, therefore the temperature is only indicative.



(a) Leakage current density (b) Material resistivity

Figure 3.10 Representation of leakage current density and material resistivity according to the temperature variation. The constant difference between the two sensors may be explained by some unknown sensor different characteristics.

The current-voltage analysis[7] results are shown in Figure 3.10. As expected, leakage current density tends to increase with temperature, whereas the resistivity subsequently decreases accordingly. The former is always lower than the value reported by Ruat et al. [2], who found instead about 70 nA/mm^2. Instead, the latter is consequently higher than values in the literature [2], which is about 0.9 10^9 $\Omega\, cm$. To explain the difference, it must be remembered that Ruat et al.'s sensors come from the *Tomsk State University* (TSU), while the ones investigated in this work from *DECTRIS AG* (see Section 2.1), therefore it is expected differences such as material quality and uniformity, electrodes material and geometry and so on. Furthermore, for the same leakage current density, the leakage current for the 100 µm strips would be approximately doubled. In any case, the leakage current values found are considerably

[7] Leakage current was measured by varying bias voltage. Then the resulting I–V curve has been linearly fitted. The current density was obtained as $J = I/A$, while the material resistivity was calculated as $\rho = RA/d$.

higher than that typical of standard silicon strip sensors used: indeed, Andrä [35] reported about 0.12 nA for single strip[8], hence a current density of about 0.3 nA/mm^2.

Furthermore, a bias of -300 V, which was firstly considered as preliminary during the first acquiring, has been therefore selected as the standard operating value for the further GaAs:Cr measurements: despite a higher voltage would increase the electric field, as shown in Ruat et al. [2], for 500 μm GaAs:Cr sensors there is no further increase of drift velocity after approximately –200 V. However, higher electric field would limit trapping as well, but at the expense of larger leakage current.

[8] The Si strips studied in Andrä's work are 8 mm long, while having 50 μm as pitch

4 Experimental setup

The characterization presented in the following chapters was carried out using different experimental setups. X-ray measurements have been performed using a laboratory X-ray tube, needed to characterize the energy calibration of MYTHEN III. Moreover, a laser setup has been employed to generate charge carriers at controlled positions inside the sensors with a precision in the order of tens of micrometres, allowing spatially resolved investigations of charge collection and charge sharing.

4.1. Laboratory X-ray setup

4.1.1. X-ray tube source

The detector characterization measurements were performed using a laboratory X-ray setup designed to provide eventually monochromatic photons. A conventional X-ray tube was used as the primary radiation source, while a set of secondary fluorescence targets was employed to obtain quasi-monochromatic X-ray radiation at selected characteristic energies. Indeed, the radiation directly emitted by the tube is therefore polychromatic. A filter wheel placed in the primary beam allowed different filters to be selected to remove the low-energy part of the bremsstrahlung spectrum, thereby hardening the radiation and increasing its mean energy. Such filtration, however, is not enough to make the beam monochromatic.



(a) (b)

Figure 4.1 Photographs of the X-ray tube where the Mo target and its secondary filter (Zr) are present; in (b) the detector is also present on the right

An X-ray tube consists of a cathode and a metallic anode (in this work, tungsten) enclosed in a vacuum chamber. Electrons are emitted from a heated cathode and accelerated towards the anode by an applied high voltage. Their interaction with the target material produces an X-ray spectrum containing two main contributions: a continuous bremsstrahlung component, originating from the deceleration of the incident electrons in the target, and discrete characteristic emission lines resulting from the ionization and subsequent electronic rearrangement of the target atoms [7] [8]

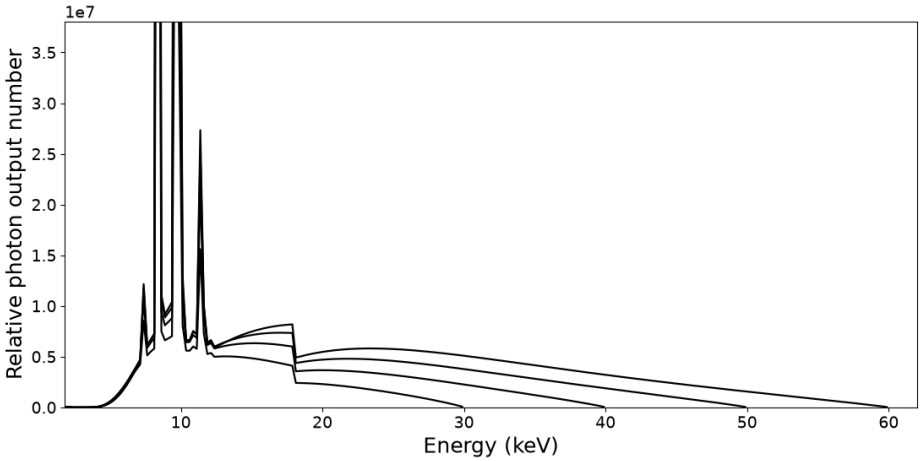


Figure 4.2 Example of W anode spectrum filtered with Zr for different kV. Spectra made using *SpekPy*[9]

4.1.2. Beam filtration and fluorescence target configuration

To get rid of polychromic, the primary beam from the X-ray tube irradiates a secondary target. The latter, according to its own elemental composition and following the electronic rearrangement after a photoelectric absorption of impinging photons, emits its characteristics photons (actually the process is analogous to the autofluorescence effects already mentioned in Section 3.2.3), eventually "filtering" the spectrum and delivering only some discrete peaks corresponding to the electrons' transitions between the atomic shells of the target material. The main drawback is the substantial reduction in photon flux, since only a fraction of the primary photons interacts in the target and only a fraction of the emitted fluorescence radiation reaches the detector. Indeed, to carry out the count rate capability analysis presented in Chapter 7, the primary beam was employed, since it was the only one able to provide a sufficiently high number of photons [7].

Nevertheless, the emitted radiation is not perfectly monochromatic, since the radiation emerging from the target presents several characteristic transitions simultaneously. Together with the dominant K_α emission, a K_β component can generally be observed.

[9] Specific Python library aimed at reproducing photon spectra coming out of several anode material X-ray tubes. Further details at Section 7.2.1

When available, an additional K_β filter is thus placed between the fluorescence target and the detector: its material and thickness are selected so that the absorption edge of the filter provides a stronger attenuation of the K_β component while maintaining a comparatively higher transmission of the desired K_α radiation. Therefore the final beam can consequently be regarded as quasi monochromatic rather than strictly [7].

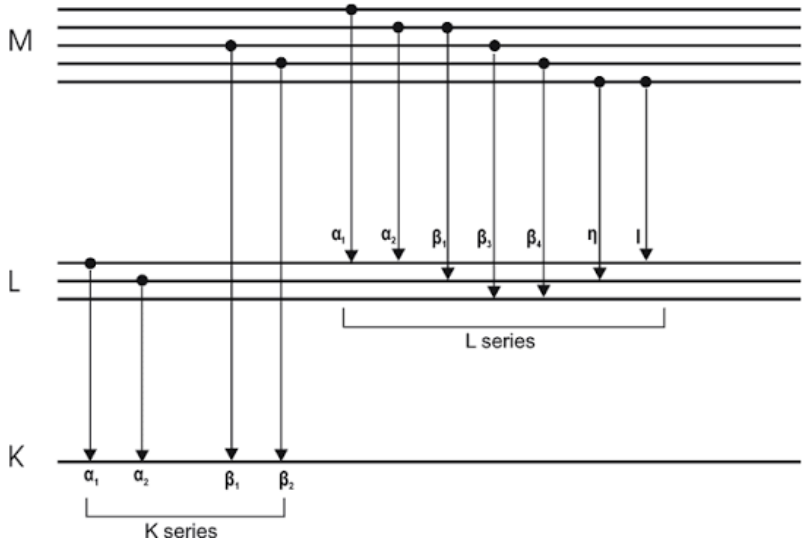


Figure 4.3 Representation of different possible electron transitions leading to K and L photons emission

In some cases, such as the high rate characterization presented in Chapter 7, the direct polychromatic beam of the X-ray tube was employed. Indeed, the secondary fluorescence configuration strongly reduces the available photon flux and was therefore unsuitable for producing sufficiently large pile up effects.

Instead, figures of merit such as the ones pertaining to noise and charge sharing (see Section 5.2) have been based on parameters extracted from the threshold scans performed with Mo fluorescence photons. Indeed, Mo has been selected as reference because its 17.5 keV fluorescence K-alpha line lies in an intermediate region of the investigated energy range, while other undesired peaks are suppressed by the secondary filter (Zr in this case, visible in Figure 4.1b)

4.2. Optic laser setup

Optical excitation gives a useful method to generate electron hole pairs in a spatially localized region of the semiconductor. As mentioned in Section 3.1.1, if the photon energy is sufficiently high for an allowed optical transition, absorbed photons then promote electrons to higher energy states and therefore generating mobile charge carriers which drift under the electric field applied to the sensor.

Compared with X-ray irradiation, the main advantage of a laser source for the present measurements is the possibility of controlling the injection position. By focusing the optical beam and translating the sensor relative to it, the detector response can be sampled as a function of position with a spatial resolution determined mainly by the laser spot size and by the positioning system.

Laser injection should nevertheless be regarded as complementary to X-ray measurements rather than as their direct equivalent, since the laser wavelength employed in this work (660 nm, or red) limited the investigation to the first few micrometres of depth within the sensor material. Moreover, due to opaque Ni backside present on the sensors, the laser thus could not be injected through the backside. Therefore, the measurements presented in this work were mainly performed in an edge-on configuration, allowing then the laser scan despite restricted to a superficial lateral region.

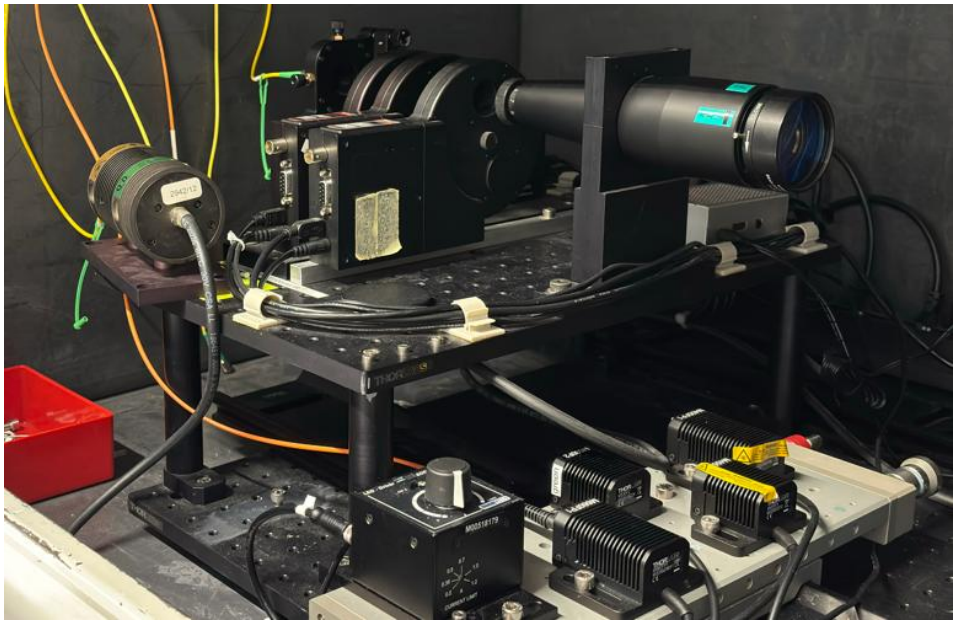


Figure 4.4 Photograph of the laser setup

Two laser wavelengths, 660 nm (red) and 1000 nm (IR), were available in the setup, while the measurements discussed in the following chapters were mainly performed using the 660 nm laser, since the GaAs is transparent at 1000 nm, although the presence of and other defects can introduce sub-bandgap absorption through deep levels (see Section 3.3.2). Indeed, a detector response to the 1000 nm laser was experimentally observed, consistent with the presence of deep levels allowing sub-bandgap photon absorption, although no quantitative results were extracted.

5 Threshold scan response and S curve characterization

SPC detectors react by counting only the single photon pulses whose amplitude is higher than a certain threshold [3]. Consequently, the number coming from the counters carries no information about the exact energy of the photons which have interacted with the sensors. Nonetheless, the only energetic information acquired is given by the fact that the counts recorded possess a pulse which was higher than the threshold, so the outgoing amounts of counts are not only integrated in time but from the (still unknown) energy upwards.

This leads to the conclusion that a proper choice of a comparator threshold level is strictly needed to characterize the measurements. The easiest way for a SPC detector to correlate the mV (or corresponding *DAC units*) to energy is by carrying out a so-called "*threshold scan*" for various discrete energies. This method is routinely employed for the calibration and characterization of single photon counting detectors developed at PSI.

Therefore, in the following sections the methods, which will be employed to characterise the MYTHEN III like the so called "*S curve*", will be discussed as well as the useful figures of merit.

5.1. Threshold scan and S curve response

For this reason, given a proper experimental setup (see Section 4.1), it is possible to perform a "threshold scan", the measurement procedure which consists in a series of in-time contiguous scans where the value of the global threshold level is increased for every fixed step. Eventually, what is obtained is the number of counts detected during the exposure time, corresponding to photons whose energy is equal to or higher than the threshold. The latter has been set globally before in the detector configuration. Therefore, its interpretation relies on the definition of integral between the current threshold level and the above energies:

$$N(E_t) = \int_{E_t}^{\infty} S(E')dE' \tag{5.1}$$

where $S(E)$ is the differential spectrum in energy.

For the negative and positive polarity configurations used for GaAs and Si the corresponding forms are respectively as follows:

$$N(E_t) = B_0 + B_1 E_t + \frac{1}{2}\left[1 + erf\left(\frac{E_t - \mu}{\sqrt{2}\sigma}\right)\right] \cdot [A + C_S \cdot (E_t - \mu)] \quad (5.2)$$

$$N(E_t) = B_0 + B_1 E_t + \frac{1}{2}\left[1 - erf\left(\frac{E_t - \mu}{\sqrt{2}\sigma}\right)\right] \cdot [A + C_S \cdot (E_t - \mu)] \quad (5.3)$$

where erf is the cumulative Gaussian distribution.

Therefore, for each channel and each calibration energy, the fit returns the following six parameters:

- *Background intercept* (B_0), which defines the constant contribution to the count rate, or the value of the linear background at zero threshold.
- *Background slope* (B_1), which describes the threshold dependence of the background contribution. This term accounts for the approximately linear background component mentioned later in Section 5.1.2.
- *Inflection point* (μ), which is the location parameter of the cumulative Gaussian component. It identifies the threshold value associated with the full-energy peak and then it is exploited for the following energy calibration procedures.
- *Noise parameter* (σ), which is used to derive the ENC (see Section 6.3), determines the broadening of the cumulative Gaussian distribution, and it is thus directly describing quantitively the noise affecting the measurement.
- *Amplitude* (A), which sets the height of the photon-induced *S curve* in the absence of the overlaying linear charge-sharing contribution.
- *Charge sharing* (C_S), that quantifies the relative magnitude of the charge sharing component and determines its linear contribution to the fitted response.

The fitting is performed using the phenomenological model proposed for MYTHEN (and generally for SPC) detectors by Bergamaschi et al. [5]. Hence the resulting *S curve* contains the detector response to the incident (ideal) monoenergetic radiation and is fitted independently for each strip. The fit is restricted to the photon induced component of the curve, above or lower the low threshold region according to the polarity, so that the response associated with the incident X-ray beam can be characterised excluding the overwhelming noise peak contribution [5].

5.1.1. Noise and charge sharing contributions

The actual *S curve* can differ noticeably from the ideal one due to some phenomena and spur effects related to the detector physical limitation, like the intrinsic noise, and even the strip width, like the charge sharing effects.

As depicted in the figure below, the step curve would represent the ideal detector response: in correspondence of the step, nothing more is counted for threshold larger than the photon energy (for the moment still in *DAC units*) and all the N_0 number of photons have been totalled.

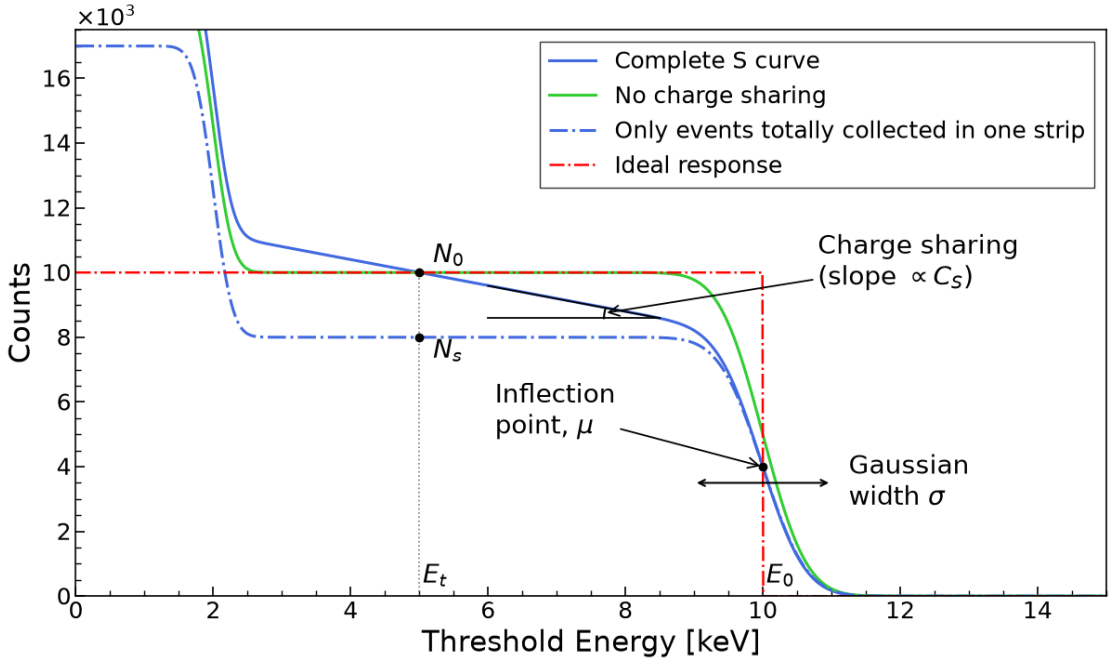


Figure 5.1 Example of simulated threshold scans for a beam of 10 keV. The detector counts $N_0 = 10^4$ during the exposure time, while the single strip "sees" only $N_s = 8\ 10^3$ completely collected events because of charge sharing. Based on [7].

Another contribution to be factored in is the intrinsic noise which determines the sigmoid shape of the *S curve* (so the reason for the name) due to the convolution of the latter with the ideal radiation integrated spectrum. Indeed, when the threshold coincides with the photons' energy, all the latter that produce a positive noise contribution are counted, whereas those with a negative noise fluctuation fall below threshold and are so they are discarded. In this sense, the threshold value corresponds to the inflection point of the *S curve*, where the slope is steepest and which, in the ideal case, coincides with the photon energy. Therefore, only for thresholds below the photon energy, those affected by a negative noise contribution are also counted.

Furthermore, the real effects of noise severely affect the *S curve* at higher values of threshold scanned (or lower for Si), with a sudden increase in the number of counts due to the continuous triggering of the comparators caused by the baseline noise. The noise counts can reach also the order of 10^7 and even get over the $2^{24} - 1$ (24 bit

counters) capacity, thus triggering an overshooting which resets the count at zero: this may happen with ease for longer exposure time [5].

Until now, the sigmoid shape can be trivially expressed by a Gaussian cumulative distribution including a steeper noise peak due to the above-mentioned baseline noise. The latter's number of counts can be estimated, assuming a gaussian noise type, by the relation:

$$N_n(E_t) \cong \frac{T}{\tau_S} D\left(\frac{-E_t}{\sqrt{2}\sigma}\right) \tag{5.4}$$

where τ is the shaping time which depends on the bandwidth of the readout chain (see Section 2.2.2). This equation is irrelevant unless the threshold value gets closer to the baseline.

In conclusion, the charge sharing phenomenon is also considered: when a photon is not absorbed in the exact centre of the strip and/or the diffusion of charge carriers' plasma clouds is predominant, then the generated pairs could be potentially collected by the closer strips, eventually leading to a decrease of the pulse amplitude seen by the strip (for further details see Section 8.1) This spur effect causes the inclination of the former *plateau* which can be now modelled after an increasing (or decreasing for Si) straight line.

5.1.2. Additional spectral and background contribution

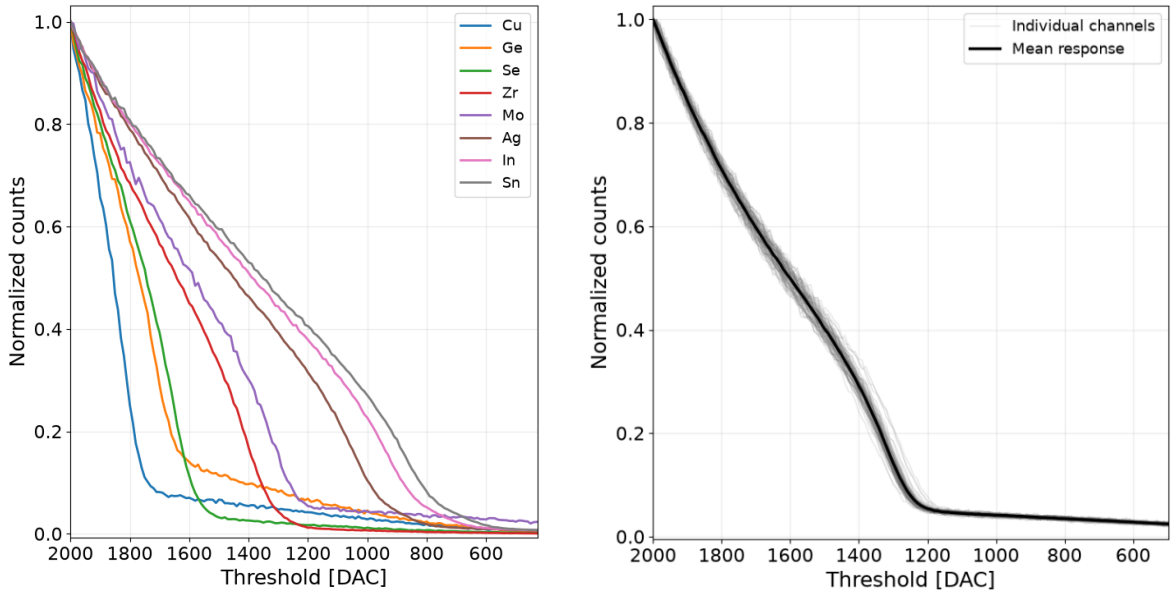
The background presence is explained by the experimental measurements set-up itself: some of the photons coming from the X-ray tube manage to reach the detector, thus polluting the summing of counts that could have been only strictly linked to the monoenergetic beam. This continuous background contribution can be introduced, approximatively, as a straight line whose intercept with the threshold axis corresponds to the maximum scattered x-ray tube photon energy. Consequently, the fitting process outcome would be potentially worse leading to more uncertainty for the estimated parameters.

Another contribution is the presence of discrete chromatic photons coming out from the irradiated target (as described in Section 4.1.2): this leads to the ideal presence of more sigmoid steps along the curve, since they will be integrated and counted accordingly, thus ending up with a worse quality *S curve* in terms of fitting. For this reason, parameters like the noise can be estimated, if using a polychromatic source like done in this work, only from K_β filtered fluorescence target.

Eventually, it is possible also consider the deleterious contribution of fluorescence photons-related sigmoid steps which distort the fitted monoenergetic component, hence possibly biasing the extracted parameters. In conclusion, examples of threshold scans are shown in Figure 5.2.

5.1.3. Effect of signal polarity on the S curve

Eventually, the growth direction depends directly on the polarity of the ASIC which in turn is setup so that is able to follow the kind of collected charged carrier: for instance, in GaAs the polarity would be negative since the collection of electrons induces a negative current and thus pulse, therefore the MYTHEN III must be set in electron collection mode or negative working polarity. Since the threshold scan consists in a sequential measurement with larger threshold edges, it will firstly cross the peak of the photopulse and only at the end it will yield the noise: indeed, that goes with an abrupt increase of counts which is depicted by the amplitude-dominant noise peak). Trivially the threshold scan behaviour and thus the *S curve* direction is the opposite for the hole collection mode employed for Si detectors [7].



(a) Threshold scans at different energies[10] (b) Threshold scans for all 50 µm channels[11]

Figure 5.2 Examples of threshold scans: (a) shows normalised threshold scans at different fluorescence energies for a representative channel, while (b) shows normalises threshold scans at Mo energy (17.5 keV) for all 50 µm channels, with the average response in black

[10] *S curves* related to Cu ($Ek_\alpha \cong 8.0\,keV$) and Ge ($Ek_\alpha \cong 9.9\,keV$) are not well defined since they lay below the lower energy limit (due to noise), however the scans have been carried out for demonstrative purposes.

[11] The depicted scan has been carried out after the trimming procedure which reduces the spread out of channel response (see Section 6.6).

5.2. Derived figures of merit

5.2.1. Gain and offset

The mentioned parameters returned by the *channel by channel* fits are not figure of merit in themselves, but they must accordingly be employed to derive quantities useful to characterize the detector.

In particular, the inflection points measured at several known photon energies are used to build the energy calibration relation [5], from which gain and offset are obtained by linearly fitting the former points with their associated energy:

$$\mu_i[DAC] = g_i\, E[keV] + O_i \tag{5.5}$$

Where μ_i is the threshold position in DAC units for channel i, E is the incident photon energy, g is the gain in terms of $\frac{DAC}{keV}$, and O_i the offset [7] in DAC. This is the standard procedure used to determine the energy calibration of MYTHEN III channels from threshold scans.

The gain is the slope of this relation and then quantifies the conversion between photon energy and DAC units and consequently is expressed in $\frac{DAC}{keV}$. The offset is the intercept of the same fit and corresponds to the extrapolated threshold position at zero energy, so it is expressed in DAC.

5.2.2. Other relevant figures of merit

The agreement of the calibration line (inflection points versus photon energies) with a linear model defines how much linear is the detector response over the energy range: this is essential because it allows the threshold axis to be converted with reliability from *DAC units* to energy, and vice versa.

Another useful quantity is the equivalent threshold dispersion [7]. At a fixed incident energy, the inflection point of each channel could have a different value because of the difference among the strips and the associated readout chains. The *root mean square* (RMS) of the resulting channel distribution then represents the threshold dispersion (typically expressed in electrons RMS or in energy units) which quantifies the residual mismatch among channels before and after calibration or trimming (this aspect will be addressed accurately in Section 6.6)

The remaining fit parameters are used for other figures of merit, which are discussed separately: the Gaussian width is converted into the equivalent noise charge (ENC) for the noise analysis (section 6.3), while the charge sharing parameter can be used to estimate the fraction of shared events (section 8.1). Furthermore, the combination of ENC, gain and offset defines the lower and upper bound of the detector's usable energy range (section 6.4), while some other parameters which will be addressed in

the following chapters, such as the Modulation Transfer Function (MTF). can give indication about the spatial resolution (section 9.1.2)

Another quantity commonly used is the *Signal to Noise Ratio* (SNR). Ideally, for photon counting detectors, the fluctuations of recorded number follow the Poissonian statistics. Therefore:

$$SNR = \frac{N}{\sqrt{N}} = \sqrt{N} \tag{5.6}$$

where N is the number of counts. In a real detector, however, the effective SNR could be reduced by contributions such as electronic noise, background radiation and so on [7]. Also, the charge sharing (see Section 8.1) may affect the SNR, especially when a segmented detector, like the MYTHEN III, can lead to a photon being counted more than once due to the charge spread out during the drift (this is called *multiplicity*, described in Section 5.3) [36].

Related to the SNR, the *Detective Quantum Efficiency* (DQE) describes how efficiently is preserved the statistics of the incoming radiation, considering also the degradation introduced by the readout electronics. The DQE is defined as follows, where f is the spatial frequency expressed in lp/mm (see Section 9.1.2):

$$DQE(f) = \frac{SNR_{out}^2(f)}{SNR_{in}^2(f)} \tag{5.7}$$

In the ideal case, the DQE tends to be equal to the *Quantum Efficiency* (QE), which is defined as the ratio between the absorbed and the impinging photons [8].

5.2.3. Effects of interstrip interpolation mode

As presented in Bergamaschi et al. 2022 [4], the *S curve* coming from the central counter in interpolation mode can be considered the "golden standard" for the calibration due to the suppression of charge sharing effects. Compared to the standard mode, only a difference of 1% in terms of gain and offset has been reported [4]. At the same time, in the standard mode the noise is overestimated by about 10%. However, the data in Bergamaschi et al. 2022 [4], has been proved for 50 μm Si sensors, so for GaAs the numbers can differ but the same trend is expected: indeed, due to the larger charge sharing expected for GaAs:Cr (see Section 8.1) as documented in Ruat et al. [2], the difference between standard and interpolation mode are expected to be even more evident.

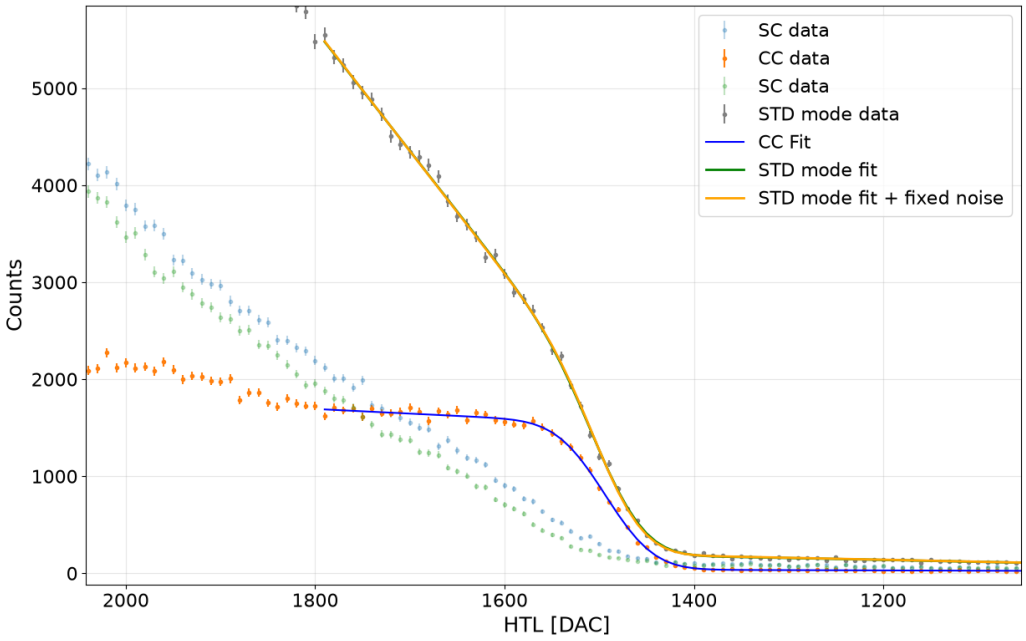


Figure 5.3 Comparison of standard mode *S curve* fitted with all free parameters with one with fixed noise parameters coming from the central counter in interpolation mode. SC stands for Side Counter, CC for Central Counter (all coming from data acquired in interpolation mode), while STD for Standard Mode.

A verification has been checked for the GaAs:Cr detector. The noise parameter coming from the central counter (CC) in interpolation mode has been fixed in the fitting of the standard mode *S curve* acquired under identical conditions. As shown in Figure 5.3, the resulting curve (yellow) overlaps with the fit curve instead having all parameters left free (green). This supports the correlation between the noise and the charge sharing in standard mode, coherently with the noise overestimation predicted above.

5.3. Optimal comparator threshold

The choice of the threshold level is a key parameter in photon counting detectors. For monochromatic X-rays, the threshold must be set to half of the photon energy. In this way, count losses or overestimations linked to charge sharing deleterious effects are mainly avoided, as well as possibly pollution coming from being too low and close to the electronic noise [4].

Moreover, as discussed in Bergamaschi et al. [4], setting correctly the threshold at half photon energy leads to getting an ideal DQE (see Section 5.2.2).

For example, if the level is set at energies higher than the ideal one, a fraction of counts (whose amplitude is also reduced due to charge sharing effects) will be neglected leading to a loss of efficiency but sharply increasing the spatial resolution since the strip size will see photons as if it were tighter. On the other hand, when the comparator threshold is set below half of the incident photon energy ($E_{th} < E_0$), the charge cloud generated by a single photon absorbed in the gap between two strips can provide enough charge to exceed the threshold in both adjacent channels. This causes a single physical event to be registered multiple times, a phenomenon known as *multiplicity* [4] that affects the detector also in terms of spatial resolution.

As demonstrated by Michel et al. [36], this increased multiplicity has two major negative effects. Firstly, it degrades the spatial resolution, since the exact point of interaction cannot be uniquely assigned when multiple detector strips are triggered. Secondly, it introduces additional statistical fluctuations in the measured number of counts, since multiple counts may originate from a single photon interaction [4] [5], then deviating the SNR from the ideal one.

5.4. Weighted mean

Since the detector parameters are extracted for several channels, a single representative value is required for each measurement. However, the uncertainties and the derived values associated with the individual channel measurements can be different. For these reasons, a weighted mean is employed: hence channels whose parameters are determined with a smaller uncertainty contribute more to the average, while less precise measurements are assigned a lower weight.

Consequently, the average can be computed as follows:

$$\overline{x_w} = \frac{\sum_{i=1}^{N} w_i x_i}{\sum_{i=1}^{N} w_i} \tag{5.8}$$

where N is the number of channels and w_i is the assigned statistical weight to the value x_i. The weight is defined as follows:

$$w_i = \frac{1}{\sigma_i^2} \tag{5.9}$$

Instead, the uncertainty referred to the weighted mean $\overline{x_w}$ is:

$$\sigma_{\overline{x_w}} = \frac{1}{\sqrt{\sum_{i=i}^{N} w_i}} \tag{5.10}$$

This quantity describes how precisely the average detector response is known. As more channels contribute to the average, this uncertainty generally becomes smaller.

However, the uncertainty on the weighted mean should not be confused with the actual variation of the detector response from channel to channel. Even when the average value is determined very precisely, the individual channels can still show significantly different gain, noise or offset values: then the *channel to channel* spread is evaluated separately, and it is defined as the standard deviation of the considered channels as follows:

$$s_{ch} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2} \tag{5.11}$$

where $\bar{x}$ is the simple average mean.

In (5.11), s_{ch} represents the spread among the channels of the considered parameter. Differently from $\sigma_{\bar{x}_w}$, it quantifies the non uniformity of the response. Therefore, both quantities are reported throughout the detector characterization.

6 Detector characterization

The measurement that will be presented in this chapter aim at characterizing the response of MYTHEN III coupled with the two prototype GaAs:Cr sensors already introduced in Section 2.1. In particular, the detector response is studied considering the effects of the feedback configurable parameters (see Section 2.2.2) on figure of merits like gain and noise.

Before starting the GaAs:Cr characterization, a complete MYTHEN III module equipped with standard Si sensors was measured as a reference to verify the analysis procedures before applying them to the GaAs:Cr sensors. Therefore, the baseline operating region is investigated for GaAs:Cr. The influence of the configurable preamplifier and shaper parameters on gain and noise is then studied, from which the corresponding operating energy range is estimated. Finally, the goodness of linearity of the energy calibration is verified and the dependence of the detector response on additional operating conditions, such as temperature, is studied, and the threshold trimming performance is evaluated to equalise the channels response.

6.1. Baseline operating point and gain dependence

As described in Section 2.2.2.2, the baseline position, despite indirectly, is controlled by the reference voltage *VDC,shaper*. Changing this parameter shifts the shaper baseline and consequently the calibration offset (the extrapolated threshold position corresponding to zero photon energy), therefore a suitable *VDC,shaper* must be found before characterizing the detector: indeed, if the baseline is too close to one end of the available threshold range, part of the useful response may become unavailable and, on the other hand, an excessive baseline shift may reduce the remaining range available for higher energy signals. Moreover, *VDC,shaper* might also affect the measured gain by varying the operating point of the shaper stage.

Due to the GaAs:Cr detector being operated with negative polarity, rather than the positive polarity used for the Si sensors, the pulse at the shaper output develops in the opposite direction. Therefore, a higher *VDC,shaper* and consequently a higher baseline is expected for the former than for the latter. The value for Si is already known but it is first verified here by using a separate, complete MYTHEN III Si detector, provided with 10 sensors, each with 128 strips with the following size: 50 μm width per 8 mm length and 450 μm thickness, hence yielding 1280 strips and thus granting reliable

statistics. Instead, for the GaAs, the same procedure is used to seek a reasonable value of the baseline.

6.1.1. Silicon module scan

The following survey of threshold scans was carried out by setting the MYTHEN III in the Standard readout configuration[12] and then scanning from 0 up to 2200 *DAC* (increasing by 10 at every step) in terms of threshold variation. Instead, the *VDC,shaper* was varied from 200 to 1400 *DAC* in steps of 20, still centred around the default value of 800 *DAC*. The results obtained with the 17.5 keV Mo K-alpha photons are shown below.

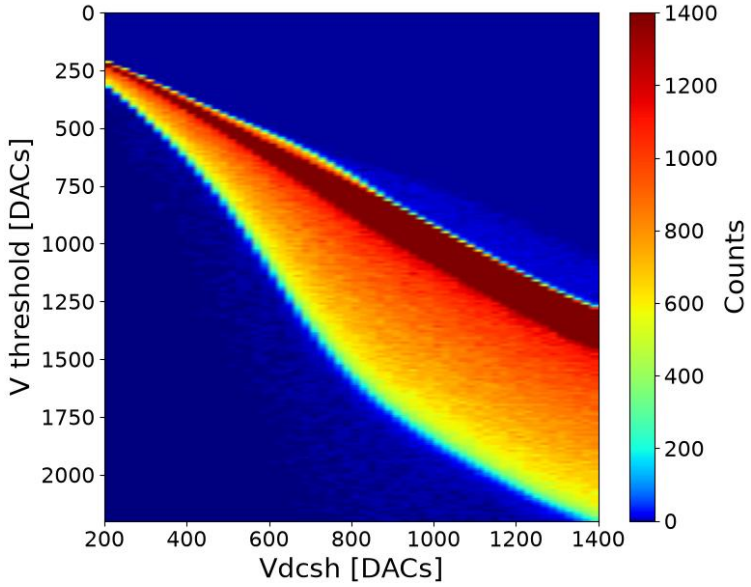


Figure 6.1 Threshold scans[13] of a representative channel (channel 803) as a function of *VDC,shaper* (indicated as *Vdcsh*)

As visible in Figure 6.1, increasing *VDC,shaper* produces a shift of the baseline towards higher threshold *DAC* values, with an approximately linear behaviour. At the same time, the difference between the baseline and the photon *S curve* increases, indicating an increase of the signal amplitude in *DAC* units (and probably also in gain). This variation becomes progressively smaller around the default *VDC,shaper* value of 800 *DAC*, where the response appears to approach saturation.

Furthermore, to verify whether the gain is affected by the *VDC,shaper* varying, energy calibrations were performed for some of the latter's values by fitting the *S curve* inflection points extracted from threshold scans (see Section 5.2.1) acquired with

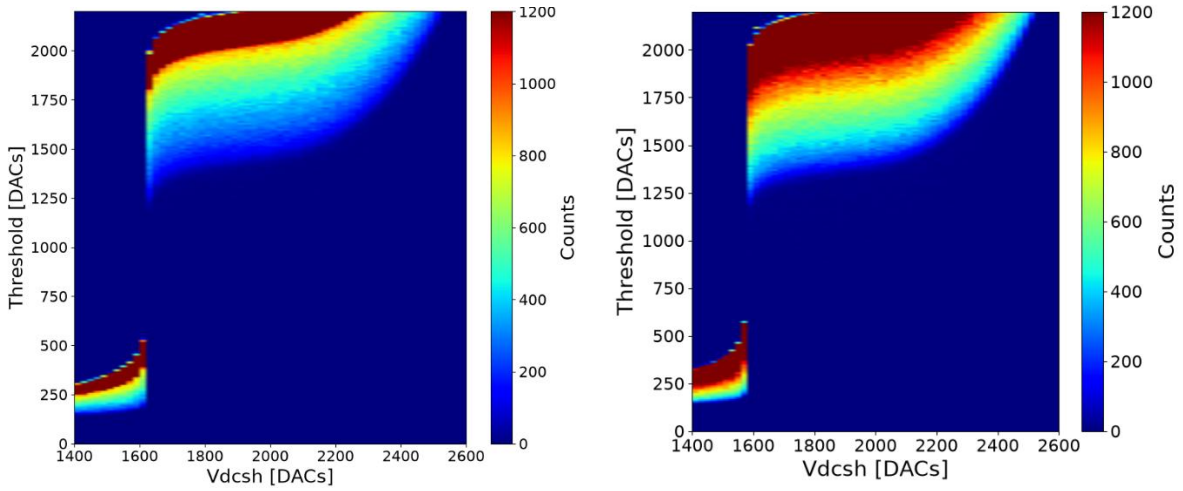
[12] Standard settings for Si: *VR,preamp = 900 DAC, VR,shaper = 1050 DAC, VDC,shaper = 800 DAC, C,preamp = 10 fF, C,shaper = 30 fF, C,AC = 450 fF*

[13] Differently from a two dimensional plot, each vertical slice corresponds to a conventional threshold scan acquired at a fixed *VDC,shaper*; instead of plotting the counts along the vertical axis as for a single *S curve*, they are here encoded by the colour scale, while threshold and V_{dcsh} are represented on the vertical and horizontal axes, respectively.

different fluorescence targets (Cr, Cu, Mo and Sn): this confirms the gain increasing until around 800 *DAC*, above which the variation becomes significantly smaller. The relatively large spread is given by differences among the readout ASICs which behave differently from each other, and this affect the uniformity of the module response.

6.1.2. GaAs:Cr scan

Like the previous section, the *VDC,shaper* has been scanned from 1400 to 2600 *DAC* in step of 20, maintaining the former threshold scan range used for the Si. The 50 µm and 100 µm strip pitch sensors have been irradiated at the same time, retaining the preliminary readout configuration[14]. Note that these settings were adopted *a priori* as a working configuration and should therefore be regarded as preliminary.



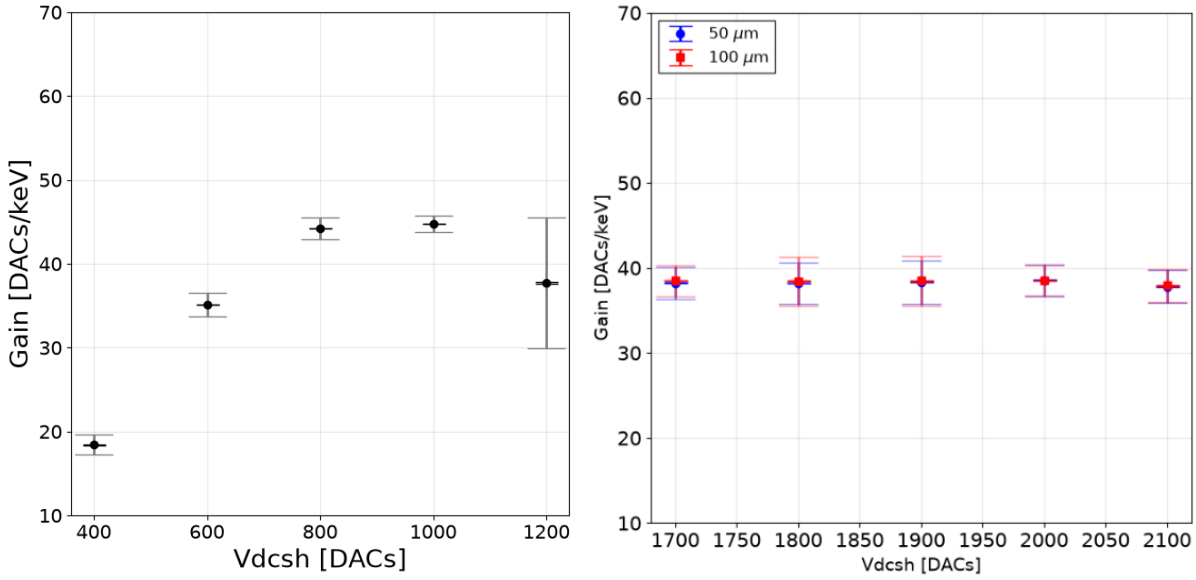
(a) Channel 501 (50 µm) (b) Channel 795 (100 µm)

Figure 6.2 *VDC,shaper* scan for both sensors' representative channels. Same method as described for Figure 6.1 (indicated as *VDC,shaper*)

The results obtained with 17.5 keV Mo $K\alpha$ photons are shown in Figure 6.2. Beside the different polarity of the *S curves* with respect to the Si ones, it is visible a non-working region for values below 1600 *DAC*, as well as a reduction of thresholds range for higher *VDC,shaper* values.

For the GaAs:Cr sensors, the gain remains nearly stable over the investigated *VDC,shaper* range, indicating that the gain is already operating in a saturation region. Therefore, *VDC,shaper* can be fixed to 2000 *DAC* for the futu re measurements. This value also retains a wider useful threshold range, which is beneficial considering the potential use of GaAs:Cr at higher photon energies than those typically accessible with Si sensors.

[14] *A priori* settings for GaAs:Cr: *VR,preamp = 2000 DAC, VR,shaper = 2400 DAC, VDC,shaper = 2000 DAC, C,preamp = 10 fF, C,shaper = 30 fF, C,AC = 450 fF. These values were selected a priori as a starting working configuration*



(a) MYTHEN III Si module (b) MYTHEN III GaAs:Cr modules

Figure 6.3 Comparison of gain vs *VDC,shaper* (indicated as *VDC,shaper*) for Si (a) and both GaAs:Cr sensors (b), with gain calculated using the weighted mean (see Section 5.4), while inner and outer error bars represent respectively mean error and 1σ channels spread

The weighted average gains shown above for the Si and GaAs 50 μm, 100 μm are respectively $44.20 \pm 0.01 \frac{DAC}{KeV}$, $38.55 \pm 0.04 \frac{DAC}{KeV}$ and $38.49 \pm 0.03 \frac{DAC}{keV}$. Considering for the latter only the 50 μm data, heir ratio close to the inverse ratio of the energies required to create an electron–hole pair in the two materials:

$$\frac{G_{Si}}{G_{GaAs}} = \frac{44.20}{38.55} \cong 1.15 \tag{6.1}$$

$$\frac{\varepsilon_{GaAs}}{\varepsilon_{Si}} = \frac{4.2}{3.6} \cong 1.17 \tag{6.2}$$

This close agreement suggests that the different electron-hole pair creation energies might explain most of the observed gain difference.

A similar comparison has been reported by Ruat et al. [2], although they also noted that the effective gain can be potentially affected by differences in the sensor electrical contact structures and material properties. Nevertheless, in the present measurements, the comparison should therefore be regarded as qualitative, since the Si and GaAs:Cr detectors were not operated under fully equivalent readout conditions due to the different semiconductor material.

6.2. Gain dependence on analogue readout settings

After finding a suitable baseline operating region, in terms of *VDC,shaper*, for the GaAs sensors in their preliminary/default readout setting, the influence of the analogue front-end settings has been investigated, with the aim of defining their effects on the gain and noise. Therefore, the following analysis presented in the next sections focuses on how the measured calibration gain varies accordingly to the adjustable feedback parameters of preamplifier and shaper, as well as which parameters possess the strongest influence. At the same time, the experimental data will serve to verify the model already presented in Section 2.2.2. Recalling this section, the MYTHEN III ASIC readout chain consists of a CSP followed by an AC coupled shaper, both equipped with adjustable feedback and responsible in determining the response in terms of energy calibration. Indeed, the gain is not computed for the single amplifying stage itself, but for the complete chain.

For each feedback configuration, several threshold scans were acquired at known photon energies, and the corresponding inflection point were fitted in order to get the gain. As introduced in Section 5.4, the result will be the weighted average among the channels. The same procedure is applied to the offset, which corresponds to the zero photon energy.

Since the interstrip interpolation mode is not available for the 100 μm sensor (see Section 2.4.2), the latter's data was acquired using the standard one. Eventually, five different targets have been irradiated to use their photons emission to calibrate the sensors: copper (8.0 keV), Selenium (11.6 keV), Molybdenum (17.5 keV), Silver (22.4 keV) and Tin (25.6 keV) (see Section 4.1.2).

6.2.1. Feedback capacitance variation

The first parameters to be broadly adjusted were the feedback capacitances, respectively the preamplifier, the shaper and the coupling one. Instead, the feedback resistances have been kept fixed. Furthermore, the baseline voltage *VDC,shaper* has been set at 2000 *DAC*, as verified in Section 6.1.2.

So, the table below summarizes the measurement combinations which have been tried: in this case thirty different measurements have been carried out, while *VR,preamp* , *VR,shaper*, *VDC,shaper* has been kept fixed, respectively, at 2400, 2000, 2000 *DAC*.

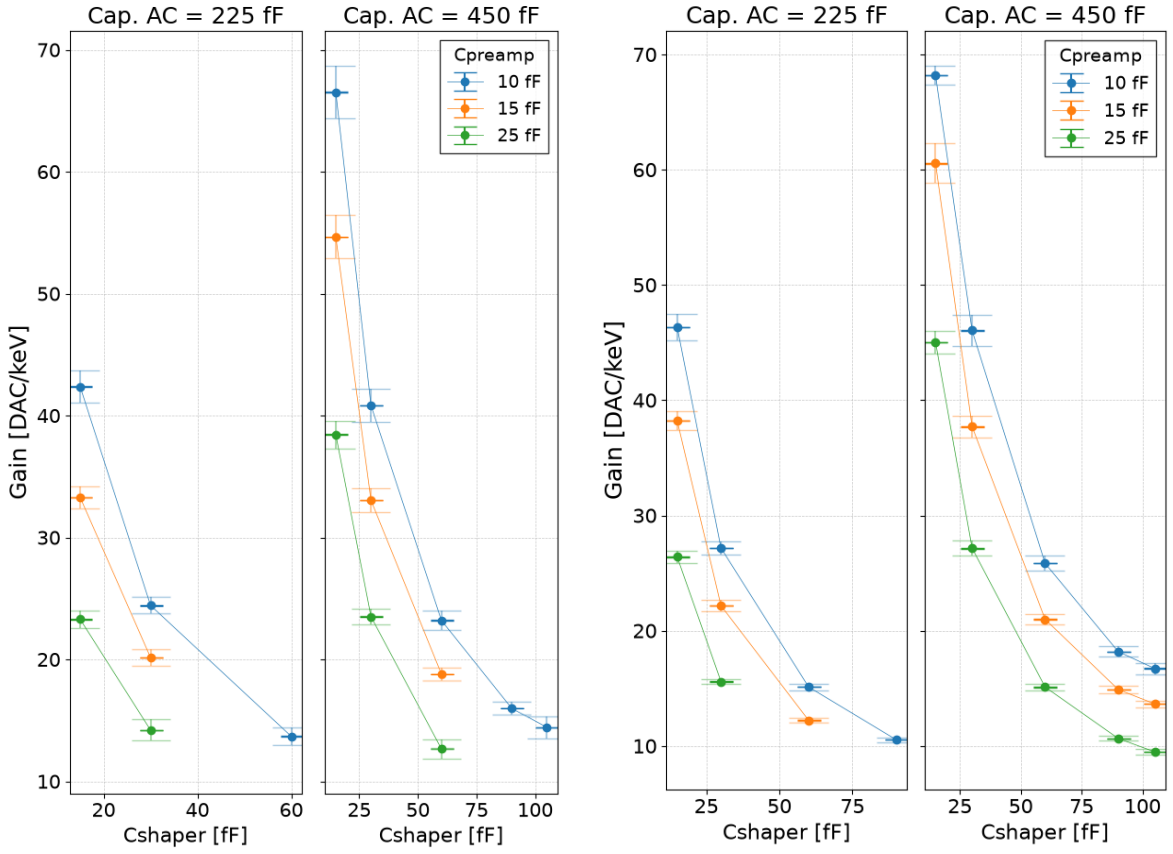
Table 6.1 List of capacitance values combination set

Capacitances [fF]	
Preamplifier	10, 15, 25
Shaper	15, 30. 60, 90, 105
Coupling	225, 450

According to the already described electronics, the gain is expected to follow a certain behaviour (see Section 2.2.2.3). Then, recalling Eq. (2.11), where parasitic capacitance contributes have been neglected [7]:

$$G \propto \frac{1}{C_{f,pre}} \frac{C_{AC}}{C_{f,sh}} \tag{6.3}$$

Hence, from the measurement surveys the following data has been collected, summarized in the below Figure 6.4:



(a) 50 µm sensor (interpolation mode) (b) 100 µm sensor (standard mode)

Figure 6.4 Gain vs variation of *C,shaper* and *C,preamp* (*C,AC* indicated as *Cap. AC*)

Firstly, a clear monotonic decrease of the calibration gain is observed when increasing the shaper capacitance, and this behaviour is clearly visible for every preamplifier capacitance. Furthermore, the gain decreases with the increase of the preamplifier capacitance and with the decrease of the coupling one. Overall, all three trends appear qualitatively consistent with the simplified front-end model introduced in Section 2.2.2.3.

Not all configurations however exhibit reliable gain results. For instance, for some of the lowest-gain configurations, approximately below 15 $\frac{DAC}{keV}$, the *S curve* is sampled by too few points to obtain a sufficient good fit: this comes down to the scarce gain.

Therefore, the corresponding gain values are affected by larger uncertainties and thus not considered reliable and eventually left out.

As already mentioned above, the measurements simultaneously show that, given a fixed shaper setting, the configuration with the lowest preamplifier capacitance provides the largest gain, and vice versa. This reproduces the behaviours observed in the Andrä's PhD thesis [7], that investigated the characterization of the Silicon version of the MYTHEN III, where the 10 fF preamplifier configuration exhibited a larger gain than the corresponding 15 fF one. Moreover, the comparison between the two AC coupling configurations shows a systematic increase of the gain when the full 450 fF capacitance is set up rather than 225 fF.

The two sensors present slightly different values despite reproducing the same expected behaviour, but this discrepancy should not be interpreted as a pitch dependence: besides the different sensor geometry, which varies the input capacitance seen by the preamplifier as well as the leakage current, the two are connected to different ASIC chips and the data come from different counting modes (standard for the 100 μm, interpolation mode for the 50 μm chip).

Eventually, the measured gain variations are weaker than the ideal inverse capacitances dependence. This is not completely unexpected, since the simplified relations describe the individual analogue stages and neglect fixed input and parasitic capacitances [7], as well as other non-idealities, that might present a "damping" effect on the gain variation. Indeed, the measured variation is smaller than the factor of two expected from the ideal relation: for example, around the reference $C_{f,\mathrm{pre}} = 15\mathrm{fF}$ and $C_{f,\mathrm{sh}} = 30\mathrm{fF}$ configuration, the measured gain increases by approximately a factor of 1.6.

6.2.1.1. Model verification

To quantify how closely the measurements follow this simplified capacitive model, the measured gains are compared to the ones predicted by the latter after normalizing both to the following reference configuration:

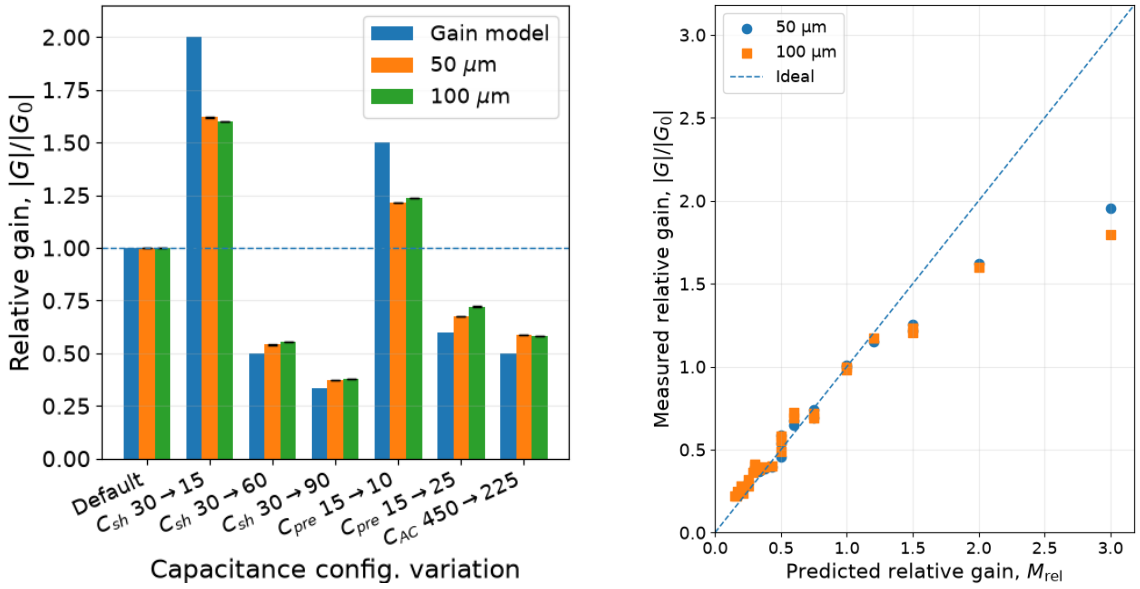
$$C_{coupling,0} = 450\ fF, C_{preamp,0} = 15\ fF, C_{shaper,0} = 30\ fF$$

Therefore, the relative observed gain and the expected one can be defined respectively as follows:

$$G_{rel} = \frac{|G|}{|G_0|} \tag{6.4}$$

$$M_{rel} = \frac{C_{AC}}{C_{AC,0}} \frac{C_{pre,0}}{C_{pre}} \frac{C_{sh,0}}{C_{sh}} \tag{6.5}$$

In this way, the unknows factors (i.e. the parasitic ones) cancel out and only the variation produced by the varying configuration is tested.



(a) Relative gain versus capacitance model (b) Verification of gain model

Figure 6.5 Comparison predicted versus observed gain

Figure 6.5(a) above compares the predicted relative gain with the observed one, with one capacitance at a time being varied with respect to the above mentioned reference configuration. What can be inferred from is that the response is systematically weaker than predicted: indeed, settings for which the ideal model predicts an increase tend to remain closer to the reference, and the same happens for gain reduction under unity. The same "damping" has been observed in Andrä's work [7], indeed *"For the combination of 60fF and 225fF, the gain decreases due to both, the larger feedback (shaper) capacitance as well as the lower coupling capacitance. However, the decrease in gain is less than expected."*

A more general comparison can be obtained, as shown in Figure 6.5(b), by plotting the measured relative gain directly against the predicted one. While some of the data lap in the surrounding of the ideal behaviour (the straight diagonal line whose points would represent $G_{rel} = M_{rel}$), for larger gain variation there is however a systematic reduction: in particular the ideal model tends to overestimate the gain increase got by reducing the capacitance, as well as the gain reduction got by their increase.

6.2.2. Feedback resistance variation

After measuring the effects of varying the capacitances, the effects of resistor variation on the signal formation have been investigated. While keeping fixed the *VDC,shaper*, the parameters responsible for adjusting the feedback resistances, namely *VR,preamp* and *VR,shaper_npol*, have been varied systematically in order to estimate their contribution to the signal formation and therefore on the gain itself. Analogous to the former, capacitance values have been kept constant: for the latter, an high gain setting has been chosen.

One noteworthy difference with respect to the capacitors is the possibility of controlling continuously through dedicated *DAC* voltages, rather than discretely switch components on and off. However, likewise the precedent analysis, the measurement survey has been carried out by setting some defined voltages, as the following Table 6.2 shows, while *C,preamp*, *C,shaper*, *C,AC* has been kept fixed, respectively, at 10, 30, 450 fF. Also the *VDC,shaper* has been kept, at 2000 *DAC*.

Table 6.2 List of voltage resistance values combination set

Voltage resistances [*DAC*]	
Preamplifier	2300, 2400, 2500, 2600, 2700
Shaper	1600, 1700, 1800, 1900, 2000, 2100

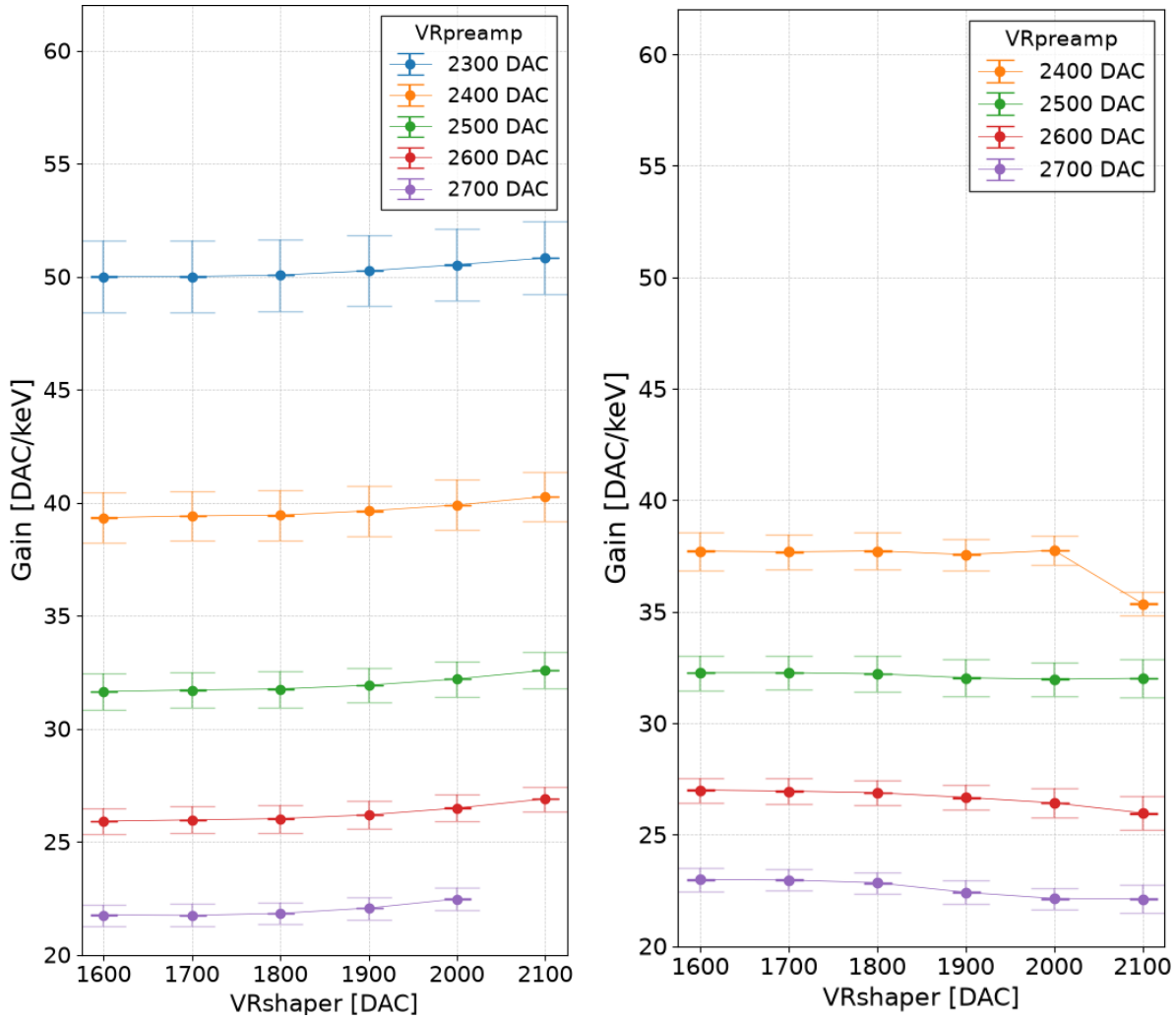
Hence, from the measurement surveys the following data has been collected, summarized in the below Figure 6.6.

Actually, the measurement comprehended *VR,shaper* values like 2200 and 2300 *DAC* and the ones lower than 1600 *DAC*, but the output gave for the mentioned setups any reliable data: one possible hypothesis is the saturation of the MOFSET, or generally the leaving of its linear working region, due to the excessive or generally unadapted voltage applied on that, then driving the stage out of its own operating point.

The 100 μm sensor, moreover, showed malfunctioning for the *VR,preamp* set at 2300 *DAC* and lower: this may be due to the major leakage current density incoming from the sensor's strips, which is as expected larger for the 100um one rather than the 50um (see Section 3.3.6). So, a larger DC current through the preamplifier feedback network can shift excessively its operating point and therefore reduce the available stable feedback range. No similar problems have been reported instead for the 50 μm sensors, except for values lower than 2300 *DAC*, thereby measurements were not feasible.

However, the ranges employed in the present measurements are considerably higher than the standard values reported instead for the Si version. While the difference attributed to *VR,shaper* is explained by the higher baseline voltage *VDC,shaper*, there is

no clear explanation for the preamplifier voltage *VR,preamp* : possible explanation cloud be a different working point of the whole readout chain or even the larger leakage current of the GaAs strip.



(a) 50 µm sensor (interpolation mode)

(b) 100 µm sensor (standard mode)

Figure 6.6 Weighted average gain versus *VR,shaper* and *VR,preamp* variations

A clear stronger dependence is observed on the preamplifier feedback control: increasing the *VR,preamp* causes a progressive shift of the gain value towards lower values, and the same trend is observed for both sensors. However, the mentioned dependence goes in the opposite direction with respect to the one reported for the Silicon sensor MYTHEN III (which is operated in positive polarity mode) characterized in Andrä's PhD thesis, where increasing *VR,preamp* leads to larger gain. This discrepance could be explained by the negative polarity mode, as well as the different *DAC* values employed and the fact that for electron collection mode the *n_pol* shaper's MOSFET transistor is used to deal with negative signals.

Nevertheless, both measurements depict that the preamplifier feedback control has a strong influence on the overall gain, despite that in this case the shaper, but also the preamplifier equivalent resistances are inversely proportional to the gate voltage applied, the opposite with respect to what it is shown in Andrä's work.

About the effects of the shaper, within the explored range, the measured gain is indeed only weakly dependent on the *VR,shaper* voltage: the gain even still lies in the error caused by the spread among the channels themselves. Indeed, as reported by Andrä's work, the gain lies in values between 13.2 and 43.5 $\frac{DAC}{keV}$ by regulating the preamplifier feedback, but the effect of the shaper on gain were almost negligible.

In conclusion, the weak dependence on *VR,shaper* does not make the shaper useless in terms of gain, since the capacitance scans depict a strong contribution on the latter. So, these experimental data are in agreement with the different roles of the feedback components described in Section 2.2.2: indeed, the capacitance determines the amplification, while the resistance modifies the temporal response and thus affecting only indirectly the measured peak amplitude via ballistic deficit.

6.2.3. Overview summary

As conclusive summary of the gain dependence on the feedback, the following Table 6.3 is depicted:

Table 6.3 Summary of feedback variation effects on gain

Parameter	Observed gain dependence as parameter increases
Preamplifier capacitance	Gain decrease
Shaper capacitance	Gain decreases strongly
Coupling capacitance	Gain increases
Preamplifier resistance (voltage)	Gain decreases strongly
Shaper resistance (voltage)	Gain remains nearly constant

6.2.4. Offset variation

As discussed in Section 6.1, the offset is mainly determined by the *VDC,shaper* parameter which affects the shaper baseline (see Section 2.2.2.2): therefore, once fixed the latter, the dependence on other investigated parameters is considered weaker that their effect on other figure of merits. However, some trends are visible:the offset increases with increasing *VR,shaper*, while it slightly decreases with increasing *C,preamp*. No clear dependence instead is observed for *C,shaper* and *VR,preamp*.

6.2.5. Linearity of energy calibration

The energy calibration presented in the current Chapter, from which the gain and offset are derived (see Eq. (5.5)), assumes a linear relation between the inflection point and the photon energies. Therefore, the former relation linearity has been investigated briefly for some readout settings to prove generally the fit goodness.

In order to try to infer quantitative results, the residual has been defined as follows:

$$\Delta E_i = \frac{\mu_i - \mu_{fit}(E_i)}{g} \tag{6.6}$$

where μ_i is the weighted average inflection point at the photon energy E_i, while $\mu_{fit}(E_i)$ is the corresponding value but obtained from the calibration. The residual is then divided by the gain in order to express it in terms of energy and allowing to compare it among different settings.

However, from Eq. (6.6) the two real figure of merits used in this section are derived, namely the maximum absolute residual and the corresponding *root mean square* (RMS), as follows:

$$\Delta E_{max} = \max_i |\Delta E_i| \tag{6.7}$$

$$RMS(\Delta E) = \left[\frac{1}{N}\sum_{i=1}^{N} \Delta E_i^2\right]^{1/2} \tag{6.8}$$

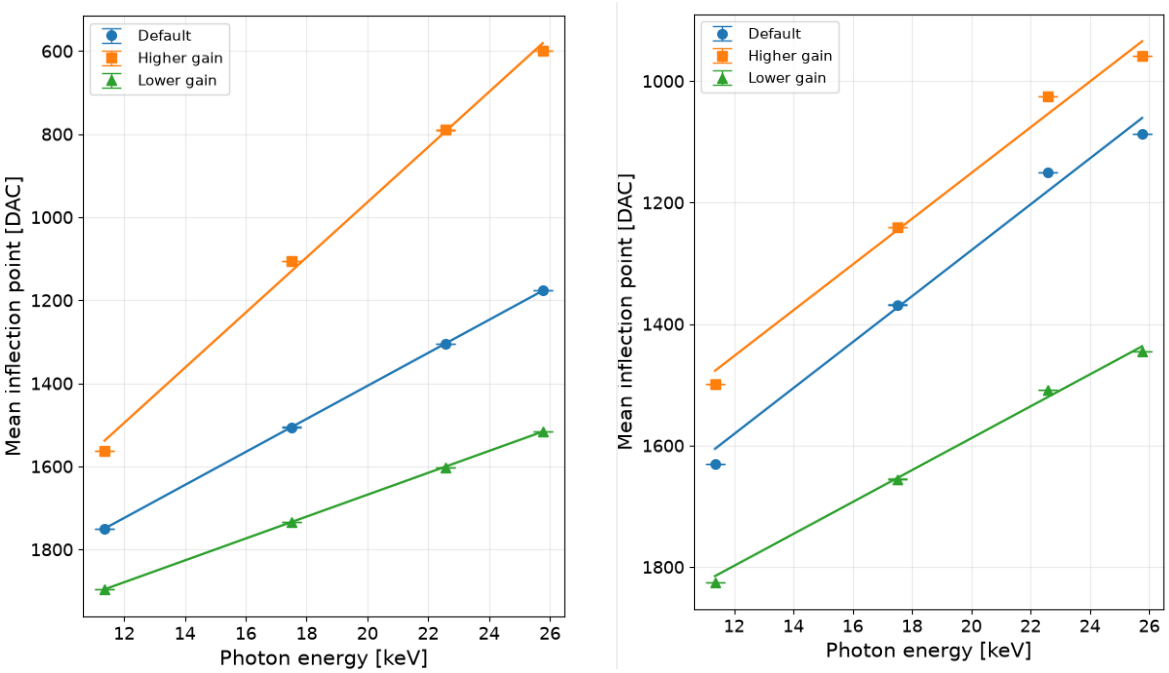
Therefore, the former describes the largest deviation occurring, while the latter the overall deviation from linearity[15].

Figure 6.7 shows, for both the 50 μm and 100 μm sensors, the linear fit of three different representative readout settings, which differ each other due to their different gain value. The inferred figures of merit are reported in Table 6.4:

Table 6.4 Quick summary of figures of merit for linearity

	50 μm [eV]			**100 μm [eV]**		
	Default	**Higher**	**Lower**	**Default**	**Higher**	**Lower**
$\boldsymbol{\Delta E_{max}}$	51	365	42	816	776	445
$\boldsymbol{RMS(\Delta E)}$	32	292	29	632	585	332

[15] The gain fit, likewise the previous sections, has been performed taking into account the error of the mean rather than the channel spread, which represent the dispersion of the response among the different strips instead of the mean value uncertainty.



(a) 50 µm sensor (b) 100 µm sensor

Figure 6.7 Energy calibration for three representative readout settings of the 50 µm and 100 µm sensor. Here only the error bar pertaining to the error of the mean is depicted.

Overall, the linear calibration gives a good description of the mean detector response over the investigated energy range. However, the 100 µm GaAs:Cr sensor shows larger deviations from the linear calibration than the 50 µm one for all the selected settings. For comparison, the reference Si detector shows an intermediate behaviour, with $\Delta E_{max} \approx 298\, eV$ and $RMS(\Delta E) \approx 188\, eV$. Moreover, the current analysis cannot identify the origin of the larger observed variations.

6.3. Noise dependence on analogue readout settings

In general, noise can be defined as any unsought fluctuation superimposed on the original signal. It affects the detector performance by degrading the energy resolution. Noise can originate from different sources, either intrinsic to the sensor or introduced during the signal-processing chain, for example in the analogue front-end electronics [7] [10].

In particular, for small signal variance detectors such as semiconductors, the noise related to the electronic chain is critical in affecting the energetic resolution [10]. Moreover, the electronic noise is also responsible for determining a lower cap on the ability of detecting the lower signal as well as for discerning ones with similar amplitudes [10].

As defined in Radeka [37], the *equivalent noise charge* (ENC) is the measure of the noise expressed as an equivalent charge at the detector input. It represents then the amount of input charge that would produce a signal equivalent, in terms of RMS, to the measured noise at the output of the analogue chain.

For SPCs, the ENC is usually expressed in electrons [15] and can be inferred from the noise parameters obtained from the fitted *S curve*, provided that the gain and the mean energy required to generate an electron hole pair are known:

$$ENC[e^-] = \frac{\sigma[DAC]}{g\left[\frac{DAC}{keV}\right]\ \varepsilon\left[\frac{keV}{e^-}\right]} \tag{6.9}$$

Where σ is the noise parameter extracted from the *S curve* fitting, g the computed gain and ε the energy required to generate an electron hole pair. Naturally, it can also be expressed in energy units by multiplying the ENC by ε.

In this work, the ENC is determined individually for each channel, and the weighted average value (like usually done for other parameters like gain, offset, etc.) is then calculated over all channels, taking into account the spread of the noise values among the strips.

The ENC obtained from the *S curve* fit represents the noise measured at the channel level and therefore does not necessarily correspond exclusively to the contribution of the readout electronics. Moreover, as mentioned in Andrä's work [7], the noise and charge sharing extracted parameters are correlated with each other in the *S curve* model. Since the charge sharing model is strongly simplified (see Section 8.1), this correlation can thus affect the extracted noise value and thus represents a limitation on the accuracy with which the noise can be determined from the *S curve* fit [7].

6.3.1. Sensor-related noise contributions

Contributions originating from the sensor itself can also affect the measured noise. In particular, leakage current can contribute also to the measured noise due to its statistical fluctuations (giving origin to shot noise), thus degrading detector figure of merits like energetic and timing resolutions. The leakage current contribution is particularly relevant for GaAs:Cr sensors, as different sensor materials can exhibit significantly different dark current intensities.

Indeed, as discussed in Greiffenberg et al. [6], which analysed two different batches of GaAs by employing a *Charge Integrating Detector*, a lower dark current was observed for the 2017 GaAs:Cr sensor (13.38 μA) compared with the previously characterized 2016 sensor (25.12 μA). The corresponding noise was also lower, respectively with an ENC of 101.65 e^- compared with 115.93 e^-. Moreover, the spatial distribution of the noise followed that of the dark current, while the noise increased with increasing integration time, most likely due to the increased shot noise associated with the dark current.

One of the major perks of SPC detectors is the capability of rejecting pulse whose amplitude is below the selected threshold; hence the latter makes possible to suppress lower-amplitudes leakage current contributions except of counting them as events. Nevertheless, the leakage current fluctuations are superimposed on the signal and therefore appear as noise on the recorded pulse. Moreover, an enough large current can even disrupt the normal operating point of the electronic chain (preamplifier mainly) though its feedback network, since the average leakage current component cause a voltage drop on the preamplifier feedback resistance, hence for higher values of the latter (see Section 2.2.2.1)

Eventually, other sources originating from the sensor are the Fano noise (see Section 3.1.7) and the interstrip capacitive coupling between neighbouring channels (Section 3.2.4), which Ruat et al. [2] deemed as the dominating factor of electronic noise. Other sensor-related effects, such as incomplete charge collection, fluorescence and charge sharing, can further affect the observed energy resolution [2], as well as the crosstalk between the ASIC electronic chains (which have not been covered in this work)

6.3.2. Electronic chain contributions

In addition to the contributions coming from the sensor, the readout chain introduces further noise, especially components like the preamplifier and the shaper. They can be generally classified into parallel and series noise sources [37], depending on they affect the signal: parallel noise sources include the white noise associated with leakage current fluctuations and the one originating from the current-one contribution of input transistor. On the other hand, the series noise includes the thermal noise of the feedback resistors and of the voltage-noise contribution of input transistor [7]. Moreover, the frequency spectrum of both the mentioned types is so broad that it is,

for the sake of simplicity, assumed as a uniform distribution, or "white". Instead, the signal spectrum is usually comprised in a more limited range of spectrum [15]. As already explained in Section 2.2.2.2, the shaper, but also the preamplifier, gives the signal a semi-Gaussian shape, acting at the same time as a filter to suppress part of the noise outside the signal bandwidth. Then the choice of the shaping time therefore affects the amount of noise reaching the comparator.

So, in the MYTHEN III analogue front-end, the overall temporal response is determined by the combined action of the preamplifier and the shaper. Their feedback networks, characterized by the feedback capacitance and resistance, determine the relevant time constants and thus contribute to defining the overall shaping time [7].

The overall contributions of the analogue front-end can be described using a simplified CR-RC model consisting of a preamplifier followed by a shaper. For this model, the total ENC is obtained by combining the parallel and series contributions quadratically [7] [38]:

$$ENC_{CR-RC}^2 = \frac{e^2}{4q^2}\left(\frac{\tau}{2}I_0 + \frac{1}{2\tau}V_0 C_{in}^2 + 2V_{-1}C_{in}^2\right) \tag{6.10}$$

Where q is the elementary charge, e the Euler's number, τ the shaping time, C_{in} the total input capacitance, I_0 the parallel noise current source, while V_0 and V_{-1} describe, respectively, the white and $1/f$ series voltage noise sources (the latter is typical of transistors). Moreover, the input capacitance is given by the sum of sensor, parasitic and preamplifier capacitance [7]. The equation shows the trade off between series and parallel noise contributions on the shaping time τ: as shown below in Figure 6.8, increasing τ reduces the series noise contribution, while increasing the parallel one. Consequently, a possible increase, for instance, of feedback preamplifier resistance (then more voltage drop expected due to leakage current) and thus the τ, does not necessarily increase the ENC. Moreover, the series contribution increases with the total input capacitance, making a low sensor capacitance desirable for reducing the ENC [38].

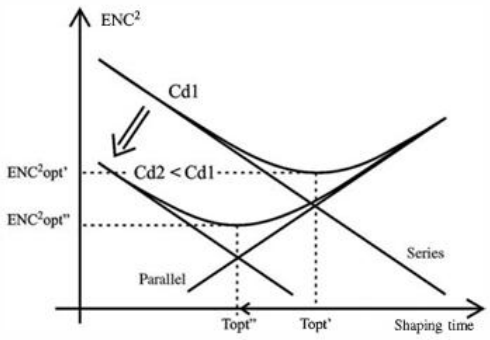
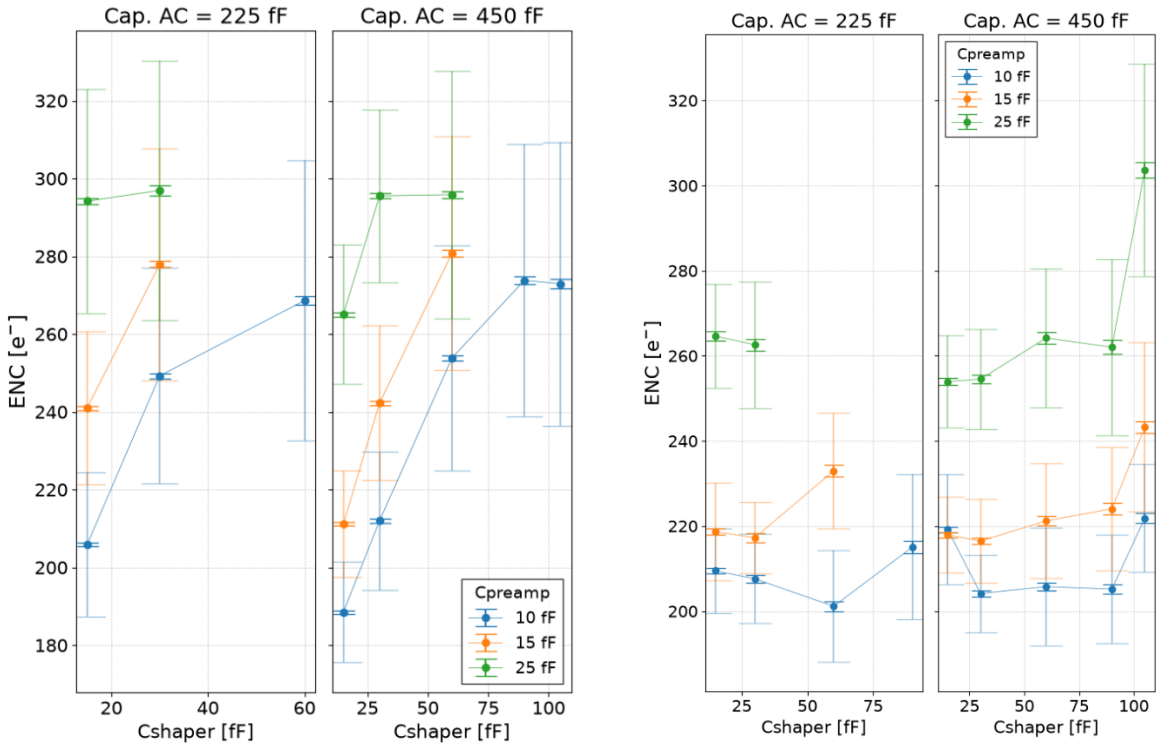


Figure 6.8 ENC variation versus τ and detector capacitance. Reproduced from [15]

6.3.3. Dependence on feedback capacitances

After working out the gain for every single channel, the ENC has been calculated using the Eq. (6.9), thus exploiting the noise parameter of the *S curve* fitting from the same data used to compute the gain.

As discussed in Section 5.2.3, the extracted noise parameter should be interpreted cautiously due to the significant amount of charge sharing. Since the 50 µm sensor was operated in interpolation mode while the 100 µm one in standard mode, the latter values are therefore potentially more affected by correlation between the fitted noise and the charge sharing contributions, especially for lower energies where the fitting is made more difficult [2]. Moreover, the noise width extracted from the *S curve* may be broadened also by fluorescence and incomplete charge collection effects [2].



(a) 50 µm sensor (interpolation mode) (b) 100 µm sensor (only standard mode)

Figure 6.9 ENC variation with *C,shaper* and *C,preamp* (*C,AC* indicated as *Cap. AC*)

In a similar fashion of the gain, the above figures depict the weighted mean ENC over the active channels, while the shaded bands correspond to the one standard deviation channel-to-channel spread: the latter is considerably larger than the uncertainty on the weighted mean, indicating a significant non uniformity among channels.

For the 50 µm sensor, dependencies on the capacitances are clearly visible: the average ENC generally increases with increasing preamplifier feedback capacitance, while it shows also a grow alongside the increase of shaper capacitance. Moreover, the 450 fF AC coupling configuration generally is affected by lower ENC values than the

corresponding 225 fF, but only when *C,shaper* is lower. These trends should be interpretated considering that the capacitance modify both the gain and the temporal response (hence the band pass), as discussed in Section 2.2.2.3. At the same time, the resulting changes in shaping and bandwidth vary the noise, hence the latter variation cannot be attributed only to gain changes alone

For the 100 µm sensor, instead, it is still visible a dependence on the preamplifier feedback capacitance, with larger *C,preamp* generally corresponding to higher ENC mean value. On the other hand the dependence on feedback shaper capacitances is clearly weaker and without a clear pattern. However, such a weak trend does not imply that the shaper has no effect on the electronic noise, since the latter's feedback causes variation of the shaping time which in turn affect the series and parallel noise, as explained in Section 6.3.2. Eventually, the absence of a dependence on C,shaper should not be interpreted quantitatively. In conclusion, the 10 fF preamp capacitance configuration can be deemed as the one characterized by the lowest noise achievable, likewise for the 50 µm sensor.

6.3.3.1. Model verification

To study the effects of gain variation on noise, each noise value for setting and associated gain have been normalized to the reference configuration (see Section 6.2.1.1). So, recalling Eq. (6.4) and(6.5), it is possible to check whether the relative noise in *DAC* would scale directly with the relative gain (this condition is represented by the diagonal in figure X). The below comparison shows that the noise width cannot be attributed strictly to the gain alone due to deviations, despite showing a general linear trend.

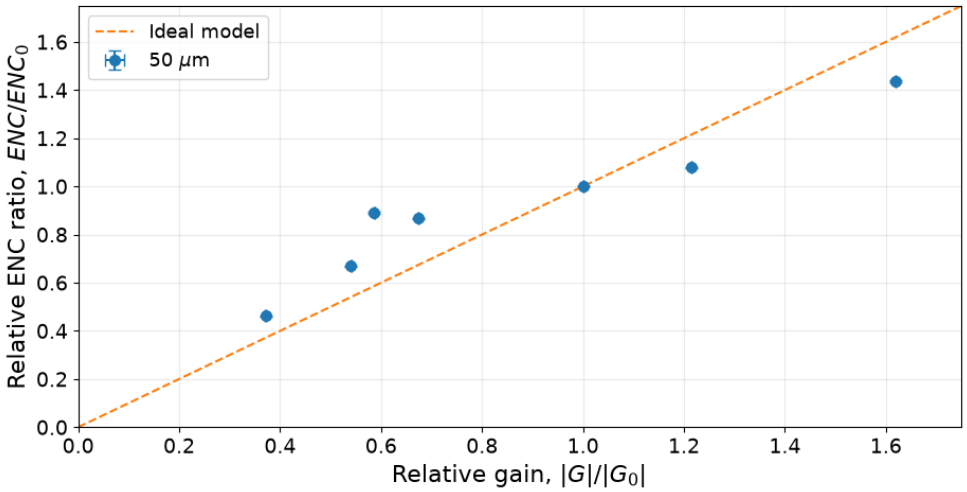
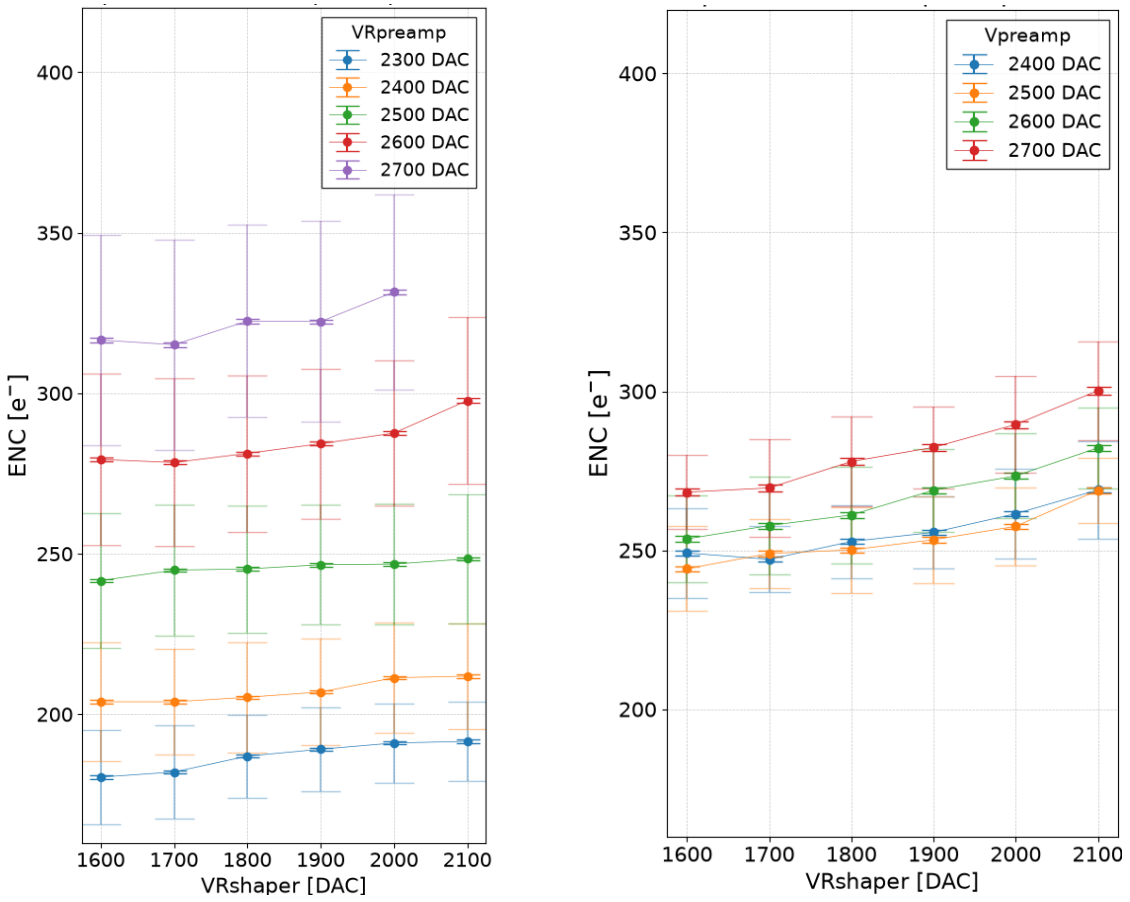


Figure 6.10 Relative ENC versus relative gain

6.3.4. Dependence on feedback resistance

In a similar fashion to before, the ENC has been computed to study its dependence on the feedback equivalent resistance variation. Moreover, the same consideration about the deleterious charge sharing effects on fit goodness can be made here.



(a) 50 µm sensor (interpolation mode) (b) 100 µm sensor (only standard mode)

Figure 6.11 ENC behaviour with *VR,preamp* and *VR,shaper*

For the 50 µm sensor, the separation between different *VR,preamp* settings is more evident, then suggesting a stronger influence of the preamplifier configuration on the average ENC. Also, a weaker but still systematic dependence on *VR,shaper* is also visible, with mean ENC only slightly increasing over the investigated range. Therefore, the shaper feedback resistance plays a minor role also on determining the noise, as its effects are moderate compared with the channel-to- channel spread.

Instead, for the 100 µm sensor the separation between different preamplifier settings is weaker and compatible, for lower *VR,preamp* , with the channel-to-channel spread. Nevertheless, an overall increase of the mean ENC with *VR,shaper* is still visible. As did for the capacitance measurements, this trend should only be treated qualitatively because of the employment of the standard mode threshold scan.

Overall, these measurement surveys confirm the MYTHEN III characterization by Andrä, where the preamplifier was found to dominate the gain and noise behaviour, while the influence of the shaper feedback setting on them is clearly weaker.

It should also be noted that the ENC is an input-referred quantity (like described in Eq. (6.9) where the noise parameter is divided by the gain): therefore, the previous observed ENC variations may originate from changes in the fitted noise parameter, as well as in the gain, or even in both quantities. Indeed, as reported in Andra's work, the noise behaviour with the capacitance variation has not been fully understood likewise for the (ideal) gain model.

6.4. Estimable usable energy range

After the gain and ENC estimations carried out and illustrated in the previous sections, the minimum and maximum energies for each readout setting have been worked out, in order investigate how the accessible energy range depends on the different readout settings.

For the minimum energy evaluation, the ENC must be considered: as explained in Bergamaschi et al. [5], for SPC detectors the threshold should be set at least five times the ENC to stay far enough from noise. Since the optimal threshold is about half the photon energy, the energy can be computed as follows:

$$E_{min} = 2E_{th,min} = 10\ ENC^2[e^-]\ \ \varepsilon\left[\frac{eV}{e^-}\right] \tag{6.11}$$

For the maximum energy, instead, the procedure has been based on extracting the latter from the linear fitting already used to estimate the gain, thus conservatively reduced by five times the ENC to consider the noise:

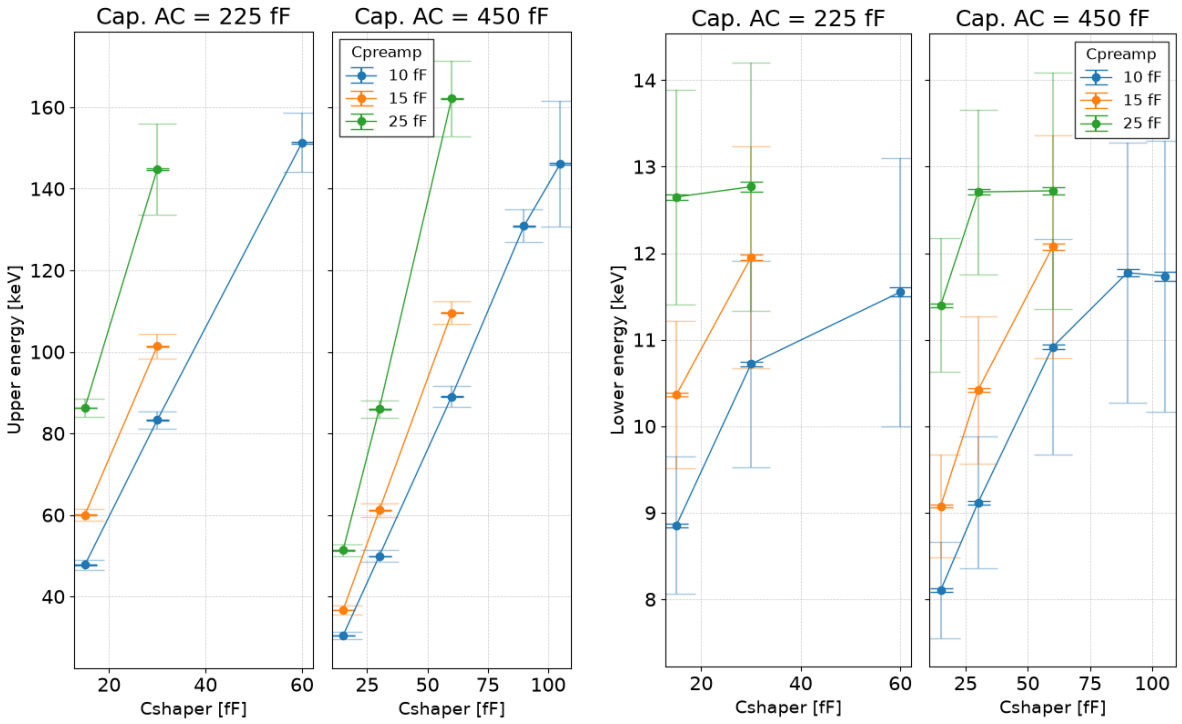
$$E_{max} = \frac{O}{g} - 5\ ENC^2[e^-]\ \ \varepsilon\left[\frac{eV}{e^-}\right] \tag{6.12}$$

It should also be noted that the upper energy limit depends on the baseline position, since *VDC,shaper* shifts the offset. Therefore, all the following results reported refer to the baseline working point selected in Section 6.1.2, with *VDC,shaper = 2000 DAC.*

Since this procedure involves an extrapolation beyond the measured fluorescence energies, the resulting values should be regarded as estimates of the accessible energy range rather than directly measured limits. Moreover, for the 500 μm thick GaAs:Cr sensors, the absorption efficiency decreases indicatively to 50 % at 50 keV, as illustrated in Section 3.3.5, so any upper limit superior to 60-70 keV provides little practical advantage for the intended range, despite being theoretically allowed by the electronics.

6.4.1. Dependence on capacitance settings

The inferred energy caps, trivially, reproduce the gain and ENC trends already discussed in the previous sections. Indeed, since Emin is directly proportional to the ENC, hence it follows the measured noise behaviour, while the upper energy cap is inversely proportional to the gain.

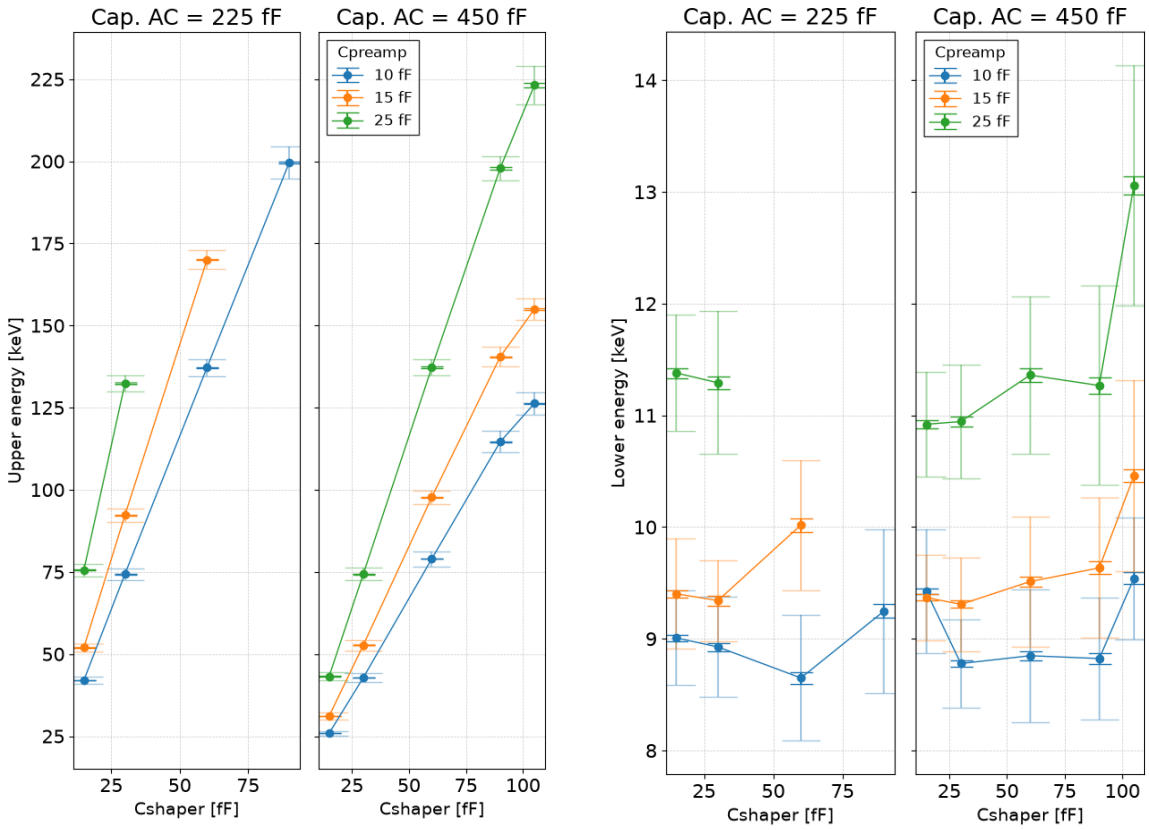


(a) Upper energy limit (b) Lower energy limit

Figure 6.12 Derived upper and lower energy limits for 50 µm sensor, according to *C,shaper*, *C,preamp* and coupling capacitance variation

Instead, the upper energy cap is governed by the gain trend: the increase of preamplifier or shaper capacitance (or the decrease of coupling one) reduce the gain allowing for the extension of the useful energy range towards higher photon energies, and vice versa.

Overall, all of figures above show how several of the investigated configurations cover the desired energy range for the intended applications. Indeed, despite the high noise pushing the minimum energy to the range 9-13 keV, these values remain below the energy region where the lower absorption efficiency of Si becomes more limited, approximately 15-20 keV (see Section 3.3.5). On the upper side, instead, several combinations drive the latter way over 60-70 keV, for which the GaAs efficiency decreases significantly. Consequently, once ensuring the upper energy coverage, configurations providing both higher gain and lower noise level are preferable.



(a) Upper energy limit

(b) Lower energy limit

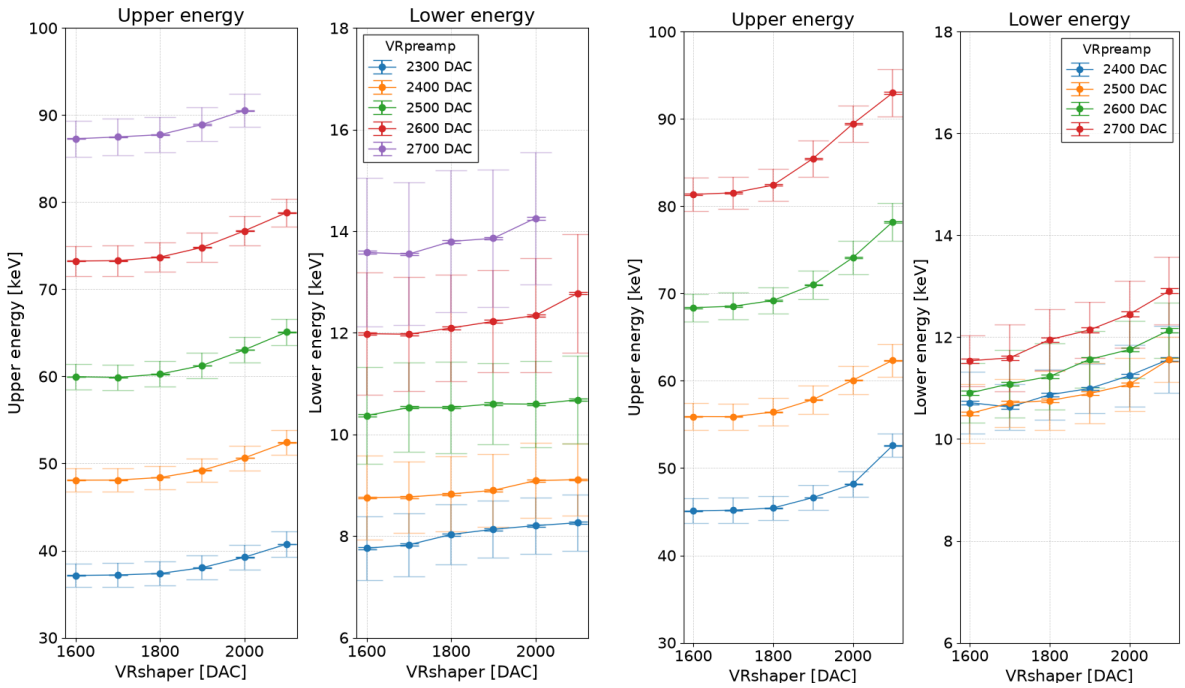
Figure 6.13 Derived upper and lower energy limits for 100 µm sensor, according to *C,shaper*, *C,preamp* and coupling capacitance variation

Eventually, what should be stressed is that, for the 100 µm sensor, the values obtained for the minimum energy should be regarded as indicative due to the imprecision already discussed before.

6.4.2. Dependence on feedback resistance settings

Coherently to the previous section, the minimum and maximum energies have been computed. In particular, *VR,preamp* has the dominant effect while *VR,shaper* affects only weakly. Indeed, over the investigated range, there are only moderate increase of both lower and upper energy caps.

From the point of view of the intended GaAs application, extending the electronic range beyond 60–70 keV brings no advantageous, so lower *VR,preamp* configurations tend to be more effective (but the complete coverage must be checked out), since they also show lower ENC values and eventually lower energies accessible. In any case, all scanned setting has lower caps below 15 keV. As before, for the 100 µm GaAs a higher minimum is expected. However, as previously discussed, the lower energy limits estimated for the 100 µm sensor should be treated with caution, since the ENC extraction in standard mode brings more uncertainty.



(a) 50 µm sensor (b) 100 µm sensor

Figure 6.14 Derived upper and lower energies limits according to *VR,shaper* and *VR,preamp* variations.

6.4.3. Comparison

Overall, the previous measurements were intended to characterize the dependence of the energy range on the readout settings rather than to determine an optimal configuration. Therefore, the performed scans identify the optimal trends in gain and noise behaviour compatible with the intended GaAs operating range depending on the various possible application.

However, a direct quantitative comparison with the Si MYTHEN III must be made with caution because of the different sensor material, polarity and in particular readout configuration. Nevertheless, comparing with Andrä's work [7], the ENC values measured for the GaAs:Cr prototype are generally higher than the lowest values reported for the Si, while showing a similar qualitative dependence on the feedback settings. Eventually, both measurements indicate that the preamplifier configuration has the dominant influence on the noise, whereas the shaper produces a considerably weaker variation.

6.5. Temperature dependence

After investigating the dependence of the response on the readout settings, an additional measurement survey has been carried out to evaluate the detector stability against temperature variations: this aspect is particularly relevant since the latter can potentially affect both GaAs:Cr sensor and the analogue front-end. For the sensor, temperature affects both leakage current and trapping/detrapping, while for the readout electronic it modifies the effective resistance of the feedback transistors and eventually the operating point of the chain [7].

The energy calibration has been performed varying the cooling system temperature, while keeping the default setting for the readout as well as sensor bias. As done in other acquisitions, those were performed using the central counter in interpolation mode (see Section 5.2.3) to estimate quantities such as gain and ENC as did in the previous sections. Unfortunately, the labelled temperature values refer to the one coming from the FPGA[16] since it was the only temperature sensor present in the detector. Therefore, these values should be considered indicative of the detector temperature rather than direct measurements of the sensor temperature.

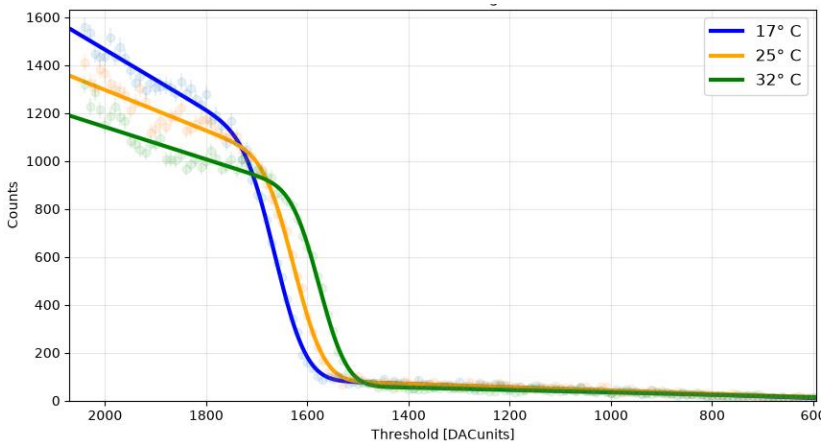


Figure 6.15 Example of fitted threshold scans, acquired at 17, 25 and 32 °C using the Mo target. The measurements were performed with the central counter in interpolation mode.

Furthermore, analysis has not been performed for FPGA temperature over 32 °C since the detector stopped working: this may be attributed to the excessive leakage current which pushes the electronics out of its optimal working point, in particular the voltage drop over the preamplifier feedback resistance might become excessive [7].

Figure 6.15 shows a visible shift of the *S curve* towards lower threshold values, as well as a decrease in amplitude, with the temperature increase. The effect has been observed for all the investigate fluorescence energies, thus indicating a variation of the detector behaviour.

[16] *Field Programmable Gate Array* (FPGA), it refers to the Linux embedded module

The resulting calibration parameters are shown in Figure 6.16: the gain increases monotonically over the temperature range, while the offset decreases, by about 35-40 *DAC*, in accordance with the shift of the *S curve* seen in Figure 6.15. The ENC behaviour, instead, stands out due to its opposite dependence, decreasing from $310.6 \pm 1.9\, e^-$ at 17 °C $241.0 \pm 2.1\, e^-$at 32 °C: this trend differs from what was expected and from the behaviour measured for MYTHEN III equipped with Si sensors [7].

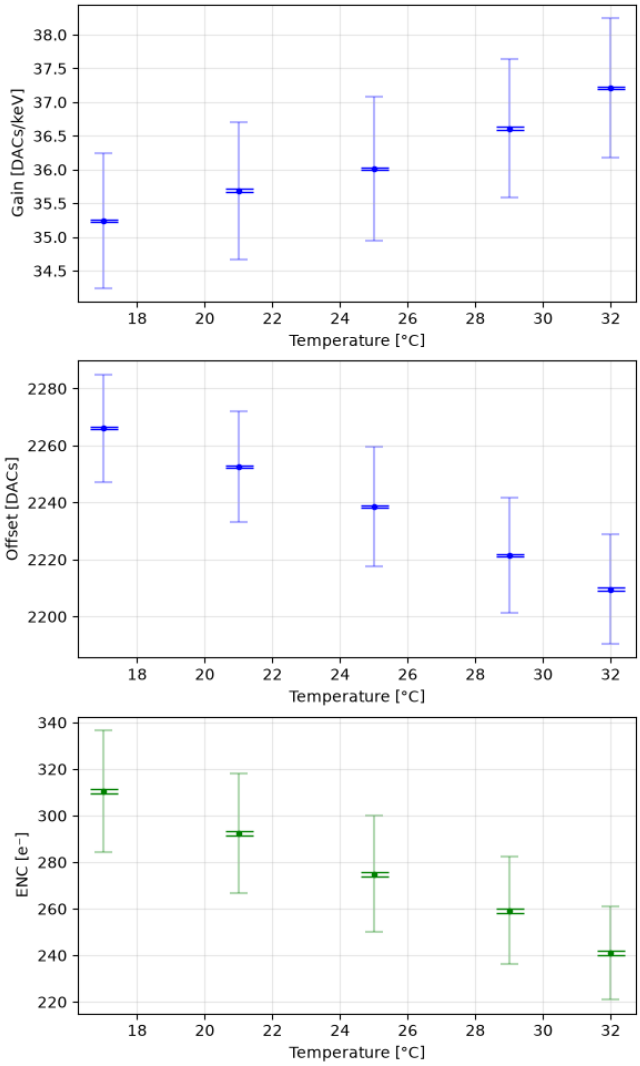


Figure 6.16 Temperature dependence of the gain, offset and ENC of the GaAs:Cr detector. The values represent the average over the selected detector channels at each FPGA temperature

As accurately described in Andrä's work [7], in fact, the MYTHEN III configuration with Si sensors shows a decreasing gain with increasing temperature, whereas the ENC increases. However, the raw noise expressed in *DAC* units remains almost constant, thus indicating that the increase of the input referred noise (see section 6.3) is caused by the reduction of gain, which in turn is mainly caused by temperature effects on the readout electronics rather than due to the silicon sensors itself.

However, a direct comparison with the present measurements must be made only with caution, since the GaAs:Cr detector is operated in negative polarity and therefore uses a different analogue configuration (where the normal feedback shaper transistor is off and substituted with the *VrfShNpol* one) [7]. Therefore, a different temperature dependence of the effective feedback resistance and shaping may contribute to the observed increase of gain. Other factors that might have some effects are the following:

- Improve of *detrapping* phenomena that might lead to a more stable response at higher temperatures, as observed by Greiffenberg et al. [1]
- Temperature dependence of the electron-hole pair generation energy in GaAs, that can be described using the empirical relationship[17] found by Bertuccio and Maiocchi [39]:

$$\varepsilon_{eh} = 4.55 - 0.00122\,T[K] \tag{6.13}$$

  Over the present range from 17 to 32 °C, assuming that the sensor temperature is approximately the measured FPGA, this corresponds to only about a 0.5%, way lower than the about 5.6 % gain increase.

Instead, the decrease of the ENC requires a separate interpretation. Recalling Eq. (6.9), the ENC is worked out dividing the fitted noise width by the gain. In the present measurements, the ENC decrease is caused also by the decrease of the former, as observed below in Figure 6.17. The opposite tendency is expected when considering the sensor leakage current, already covered in Section 3.3.6, which would increase the parallel noise contribution rather than reduce it.

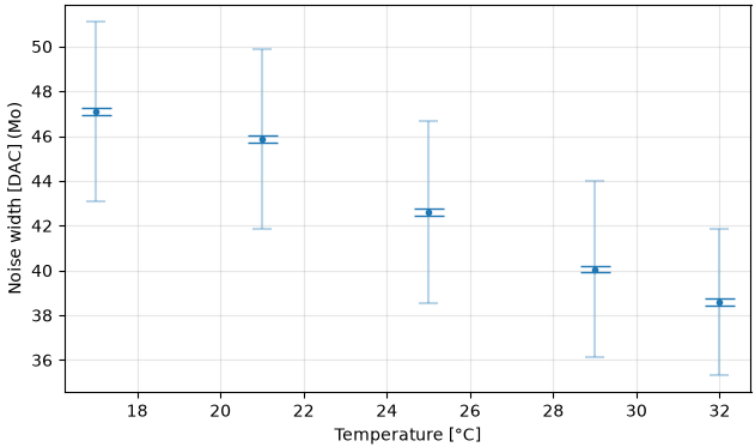


Figure 6.17 Noise parameter σ with temperature

Indeed, considering the Eq. (6.10), the ENC depends on the quantities:

$$ENC^2 \sim I_0\tau + \frac{V_0 C_{in}^2}{\tau} + {}^1\!/_f \tag{6.14}$$

[17] valid for the temperature range 230–320 K

the temperature affects the feedback circuitry, hence varying the overall shaping time constant and the balance among the contributes.

Therefore, the observed reduction of the fitted noise width indicates that the leakage current increase is not the dominant factor, since changes in the analogue gain due to the possible factor listed before, as well as the shaping in negative polarity mode, might dominate the *S curve* noise width over the temperature range.

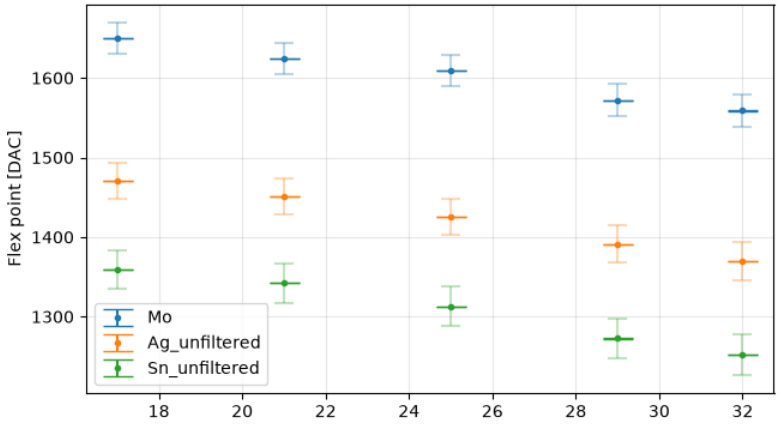


Figure 6.18 Inflection point μ for different targets and temperatures

For the sake of completeness, the inflection point trend is also depicted in the figure above. The observed behaviour agrees with what found and described in Andrä's [7], for which the inflection point tends to decrease in terms of *DAC* despite the opposite polarity.

6.6. Threshold trimming

Due to variations in the electronics, the MYTHEN III comparators do not have the same effective threshold even when the same global threshold voltage is applied. This therefore leads to *channel to channel* spread of the *S curve*. As already mentioned in Section 2.2.1, to make up for these variations, every comparator is provided with six *trimbits*, able to apply $2^6 = 64$ possible correction values: the trimbits are capable of introducing a small local shift of the comparator threshold, eventually equalizing the response among the channel if properly adjusted [3].

The correction magnitude depends on the *Vtrim* global reference voltage, determining the strength of the correction common to all channels. Therefore, while the *trimbit* introduces an increasing shift of the threshold, its size is determined by *Vtrim*. As described in Andrä's [7], the *trimbit* width decreases with increases *Vtrim* value: indeed, if the *trimbit* is too large, only a fraction of the 64 values can be exploited before saturating, while on the opposite case the 0-63 range might not be enough to compensate for the threshold dispersion.

6.6.1. Trimming procedure

For the GaAs:Cr sensors, the trimming procedure differs from the one described in Andrä's work [7], where the target is reached by comparing instead the number of counts at the inflection point while scanning both *Vtrim* and *trimbits*. Firstly, a proper value of *Vtrim* has been looked for by performing several scans (for each of 64 *trimbits*) where each channel was fitted and the inflection point extracted. This has been repeated for several candidate values of *Vtrim*, until the best one has been chosen, corresponding to $V_{trim} = 1800\ DAC$. This last one was found to provide an almost linear dependence between inflection point shift and *trimbit* value increase (the other ones showed either saturation or no reasonable increase). However, this value is higher than the range of *Vtrim* used for Silicon that are in the order of 800-1000 *DAC*: this might be explained by the different polarity in which the MYTHEN works for GaAs:Cr. Results are plotted in Figure 6.19.

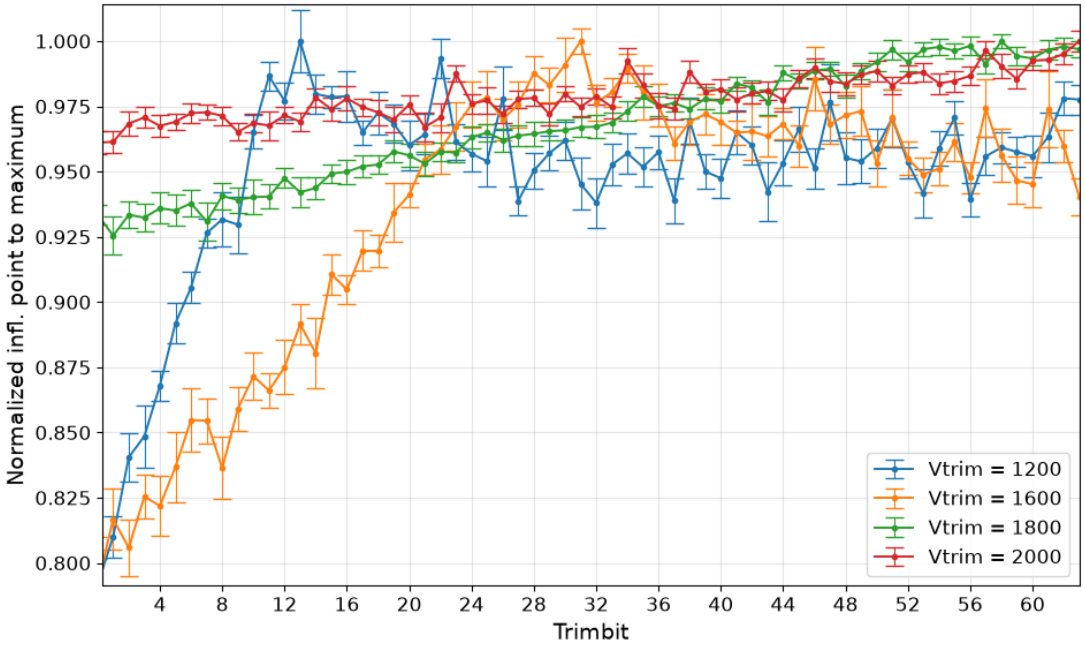


Figure 6.19 Normalized inflection points for representative channel 488 (50 µm sensor) according to different *Vtrim* values.

Once the *Vtrim* has been selected, threshold scans have been acquired for each *trimbit* value from 0 to 63, with a common reference set to *trimbit* 32, to have also a inflection point reference. Therefore, for each channel i, the trimbit whose inflection point was closer to the reference one was chosen:

$$TB_i^* = \underset{TB\in\{0,\dots,63\}}{\arg min}\ |\mu_i(TB) - \mu_{ref}| \tag{6.15}$$

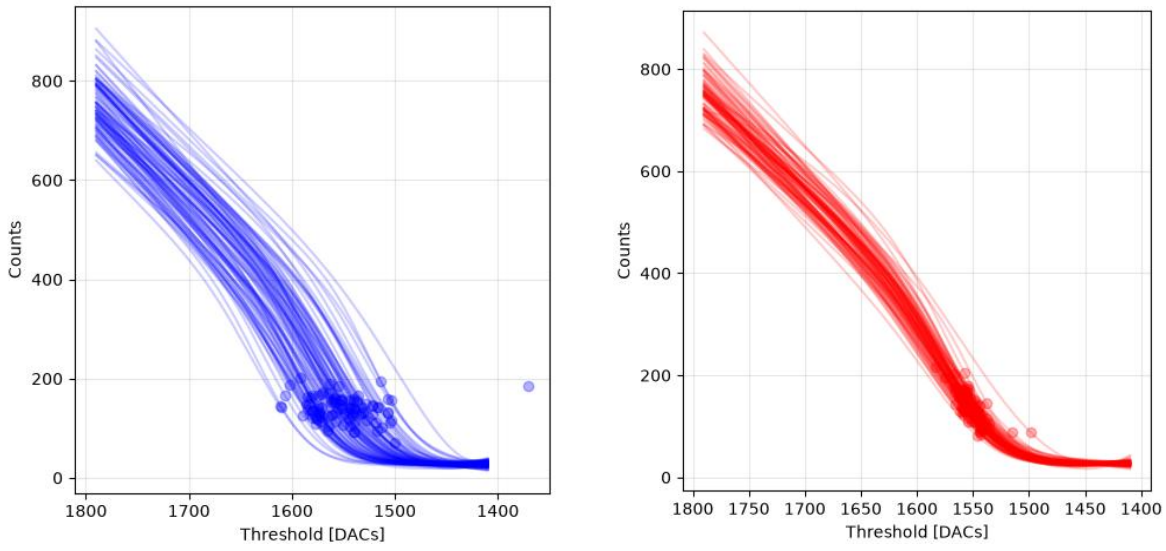
where $\mu_i(TB)$ is the inflection point for channel i at the *trimbit* TB, and μ_{ref} the inflection point of *S curves* obtained while scanning with *trimbit* set to 32.

Eventually, the best *trimbit* for each channel has been selected and then another threshold scan was performed to confirm the choice. The entire procedure has been repeated for the three counters separately, to leave out any potential inter comparator crosstalk [7]. All threshold scans have been performed by using only the Molybdenum filtered target ($E_{K\alpha} \cong 17.5\ keV$).

6.6.2. Trimming performance

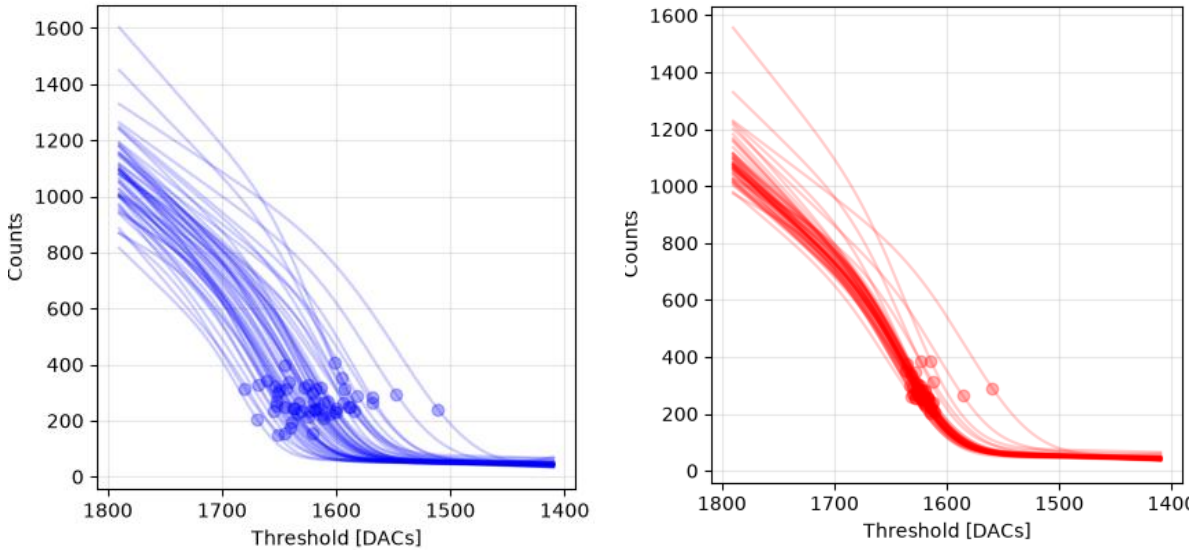
Before applying the trimming correction, a considerable spread of the *S curves*' inflection points is observed. After loading the optimum *trimbit* for each channel, the *S curves* become more aligned around the reference, as shown in the Figure 6.20 and Figure 6.21: some channels result still misaligned though, and the explanation might rely on some possible excessive deviation of the electronics behaviour for some channels and therefore a not sufficient range of *trimbits*.

For the data depicted in Figure 6.20 and Figure 6.21, the Gaussian distribution mean and standard deviation (between parenthesis) are summarized in Table 6.5, where it is clear that the trimming keeps the average threshold at the reference position (the mean inflection point looks like unchanged) while decreasing the channel to channel spread out.



(a) Before trimming (b) After trimming

Figure 6.20 *S curves* fitted before (a) and after (b) trimming procedure for sensor 50 µm, where the dots indicate the inflection point position



(a) Before trimming (b) After trimming

Figure 6.21 *S curves* fitted before (a) and after (b) trimming procedure for 100 μm sensor, where the dots indicate the inflection point position.

Table 6.5 Summary of trimming procedure for both sensor, counter 0[18]. Values represent the mean inflection point in DAC units, while the standard deviation of the channel to channel distribution is reported in parentheses.

	50 μm sensor [DAC]	**100 μm sensor [DAC]**
Before trimming	1546.48 (27.02)	1428.56 (31.09)
After trimming	1546.61 (0.95)	1428.73 (0.87)

Although the flex point are measured in *DAC*, the threshold dispersion is expressed in equivalent electrons, then allowing for relating the spread to the input charge (which would produce an equivalent threshold shift) of the detector rather than to an arbitrary digital unit, making also them independent of the gain[19] setting used during the measurement, making easier a comparison with other results [7]. The conversion is simply done by dividing the *DAC* value by the gain and by the energy required to generate an electron-hole pair. The Table 6.6 summarizes the converted results, which are in line with the standard MYTHEN III Si detector studied in Andrä's work [7], for

[18] For the sake of brevity, only the counter 0 results have been shown

[19] For these surveys, the following "standard" GaAs:Cr feedback setting has been used: *vrpreamp = 2400, vrshaper = 2000, vdcsh = 2000 [DAC], cpreamp = 10, shaper = 30, c_couplng = 450 [fF]*

which the threshold dispersion had been reduced from 122.5 ± 2.4 to 5.6 ± 0.1 electrons (silicon 50 μm width strips).

Table 6.6 Summary of threshold dispersions found, expressed in equivalent electrons

Counter	Sensor pitch [um]	Before trimming [e^-]	After trimming [e^-]
0	**50**	170.3 ± 12.4	5.9 ± 0.4
0	**100**	195.4 ± 18.8	5.5 ± 0.5
1	**50**	132.5 ± 9.7	5.8 ± 0.4
1	**100**	196.2 ± 18.9	4.2 ± 0.4
2	**50**	162.2 ± 11.8	5.7 ± 0.4
2	**100**	200.2 ± 19.3	5.2 ± 0.5

6.7. Characterization summary

Overall, the measurements presented in this chapter allowed to characterize the main response of the GaAs:Cr detector under different readout and operating conditions. The performed scans showed how gain, noise and then the accessible energy range depend mainly on the preamplifier configuration, while a weaker dependence is generally observed for the shaper settings. Moreover, the temperature measurements showed an additional dependence of the detector response, in an opposite direction with respect to what was initially expected. Finally, the dedicated trimming procedure reduced the *channel to channel* threshold dispersion to only a few equivalent electrons for both sensors.

7 High rate behaviour

The high rate behaviour of the MYTHEN III GaAS:Cr prototype has been studied in order to determine how the readout settings affect the capability of the detector to deal with higher flux of photons. For MYTHEN III the behaviour can be described by a paralyzable counter model, since the preamplifier and shaper remain continuously sensitive to the incoming charge even when overwhelmed [3].

Differently from monochromatic beam measurements previously performed for the characterization of MYTHEN III, the present has been carried out using a direct polychromatic emission of X-ray tube. Consequently, the main goal is to investigate the dependence of the high rate behaviour on the analogue readout settings.

In this chapter, after the theoretical introduction, particular attention is given to the loss of counting efficiency, to the deformation of the threshold scans due to pile up and to the effects of the feedback parameters. At the end, the high rate behaviour will be considered together with the gain and ENC characterisation presented in the previous chapter, to provide an overall summary of the characterization process.

7.1. Pile up description

The count rate capability is the maximum photon rate that can be handled by the detector can manage while maintaining a linear counting behaviour without underestimations or losses of efficiency: in the MYTHEN III characterization, the incident photon rate at which the count rate efficiency decreases to 90% is used as a practical reference value [3] [15].

From the point of view of the analogue chain, the main limitation at high photon flux is signal pile up: if two photons arrive within a time interval comparable with the duration of the single pulse, their signals overlap each other and can no longer be identified as two independent events [15]. As key determinant causes, can be discerned the followings:

- Charge collection time in the sensors, estimated in ns for semiconductors such as Si and GaAs with the thickness in the order of hundreds of micrometres, which set how quickly the charge is induced to the collecting electrodes (see Section 3.1.5). However, under normal operating conditions it is noticeably

shorter than the shaping time and therefore does not represent the dominant limitation [6].

- ASIC Shaping time: deemed as the most predominant factor, determines the rise and fall time of the analogic pulse and thus the time windows where an overlay with a second photon with the first is likely, this is called "piling up" and leads to photon counts loss since the second photon pulse cannot over cross the comparator threshold [3].
- Comparator response and recovery, which determines when threshold can fire again and thus record a new pulse. As the charge collection time, it is negligible when compared to the ASIC shaping time.

Two ideal models are commonly used to describe the detector response during dead time, namely the non-paralyzable and paralyzable models [15]: the MYTHEN III behaviour under higher flux can be described using the latter [3], because of the preamplifier and shaper remain continuously sensitive to incoming charge. Therefore, the comparators will not be triggered due to the signal enduring over the threshold, eventually recording no further counts throughout the dead time duration despite the impinging of new photons which prolong the blindness period.

For a paralyzable SPC detector, the model can be described as follows [15]:

$$\phi_{meas} = \phi_{real} e^{-\phi_{real}\tau_d} \tag{7.1}$$

where ϕ_{real} is the incident photon rate, ϕ_{meas} the observed count rate and τ_d the effective dead time. The corresponding count rate efficiency is derived simply dividing (7.1) by ϕ_{real}.

7.2. Experimental conditions and readout settings

The measurements have been carried out using the direct beam produced by the X-ray tube, described in Section 4.1, operated at 40 kVp. The Zr hard filter has been selected in order to strongly reduce the lower energies photon components of the emitted spectrum. Contrary to the fluorescence target measurements, no secondary target has been employed otherwise the photon rate would have been too low to cause noteworthy piling up effects. On the other hand, the polychromatic spectrum was the only available alternative to characterize the detector's behaviour.

The photon flux was progressively increased by varying the X-ray tube current while maintaining the same voltage and irradiation geometry: for constant voltage, the X-ray tube photon flux has been assumed as scaling proportionally with the tube current.

The following different mix of combinations of readout parameters have been tried out, as shown in the following Table 7.1 while the *VDC,shaper* and coupling capacitance have been kept fixed, respectively, to 2000 *DAC* and 450 fF.

Table 7.1 Overview of the varying feedback combinations used to investigate the high rate behaviour

Feedback parameters	Preamplifier	Shaper
Capacitance [fF]	10, 25	30, 60
Voltage (res.) [DAC]	2400, 2500	1800, 2000

Since the direct beam did not provide a sufficiently uniform irradiation over the complete detector area, the best option was to restrict to one representative channel for each sensor, which is enough for investigating the relative dependence on the readout settings.

7.2.1. X-ray spectrum simulation and effective reference energy

Since a unique photon energy cannot be assigned to the direct X-ray tube beam, the expected spectrum after filtration was first simulated using the Python Library *SpekPy* [40] [41], from which the mean photon energy was estimated and thus adopted as effective reference energy for the analysis. The use of the simulated spectrum is particularly important in this case because the detector threshold cannot be directly associated with a single incident photon energy, as would have been instead possible with a monochromatic beam.

Moreover, the simulation has been used to compute the best measurements framework in terms of kVp[20] and available hard filters[21] to keep the effective mean energy reasonably close to the Mo fluorescence energy, broadly used in the previous measurement as reference for the detector characterization.

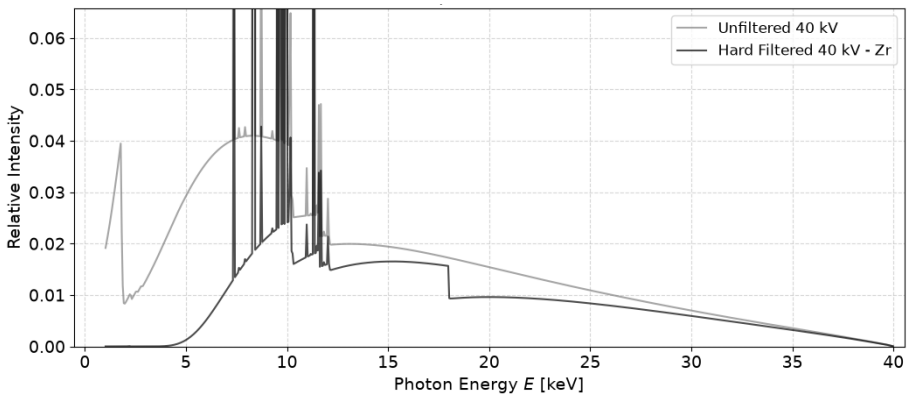


Figure 7.1 *Spekpy* simulation of W anode, Zr hard filter X-ray spectrum with kVp set at 40 kV. It is visible the Zr K edge at about 18 keV as well as the W L-photons lines.

[20] kVp (kilovolt peak) is the maximum voltage applied to the X-ray tube and therefore determines the maximum photon energy of the *bremsstrahlung* spectrum

[21] Hard filters are additional absorbing layers placed in the beam to remove lower energies photons, hence hardening the spectrum

As shown in Figure 7.1, the Zr hard filter shapes the low energy range of the spectrum, while the remaining distribution still contains both the continuous bremsstrahlung contribution and the tungsten L shell characteristic emission lines (associated to the L edges) [40].

For each tube current and readout configuration, a complete threshold scan was acquired. Since the incident spectrum was polychromatic, the count rate was evaluated at a common threshold corresponding to half of the effective mean photon energy determined from the spectral simulation:

$$E_{threshold} = \frac{E_{mean}}{2} \tag{7.2}$$

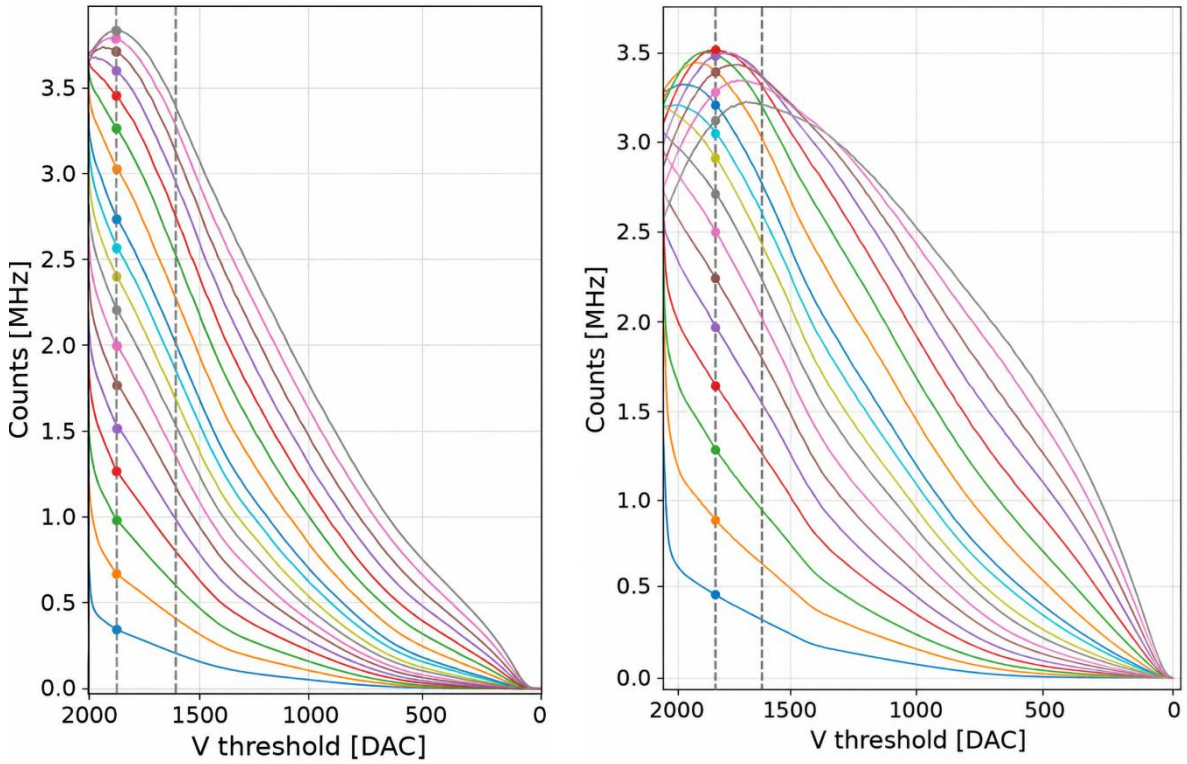
This choice follows the SPC criterion used for monochromatic MYTHEN III rate measurements (see Section 5.3), where the lowest comparator threshold is set at approximately half of the photon energy [3]: in this case, $E_{\mathrm{mean}}/2$ does not represent an exact threshold in the same way as $E_\gamma/2$ for monoenergetic photons.

Another consequence of the polychromatic irradiation is that the possible employment of the three comparators to track the piling up cannot be applied. Indeed, for the monochromatic beam, the first threshold is positioned at approximately $E_\gamma/2$, while the second and third thresholds are placed between the amplitudes corresponding to one and two photons and to two and three photons, respectively, hence allowing for identify up to three photons [3]. Instead, with a continuous distribution of energy, the amplitude produced by a single higher energy photon might overlap with the ones produced by two or more lower energy photons. Therefore, the photon multiplicity cannot longer be uniquely determined from the signal amplitude. Eventually, for this reason, the present characterization was performed using only one single comparator response.

7.3. Pile up effects on the threshold scans

A first indication of pile up, despite being qualitative, can be observed from the evolution of the threshold scans when the current increases. The plots below depict a series of threshold scans for the representative channels chosen.

As shown in Figure 7.2, for low flux the curves resemble the expected *S curve* shape, and the number of counts increases proportionally with the current. However, at higher fluxes the shapes get deformed increasingly, becoming increasingly deformed and broadened even at lower threshold values. Moreover, the recorded rate progressively saturates and then decreases.



(a) Channel 488 (50 µm sensor) (b) Channel 795 (100 µm sensor)

Figure 7.2 Threshold scans for different currents, standard feedback settings[22]

The progressive deformation can be understood from the temporal overlaying of pulses: as the rate ramps up, pulses increasingly overlap themselves before the electronic chains has recovered to its baseline [3] [15]. Hence, the resulting stacked up pulses reach amplitudes which are larger than ones produced by single photons. Therefore, for negative polarity mode, they cross threshold values progressively farther below the baseline. Consequently, the former *S curves* broaden and extend towards lower *DAC* values as the flux increases, and at the same time the noise peak progressively vanishes, since the analogue signal increasingly remains away from the baseline because of pile-up, thus masking the threshold crossings caused by baseline noise.

7.3.1. Pile up simulation

To verify whether the observed *S curve* deformation was compatible with the pile up model, a dedicated simulation was developed. For each photon event, the incident energy was randomly sampled from the previously simulated spectrum of the filtered X-ray tube in Figure 7.1.

[22] The standard settings refer to the following feedback configuration: *cpreamp 10 fF, C,shaper 30 fF, C,AC 450 fF, vrpreamp 2400, vrshape_npol 2000, vdcsh 2000*

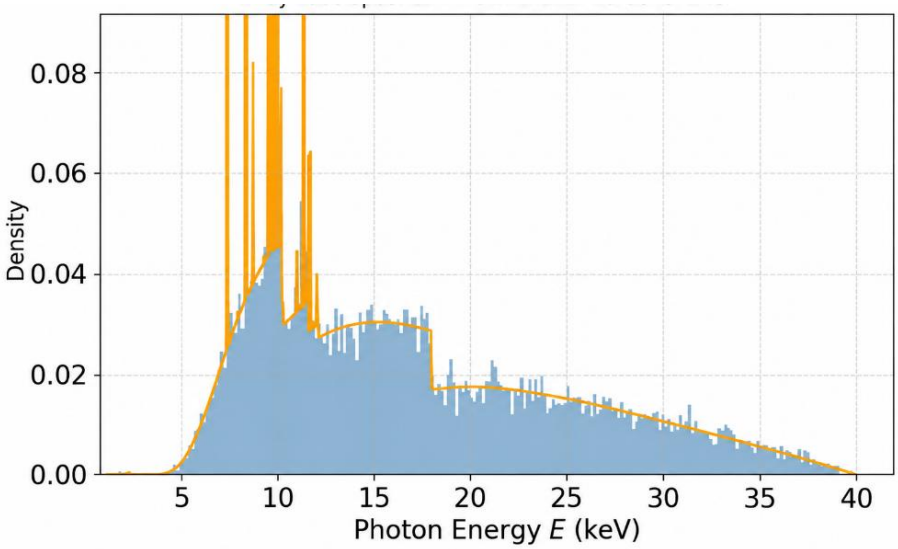
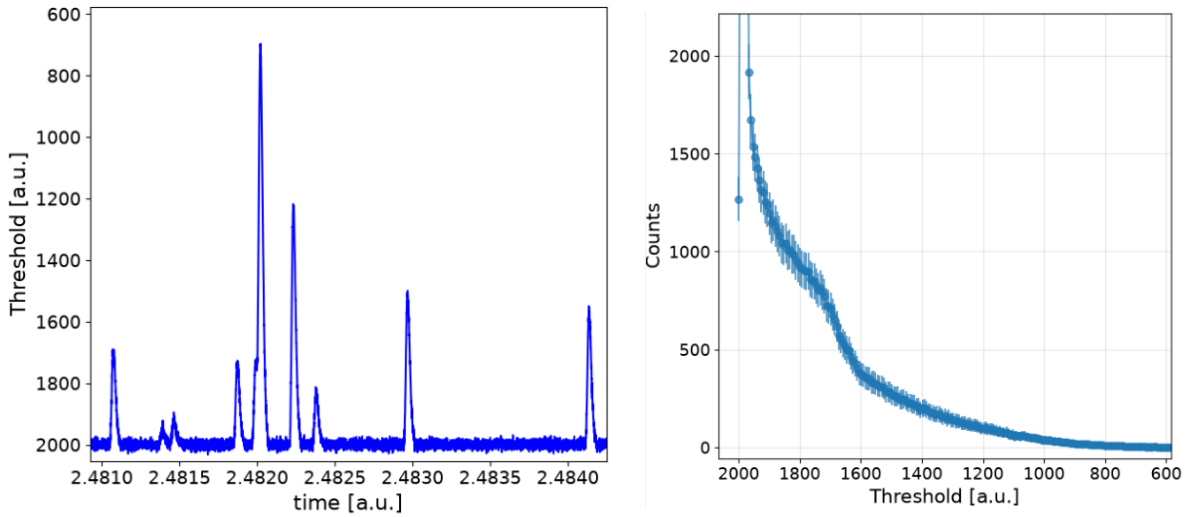


Figure 7.3 Sampled version of Figure 7.1 by using a Montecarlo algorithm

The interaction position with respect to the strips was also sampled assuming a uniform spatial distribution, and a simplified geometrical model involving the irradiated strip and its two nearest neighbours was used to account for charge sharing. Consequently, the signal amplitude recorded by a single strip is dependent on both the sampled photon energy and the fraction of charge assigned. Moreover, gaussian distributed electronic noise was added to the simulated temporal signal. Moreover, for increasing photon rates, the generated pulses start superimposing in time, so that photons arriving before the previous signal had recovered to the baseline end up producing piled up signals.



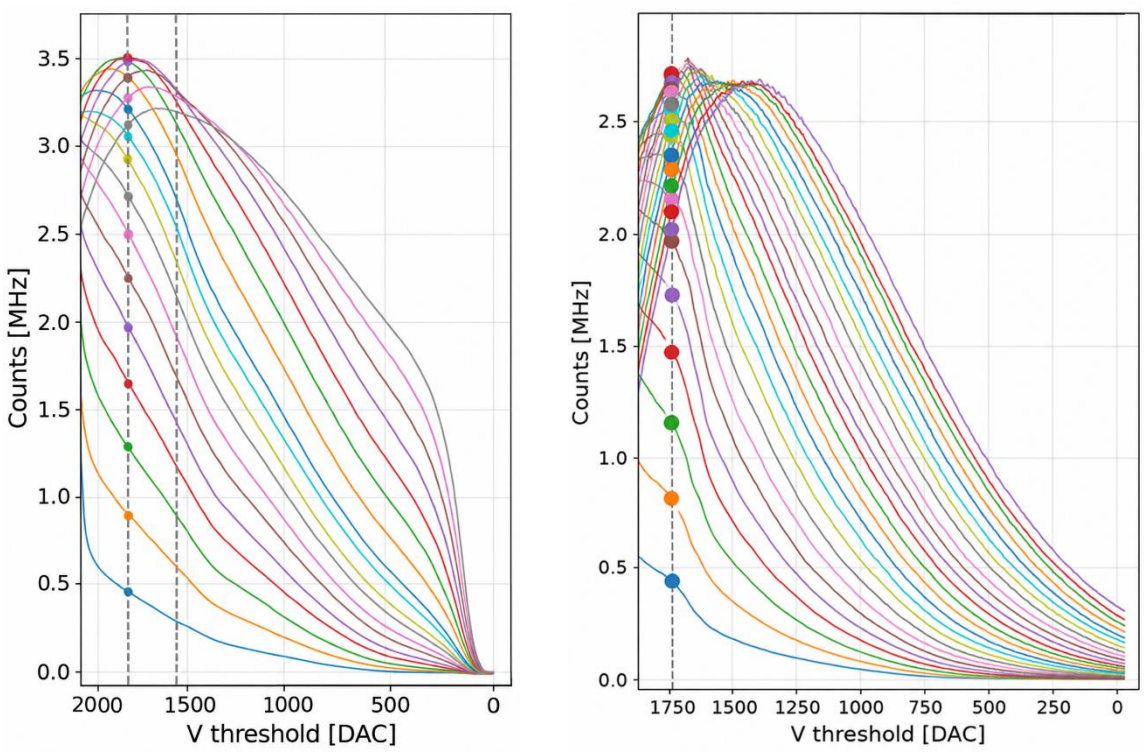
(a) Simulated pulses in time

(b) Simulated threshold scan

Figure 7.4 Simulation of pulses (a) of a polychromatic source and derived threshold scan (b)

The resulting distribution of pulse amplitudes was then used to reconstruct the corresponding threshold scan. The goal of the simulation was to reproduce the main features affecting the measured *S curve* with sufficient accuracy for the qualitative pile-up study. In the figures below it is possible to see an example of pulse shaping in time, as well as the *S curve* associated with a polychromatic source threshold scan. Despite the simplified treatment of detector response, the simulated curve reproduces the overall shape of the real measured threshold scans.

Figure 7.5 compare the experimental data with the simulated one at different virtual photon rates. The overall shapes are similar: when the simulated photon rate is progressively increased, the model reproduces the same enlargement and deformation of the *S curve* towards lower threshold values like in the experimental measurements (Figure 7.5a). The only difference, noticeable on the real measurements, consists in an abrupt decrease for threshold close to zero: this is explained by the fact that comparators does not work properly at too low threshold values. Therefore, the agreement supports the interpretation of the seen deformation as being mainly associated with pile up phenomena.



(a) Real threshold scans.

(b) Simulated threshold scans

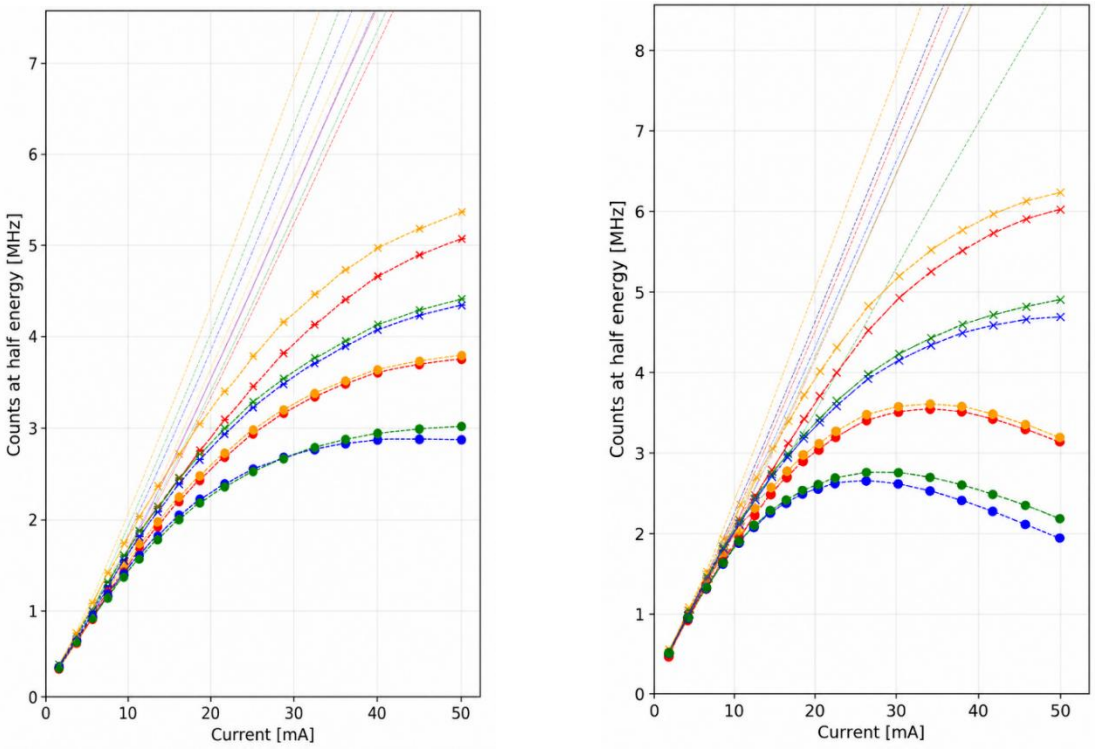
Figure 7.5 Comparison between real data (a) and simulated threshold scans (b). The dots represent the threshold at which the counts are extracted.

7.4. Count rate performance and dependence on readout settings

For MYTHEN III, it is used to consider, as count rate figure of merit, the flux at 90% of the efficiency [3]. In the present measurements, however, an effective relative efficiency has been employed, due to lack of information about the real flux and due to its polychromatic spectrum. Therefore, the low current responses (assuming negligible piling up for them) has been linearly fitted in order to extrapolate the flux for higher currents. Eventually the ratio between the measured and the extrapolated rates has been utilized to quantify the relative deviation from the linearity:

$$\eta_{rel}(I) = \frac{R_{meas}(I)}{R_{linear}(I)} \tag{7.3}$$

The following plots depict, for both representative channels, the measured count rate variation with X-ray tube current, with the count extracted at the threshold corresponding to 14.32/2=7.2 keV, for the feedback combinations illustrated in Table 7.1.



(a) Channel 488 (50 μm) (b) Channel 795 (100 μm)

Figure 7.6 Measured counts versus tube current for both sensors at different settings used. VR,shaper has been kept fixed at 2000 *DAC*. The straight lines represent extrapolated real flux

Table 7.2 Legend for Figure 7.6, showing different setting used

	C,preamp	C,shaper	VR,preamp
--●--	10	30	2400
--×--	10	30	2500
--●--	10	60	2400
--×--	10	60	2500
--●--	25	30	2400
--×--	25	30	2500
--●--	25	60	2400
--×--	25	60	2500

Firstly, the clearest dependence is observed for the preamplifier feedback resistance. Indeed, for both representative channels, the increase of *VR,preamp* (corresponding to decreasing the resistance value and the gain) basically reduces the counting losses and thus the piling up. At the same time, the increase of *C,preamp* leads to a major effect of piling up.

The observed behaviour can be understood from the simple definition of shaping time constant. Considering the preamplifier constant, $\tau_{pre} \sim R_{pre} C_{pre}$ (see Section 2.2.2.1), with the decrease of the tow quantities the pulse recovers faster and then reducing the probability of overlap with the subsequent ones.

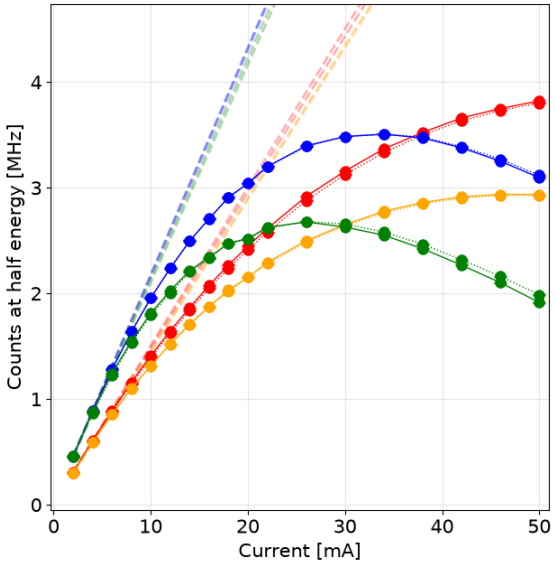


Figure 7.7 Measured counts versus for both channel by varying *VR,shaper*

	Channel	C,preamp	VR,shaper
—●—	488	10	1800
···●···		10	2000
—●—		25	1800
···●···		25	2000
—●—	795	10	1800
···●···		10	2000
—●—		25	1800
···●···		25	2000

Table 7.3 Legend for Figure 7.7. *C,shaper* and *VR,preamp* have been kept fixed at 30 fF and 2400 *DAC*, respectively

On the other hand, as shown in Figure 7.7, clearly weaker variations are obtained by changing the *C,shaper* and, for some settings, its effect is even negligible. Moreover, as shown in the two plots below, where the *VR,shaper* variation has been investigated for 10 fF and 25 fF *C,preamp* values, the high rate behaviour seems unaffected by the variation of the feedback shaper resistance. Therefore, at least within the investigated range, the variation of shaper parameters produces only minor, if not negligible, change, eventually indicating a low sensitivity of the shaper in influencing the behaviour at higher fluxes.

To provide a quantitative figure of merit for the observed trends, the effective rate corresponding to 90% relative efficiency has been extracted for each investigated configuration according to the procedure described at the beginning of this section. The resulting values are summarized in Table 7.4.

Table 7.4 Effective rate counts found for the tested feedback combinations[23] (higher is better)

C,preamp [fF]	C,shaper [fF]	VR,preamp [DAC]	Effective rate at 90% [MHz] – channel 488 (50 μm)	Effective rate at 90% [MHz] – channel 795 (100 μm)
10	**30**	**2400**	2.21	1.91
10	**30**	**2500**	2.88	2.60
10	**60**	**2400**	2.11	1.73
10	**60**	**2500**	2.88	2.27
25	**30**	**2400**	1.70	1.48
25	**30**	**2500**	2.36	1.90
25	**60**	**2400**	1.70	1.46
25	**60**	**2500**	2.24	1.85

The inferred values obtained in this study are in the same order of magnitude as the ones reported for MYTHEN III 50 μm pitch-Si detector in Andrä et al. [3] which varied, in terms of flux at 90% efficiency, from 0.45 ± 0.05MHz/strip for highest gains to 2.8 ± 0.6MHz/strip for fastest settings. Nevertheless, a direct quantitative comparison should be avoided, since the latter values were obtained with a monochromatic 12 keV beam and an independently monitored incident photon flux.

[23] The shaper variation has been neglected; the *vdcsh, C,AC* have been kept fixed at 2000 DAC and 450 fF, respectively

7.5. Overall performance and readout trade-offs

Overall, the combination of gain, noise and high rate measurements show that the preamplifier stage provides the most control over the detector response. The summary of setting influence, considering also the characterisation described in the last chapter, is reported in Table 7.5. Within the investigated ranges, decreasing *C,preamp* appears as an almost favourable condition, as it increases the gain, reduces the ENC and improves the high rate behaviour.

Instead, a different performance is observed for *VR,preamp* , which shows a clear trade-off between signal formation and rate capability. Indeed, lower values lead to larger gain and lower ENC, whereas higher ones improve the high rate response. This behaviour is, moreover, coherent with the trade-off previously observed for MYTHEN III, where faster settings have been found to reduce the dead time at the expense of larger noise [3].

Table 7.5 Overview of readout parameter variation for the explored range variation

	Gain response	**ENC response**	**High-rate response**
Decrease of *C,preamp*	Increases	Decreases	Improved count rate capability
Decrease of *VR,preamp*	Increases	Decreases	Reduced count rate capability
Decrease of *C,shaper*	Increases	Decreases	Weak improvement / limited dependence
Variation of *VR,shaper*	No clear change	No clear change	No clear dependence observed

At the same time, the shaper parameters show only a relative slight influence on every figure of merit listed before. Although the decrease of *C,shaper* increases the gain and reduces the ENC, its effect on the pile up prevention remains small, whereas *VR,shaper* seems to have only negligible effects.

8 Charge transport and signal induction effects

The transport of charge carries inside a strip semiconductor sensor affects both spatial distribution of the latter collection and the how the signal is induced on the various electrodes. As already discussed in Section 3.2.2, the measured signal depends on the carrier motion through the weighting field, as described by the Shockley- Ramo theorem.

In the following, two phenomena are investigated, which are particularly relevant for the GaAs:Cr microstrip sensors: the charge sharing among neighbouring strips and the crater effect associated with incomplete signal induction due to the short holes' lifetime.

8.1. Charge sharing

After the photon absorption in the semiconductor sensors, the charge carries cloud starts drifting along the electric field with a certain transverse diffusion, whose gaussian spread is expressed by (3.3). When its size becomes comparable with the strip width, then some of the charge will be collected also by the surrounding channels: this phenomenon is called "*charge sharing*" [11].

As discussed in Bergamaschi et al. [11], the charge transport can be approximated using a simplified *linear charge collection model*: for a strip, the model assumes a central region where the complete charge is collected by the strip, while in the latter's boundaries the charge is shared linearly between the neighbouring channels. So, the following geometrical charge sharing parameter can be defined, where d is the region width where the sharing happens and p the strip's pitch:

$$\alpha = \frac{d}{p} \tag{8.1}$$

Therefore, assuming a uniform distribution of photon absorption along the strip pitch, the above quantity also represents the fraction of events occurring in the charge sharing region, in which the signal measured by one single strip varies approximately linearly from the full charge collection to zero, while the complementary is collected

by the neighbouring strip. Indeed, the approximately linear part of the *S curves* is generated by the charge sharing contribution.

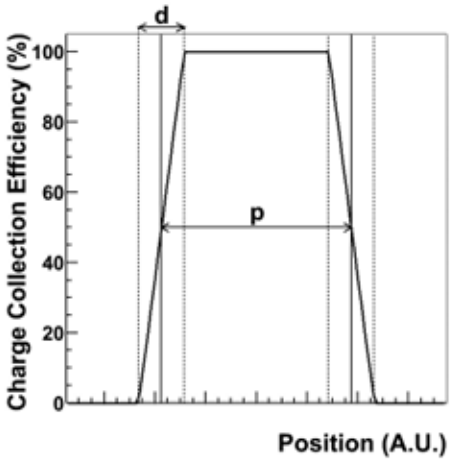


Figure 8.1 Charge collection efficiency Q/Q_0 for a strip, assuming $\alpha = 0.24$. Reproduced from Bergamaschi et al. [11]

The charge sharing parameter extracted from the *S curve* fitting (see Section 5.1) gives indication, in percentage, about the amount of charge sharing in the single strip. For the following section, the charge sharing contribution has been computed by using the below relations, which give the same results:

$$C_S[\%] = C_{s,fit}\frac{\Delta E}{\Delta E \cdot C_{s,fit} + A} \tag{8.2}$$

$$C_s[\%] = \frac{N_0 - A}{N_0} \tag{8.3}$$

where ΔE is the difference between the inflection point and its half energy in *DAC units, A and* $C_{S,fit}$ are the amplitude and charge sharing parameters extracted from the *S curve* fitting, respectively, and N_0 is the measured counts (hence at half energy, as depicted in Figure 5.1).

However, as discussed in *Bergamaschi et al. 2015* [11], the linear charge collection model represents a simplified parametrization of the actual charge distribution and then deviations from this approximation can become relevant when the charge sharing contribution is large.

8.1.1. Comparison with silicon and previous GaAs:Cr measurements

Before investigating the GaAs:Cr sensors, the charge sharing extraction procedure was first applied to the "reference" MYTHEN III Si sensors, which have a 50 μm strip and a thickness of 450 μm and have been operated at a bias voltage of +200 V. Thus, the extracted charge sharing has been estimated at 39.5 ± 0.1 % with $\sigma_{ch} = 1.5\%$[24], consistently with previous measurements reported by *Ruat et al.* [2] for a silicon sensor of the same physical size, although the bias voltage was set at +120 V. Therefore, this agreement provides a reference for the GaAs:Cr results, for which a larger charge sharing is expected due to the higher electron diffusion coefficient: although the higher electron mobility in GaAs reduces the transit time, the stronger diffusion still results in a broader charge cloud at the collecting electrodes [2].

Instead, a different behaviour was reported for the GaAs:Cr sensors. *Ruat et al.* [2], as expected, observed a larger charge sharing than the one for silicon, reaching even about 70-80% for the 50 μm strip pitch sensor. For the 100 μm, instead, the measured charge sharing was in the order of 50%, a reduction close to the factor of two expected from the simple geometrical scaling. As anticipated before, this can be mainly attributed to the different diffusion coefficient values, although the different X-ray absorption depth in the two materials could be an additional contribution to the difference.

The measurement carried out in this work for the GaAs:Cr microstrip sensors reproduce the large charge sharing already observed at least for the 50 one, with values of $81.4\% \pm 0.1$ % ($\sigma_{ch} = 2.7\%$). However, an unexpected behaviour has been found for the 100 μm one, for which a charge sharing of 74.6 ± 0.1 % ($\sigma_{ch} = 2.5\%$) has been obtained. Therefore, unlike *Ruat et al* [2]., only a slight reduction has been observed for the GaAs:Cr sensors investigated in this work when doubling the pitch strip from 50 μm to 100 μm.

Considering the simplified linear charge collection model and assuming for simplicity $C_s = 80\%$, the mentioned results would correspond to effective sharing region width of 40 μm and 80 μm for the 50 μm and 100 μm sensors, respectively.

8.1.2. Laser scan validation of the linear charge collection model

The large charge sharing, for both sensors, has been further studied by using the 660 nm laser scan described in Section 4.2. Due to the opaque Ni backside metallization, the sensor has been illuminated in an edge-on configuration with the laser position progressively displaced along the strip width direction.

For each laser position, an *S curve* has been acquired, and its inflection point consequently extracted by fitting. Since the laser intensity, as well as the readout

[24] This quantity, σ_{ch}, refers to the channel to channel spread

configuration, was fixed, the position of the inflection point is then proportional to the signal amplitude and so the amount of charge collected by the single strip[25]. Consequently, the variation of the inflection points as a function of the laser positions gives a spatial measurement of the charge sharing.

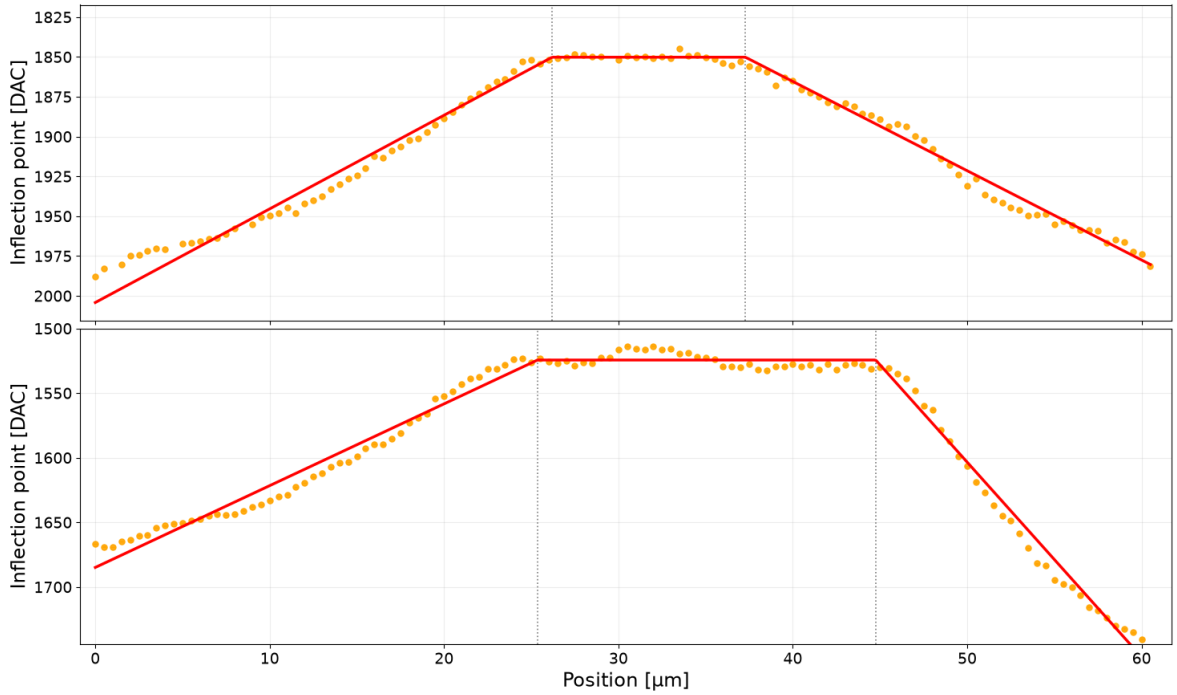


Figure 8.2 Inflection point extracted versus scan position along the strip width, as well as the trapezoidal. The asymmetrical slope may depend on laser focusing and mechanical motors uncertainty

As shown in Figure 8.2, the resulting response shows an approximately trapezoidal profile for both strip pitches. A central plateau is observed where the signal collected by the considered strip remains approximately constant, while the response decreases linearly on both sides as the laser moves to the neighbouring strips. This result is therefore coherent with the linear charge collection model introduced in the previous sections.

The fitted plateau widths are 11.08 ± 0.03 µm and 19.40 ± 0.01 µm for the 50 µm and 100 µm sensors, respectively (with the error estimated from the trapezoidal fit). Eventually these values confirm the 75-80% charge sharing found in the previous measurements. In particular, the unexpectedly large charge sharing observed for the 100 µm sensor is also reproduced by the spatial laser measurement, indicating that it is unlikely to come only from the parametrization of the *S curve*. Nevertheless, the laser

[25] For a fixed laser position, successive excitations produce a highly reproducible charge injection in the same region, differently from X-ray photons for which individual photon interactions occur at different positions and depth.

measurement should be regarded only as a complementary validation, since the scans were carried out only on the side of the sensors and laser generated carriers distribution in space may differ from the one coming from the X-ray absorption. Furthermore, the GaAs is strongly absorbing at 660 nm photons, with an absorption coefficient [42] on the order of $3.3\,10^4\ cm^{-1}$ [26], so the laser generated carriers are confined to a shallow region close to the first illuminated layers.

These numbers find also support when computing the transverse diffusion: recalling Eqs. (3.2) and (3.3), and considering for 50 µm Si d = 450 µm, T = 300 K and V_b = 200 V, the σ of transverse diffusion is about 7.4 µm, corresponding to a $FWHM = 2.355\sigma \cong 17.0\ \mu m$. This result is in the same order of magnitude with the about 40% of charge sharing (corresponding to a 20 µm region) initially found for Si in this work (and confirmed by Ruat et al. [2]). On the other hand, considering for the 50 µm GaAs:Cr $d = 500\ \mu m$ and $V_b = -300\ V$, which correspond to an electric field $E = \frac{300}{0.5} = 6\frac{kV}{cm}$, the linear drift velocity Eq. (3.2) cannot be applied anymore due to saturation [8] (as verified in Ruat et al. [2] where the linear region cap has been indicated at -200 V). Therefore, using the saturated drift velocity corresponding to $E = 4\frac{kV}{cm}$, the σ of transverse diffusion is about 14.8 μm, corresponding to a $FWHM = 2.355\sigma \cong 35.0\ \mu m$. Also, the latter is in the same order of magnitude of 80 % charge sharing (as shown in Figure 8.2(a) it corresponds to a 40 µm charge sharing region). However, this does not elucidate the strange numbers observed for the 100 µm GaAs:Cr, eventually indicating that the transverse diffusion does not explain everything.

8.1.3. Discussion of the pitch dependence

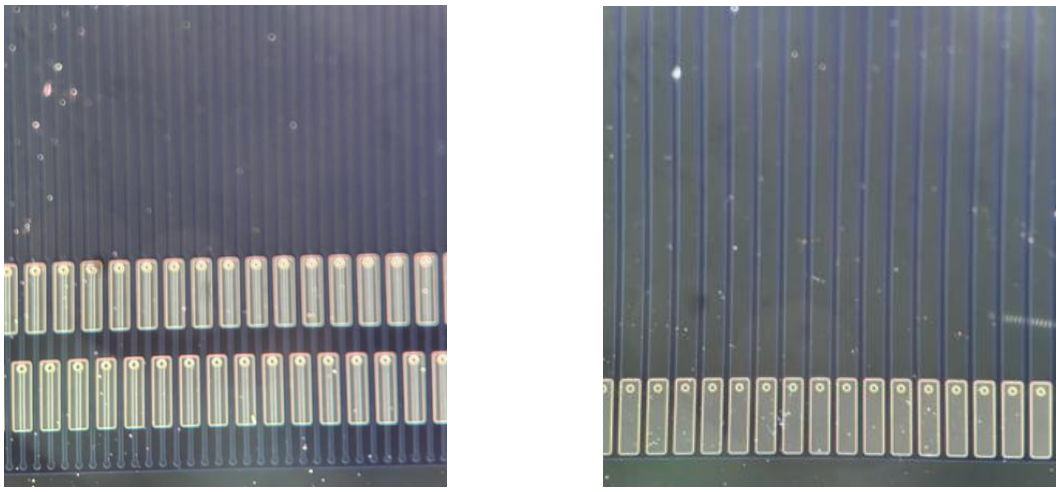
The agreement between the X-ray *S curve* analysis and the independent laser scan indicates that the unexpectedly large charge sharing fraction of the 100 µm sensor is not simply a consequence of the *S curve* fitting procedure. A possible explanation would arise from the differences in the electrode geometry, as shown in Figure 8.3. The sensors investigated in this work shows a different strip metallization different from the one of the sensors previously characterized by *Ruat et al.* [2] as observed by optical microscopy.

Since the electric field, as well as the weighting field, depends on the electrode width and the interstrip geometry, the simple geometrical scaling to anticipate and explain the amount of charge sharing would not be enough.

Indeed, a dependence of charge sharing on electrode geometry has been reported for other segmented high Z semiconductor detectors. Altıngün and Kalemci [43] investigated orthogonal-strip CdZnTe detectors with different electrode widths and showed, both experimentally and through simulations, that changes in the electrode

[26] From which it is possible to estimate the characteristic absorption depth $\delta = \frac{1}{\alpha} \cong 0.30\ \mu m$

geometry, such as a different layout or configuration, modify the charge sharing behaviour. Moreover, Kalemci and Matteson [44] showed for CdZnTe strip detectors that the charge sharing region depends also on the interaction depth and photon energy, and that part of the electron cloud can drift towards the inter-anode region, leading to incomplete charge collection. Eventually, although the mentioned studies focus on CdZnTe, they demonstrate the potential influence of the electrode layout on the charge sharing.



(a) 50 µm version (b) 100 µm version

Figure 8.3 Zoom on the GaAs:Cr sensors studied in Ruat et al. [2]. The electrode metallization appears different from the one of the sensors studied in this work, depicted in Figure 2.1

8.2. Crater effect

In addition to charge sharing, the GaAs:Cr exhibits also another position dependent response called "crater effect", which originates from the incomplete signal induction, on the electrodes, caused by the poor transport properties of holes in chromium compensated GaAs and its attitude to trap holes and therefore abruptly reducing their lifetime (see Section 3.3.4).

Indeed, Greiffenberg et al. [6] determined a hole lifetime of approximately 2.5 ns for GaAs:Cr batch produced in 2017, corresponding to only about 30 µm of drift length under a -300 V bias across 500 µm thick sensor, alongside a hole mobility of 200 $cm^2V^{-1}s^{-1}$. Another older batch had shown an even shorter lifetime of about 1.4 ns. Therefore, holes, if generated far enough from the electrode, may end up being trapped before completing their induction.

8.2.1. Physical origin

As discussed in Section 3.2.2, according to the Shockley-Ramo theorem the signal measured by the electrode is produced during the charge carrier motion though its weighting field, with both electrons and holes contributing for the collecting strip and the surrounding ones.

In particular, the mentioned short holes lifetime prevents them from completing their induction on the electrodes, thus ending up in a reduce signal and, under specific conditions, a residual signal of opposite polarity in the neighbouring electrodes [6]. Figure 8.4 shows the weighting potential purposely computed for a microstrip sensor, with more details listed in Appendix B.

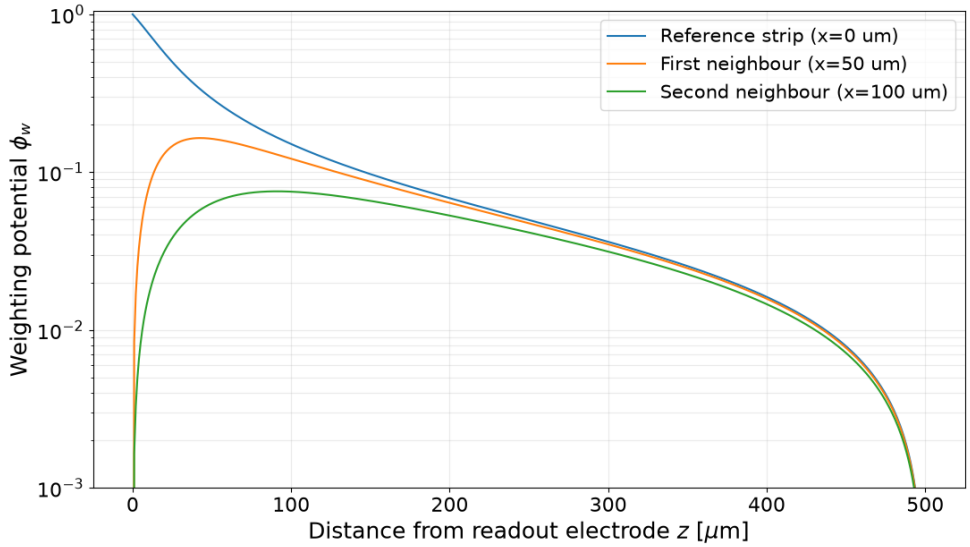


Figure 8.4 Weighting potential computed for strip geometry, based on the solution derived by Heubrandtner et al. [45]. Details of the calculation are reported in Appendix B.

For the collecting electrode, an interaction occurring close the latter, due to electrons, does not produce the complete signal, unless the remaining contribution is provided by holes drifting towards the opposite. Due to latter's' shorter lifetime, a relevant fraction of holes is trapped before completing the induction, and eventually the measured signal is reduced.

At the same time, the motion of the electrons induces also a signal of opposite polarity in the neighbouring electrodes through their weighting fields. Consequently, under complete charge transport the holes motion would compensate the electrons one, ending up with a net zero induction on the surrounding strip. In chromium compensated GaAs, however, hole trapping stops their contribution and then a residue, opposite in polarity, signal remains in the neighbouring channels. Hence, the name *"crater effect"* derives from the characteristic spatial pattern where a reduced

signal in the collecting electrode is surrounded by opposite in polarity induce signals in the surrounding ones.

Therefore, as accurately described in Greiffenberg et al. [6], a strong dependence on the interaction depth is introduced: indeed, photons absorbed close to the hole collecting backside electrode give no signs of crater effects. This behaviour has been quantitatively demonstrated by Greiffenberg et al. [6] exploiting an angle-on pencil beam measurement, eventually confirming the described behaviour.

8.2.2. MYTHEN III qualitative evaluation

Although the crater effect was originally characterized using the information provided by the charge integrating detector readout [6], it originates in the sensor and therefore it would be also expected in SPC detectors. In the latter case, as anticipated by Greiffenberg et al. [6], this effect could potentially lead to unreliable counting behaviour due to an effecting modification of the threshold response.

The measurement in this work has been carried out by focusing the 660 nm laser (see Section 4.2), on a single representative strip, in two different positions along the thickness direction. Evidence of the crater effect phenomenon is observed in the threshold scans as *S curve* like responses appear on the opposite side of the electronic baseline, as shown in Figure 8.5 where there are kind of *S curves* developed also towards higher threshold levels. Therefore, the opposite polarity signal is consistent with a residual induced signal in neighbouring strips.

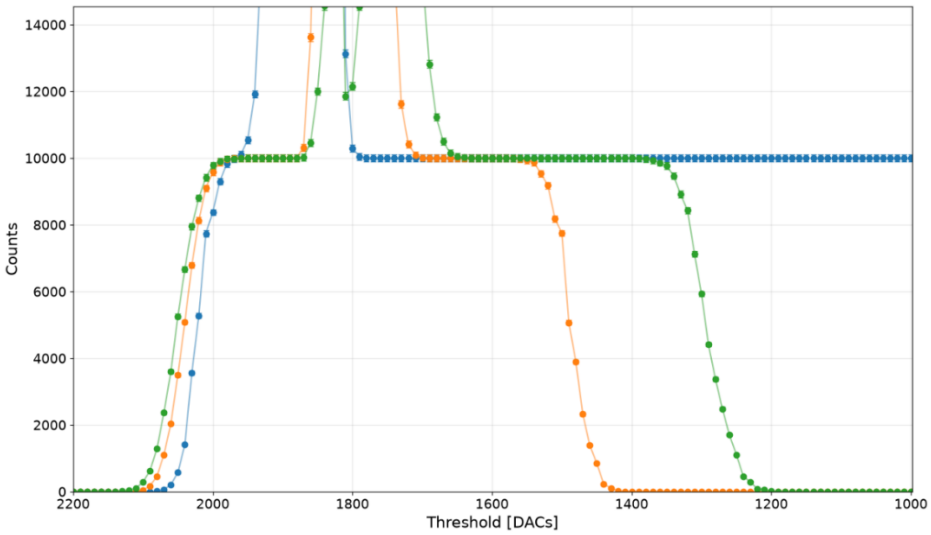


Figure 8.5 Example of "double" *S curves* (zoomed) observed in neighbour strips and linked to crater effect, scans performed with laser beam focused close to the readout electrodes, hence far enough from the backside electrodes (which collects holes)

Overall, the observed opposite polarity threshold response and its dependence on the interaction position are qualitatively consistent with the crater effect previously reported for GaAs:Cr sensors. However, further and even quantitatively measurements could be carried out to estimate the amount of deleterious effect brought about by the crater effect, and whether the results are in line with the one obtained in Greiffenberg et al. [6] by using a charge integrating pixel detector.

9 Spatial response

After the characterization of the detector response, as well as the optimization of threshold dispersion by trimming and the investigation of the charge transport properties, the spatial performance of the two 50 μm and 100 μm GaAs:Cr sensors has been investigated.

The spatial resolution describes, as already introduced in Section 3.2.3, the capability of the detector to discerning two different photon interaction occurring at different positions. For the sensors studied in this work, the physical strip pitch represents the natural spatial sampling unit: however, the effective response is affected by factors like charge sharing effects and the correct threshold choice. Moreover, MYTHEN III offers an interpolation mode that could exploit the charge sharing in order to improve the spatial information even below the electrodes pitch (see Section 2.4.2) [4].

The following sections first introduce the mathematical quantities used to quantify the spatial response, namely the *Point Spread Function* (PSF) and the *Modulation Transfer Function* (MTF). Then, the spatial resolution is evaluated in both standard and interstrip interpolation modes, followed by the discussion of charge sharing role.

9.1. Mathematical description

9.1.1. Point Spread Function

The *Point Spread Function* (PSF) describes the spatial response of a detector channel to a point-like photon source as a function of position [4]. For a microstrip detector, it can be interpreted as the probability of a photon, interacting at a certain position, to be assigned to a specific strip or counter.

Considering an ideal detector, neglecting charge sharing effects, the response of a strip can be approximated by a rectangular function with pitch p [8]:

$$PSF(x) = \begin{cases} 1, & |x| < p/2 \\ 0, & |x| > p/2 \end{cases} \tag{9.1}$$

In a real detector, however, the noise and the charge sharing deviate the response from a box-like to a trapezoidal one, even when setting a correct threshold at half photon energy. Moreover, the charge spread out makes the effective spatial response depending on the threshold level, since the signal collected by the strip progressively

decrease close to its boundaries (see Section 8.1.2) and considering that a photon is only counted when its signal exceeds the comparator threshold [4].

As described in Bergamaschi et al. [4], the PSF is not directly measured since the X-ray beam has got a finite width, therefore the measured curves would represent the convolution between the intrinsic detector PSF and the beam profile. Therefore, it is easier to scan by moving, while irradiating, a tungsten edge thus obtaining the Edge Spread Function (ESF), from which the PSF can be simply derived:

$$ESF(x) = \int PSF(x)dx \tag{9.2}$$

In this work, instead, a focused laser beam has been employed to scan directly across the strips (see Section 4.2), obtaining directly the PSF instead of deriving it. However, since the laser spot has a finite width, the measured spatial response represents the convolution between the intrinsic detector PSF and the laser beam profile. But if the beam sport is narrow enough compared with the strip response, the measured curve then provides a reasonable approximation of the detector PSF.

9.1.2. Modulation Transfer Function

The *Modulation Transfer Function* (MTF) is used to compare quantitatively the spatial responses obtained by different detectors, describing the latter's capability of preserving spatial modulations at different spatial frequencies, hence quantifying the amount of contrast transferred.

Therefore, the MTF is calculated [8] [46] considering the module from the *Fourier Transform* of the PFS as follows:

$$MTF(f) = \frac{\left|\int_{-\infty}^{+\infty} PSF(x)e^{-i2\pi fx}dx\right|}{\int_{-\infty}^{+\infty} PSF(x)dx} \tag{9.3}$$

For a box like ideal PSF, the latter's *Fourier Transform* is simply as follows:

$$MTF(f) = \frac{\sin(\pi pf)}{\pi pf} \tag{9.4}$$

with the first zero occurring at $f_0 = 1/p$. Therefore, ideal strip widths of 50 μm and 100 μm would correspond respectively to first minima of 20 lp/mm and 10 lp/mm, where *lp* stands for *"line pairs"*, thus indicating the capability of discerning separately a bright and dark line (higher is better).

However, the highest spatial frequency that can be sampled without any ambiguity is the *Nyquist Frequency*, defined as:

$$f_N = \frac{1}{2p} \tag{9.5}$$

Indeed, frequency above (9.5) may produce aliasing and therefore could not be reconstructed from the sampled signal. For instance, evaluating Eq. (9.4) at f_N:

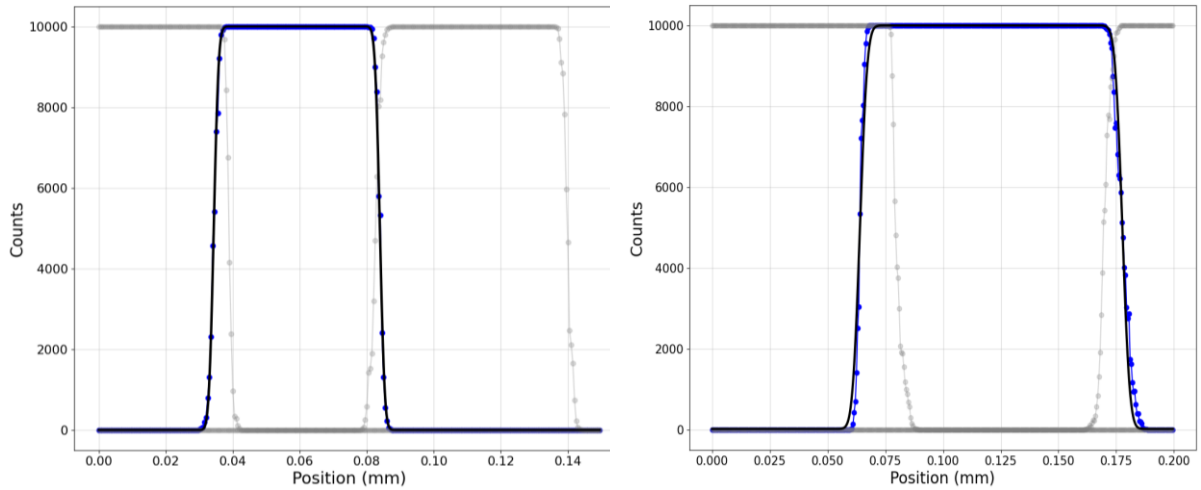
$$MTF(f_N) = \frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}} = \frac{2}{\pi} \cong 63.7\% \tag{9.6}$$

Therefore, at f_N the transferred contrast would still be the 63.7% but higher frequencies cannot be uniquely reconstructed from the sampled signal and can lead to aliasing.

9.2. Spatial response

9.2.1. Standard mode

The spatial response was first investigated in standard counting mode for both sensor pitches. For 50 µm and 100 µm pitch, the ideal first minima of the MTF are expected at approximately 20 lp/mm and 10 lp/mm. After trimming the two sensors the threshold has been set globally at half of the equivalent deposited energy produced by the laser[27].



(a) Channel 471 (50 µm) (b) Channel 797 (100 µm)

Figure 9.1 Measured PSF of two chosen representative channels (in blue) as well as their neighbour strips (in light grey)

[27] Moreover, the red (660 nm wavelength) laser frequency was set at 1 MHz, with an exposure time of 0.1 seconds: for this reason the counts climbs up to 10^4 in Figure 9.1.

The two PSFs depicted in Figure 9.1 deviate from an ideal box like shape due to the reason explained in Section 9.1.1. To obtain the MTF, the counts have been fitted with the following function, which also considers the smoothness of their edges:

$$PSF_{fit}(x) = PSF_{box}(x) * G_\sigma(x) = B + \frac{A}{2}\left[\mathrm{erf}\left(\frac{x - x_1}{\sqrt{2}\sigma}\right) - \mathrm{erf}\left(\frac{x - x_2}{\sqrt{2}\sigma}\right)\right] \quad (9.7)$$

Where $x_1 = x_c - \frac{w}{2}$ and $x_2 = x_c + \frac{w}{2}$ represents the ideal box like edges (with x_C the box response centre and w its width), A the plateau height with respect to the background, B the background, while σ determines how much the edges are smoothed (due to charge sharing, finite laser beam width…).

Actually, Eq. (9.7) derives from a convolution between the ideal box like response $PSF_{box}(x)$ and a Gaussian broadening $G_\sigma(x)$, respectively:

$$PSF_{box}(x) = B + A\ rect\left(\frac{x - x_c}{w}\right) \quad (9.8)$$

$$G_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (9.9)$$

Therefore, after fitting the results with Eq. (9.7), the fit has been decomposed in its own spatial frequency by using the *Fourier Transform*[28], thus obtaining the following plot:

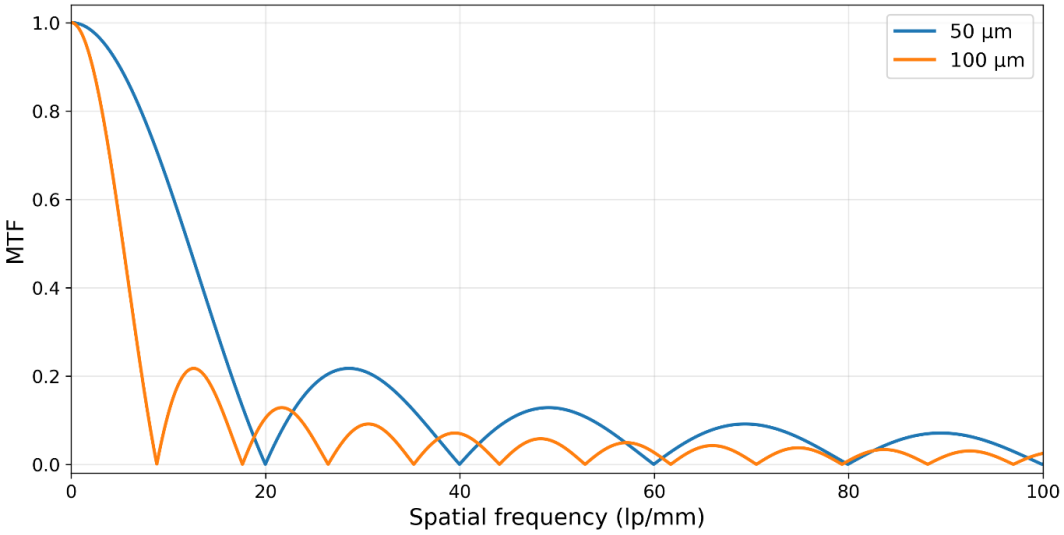


Figure 9.2 MTF derived from PSF pertaining to channel 471 (50 µm) and 797 (100 µm) shown in Figure 9.1

[28] When dealing with the fitted PSF, the background should be subtracted after performing the *Fourier Transform* as follows: $MTF(f) = \frac{|\mathcal{F}\{PSF(x)-B\}|}{|\mathcal{F}\{PSF(x)-B\}|_{f=0}}$

Instead, the results are summarized in the following:

Table 9.1 Summary of spatial frequencies found for standard mode

	1st MTF min. [*lp/mm*]	**Nyquist Freq. [*lp/mm*]**
Channel 471 (50 μm)	19.98	9.99
Channel 797 (100 μm)	8.81	4.40

The results are in line with what is expected from the strip geometry, although some differences may derive from factors (especially 100 μm sensor) such as the charge sharing and so on. However, the results for 50 μm sensors are in line with the results shown in Bergamaschi et al. [4].

9.2.2. Interpolation mode (only 50 μm sensor)

In an analogue way of the previous section, the spatial response of 50 μm sensor, using the interstrip interpolation mode, has been investigated. This mode permits to obtain additional spatial information by considering also the photons whose charge is collected also by the next strip, therefore behaving as is the strips were tighter (see Section 2.4.2).

To carry out the measurement, the *High Threshold Comparator* (HTC), which identifies the main strip and whether it crosses the threshold, was the one set close to half of the (laser equivalent) photon energy. At the same time, the *Low Threshold Comparator* (LTC) was set just above the noise level[29]. Recalling Section 2.4.2, depending on whether the charge is also detected by the left or the right neighbour, or neither of them, the event is assigned to the left (LC), right (RC) or central counter (CC), respectively [4].

Eventually the physical strip ends up virtually divided into substrips. For this reason, an increase in terms of spatial resolution is therefore expected, as previously proved with a 50 μm pitch silicon sensor in Bergamaschi et al. [4], where the average spatial resolution was improved from approximately 20 lp/mm in standard mode to about 30 lp/mm using the MYTHEN III interpolation logic.

[29] A too low LTC would be too close to the noise, generating false coincidences. On the other hand, a too high LTC would not allow for identifying correctly the charge sharing events.

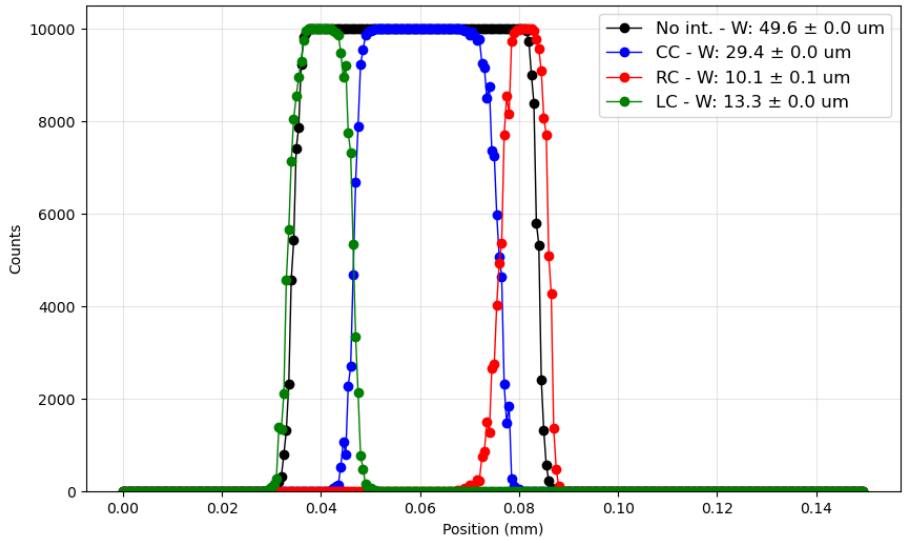


Figure 9.3 PSF of channel 471 in interpolation mode. The standard mode (black) is left as reference. Legend shows also the PSFs width

In Figure 9.3 the three interpolation counters show narrower PSFs with respect to the standard mode one: despite some difference, their combined widths are comparable to the standard mode. Furthermore, the MTF modules have been derived as done for the standard mode and the results are depicted in Figure 9.4.

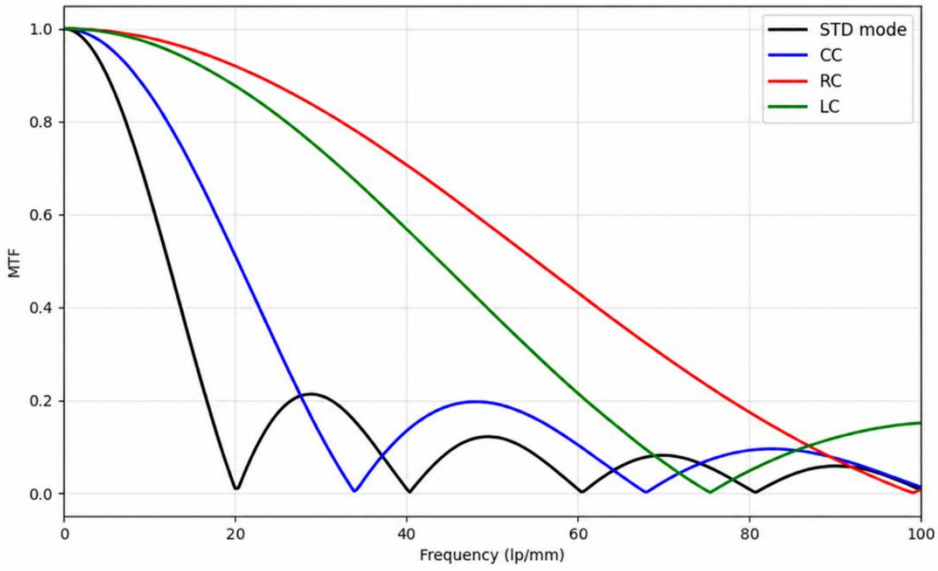


Figure 9.4 MTF derived from PSF pertaining to channel 471. For the sake of comparison, also the standard mode one has been depicted.

As shown above, the 1st minimum of CC occurs at 33.85 lp/mm, which is noticeably higher than the previous result for the standard mode (19.98 lm/mm). However, the MTFs of the two side counters do not overlap, despite a symmetric response being expected [4]: a possible explanation is the residual threshold dispersion of the low threshold comparators. Indeed, trimming has been carried out at Mo energy, so for corresponding *DAC* values different than the one used for the low threshold in

interpolation mode [7]. Therefore, the spatial regions assigned to the left and right counters are not exactly symmetric, leading to different PSF widths (Figure 9.3) and therefore to different MTFs (Figure 9.4).

Comparing the latter to Bergamaschi et al. [4], the results are qualitatively in agreement with the expected behaviour of the MYTHEN III interpolation logic: the CC exhibits a first MTF minimum at a considerably higher spatial frequency than in standard mode, while RC and LC have even higher first-minimum frequencies, corresponding to even narrower spatial regions close to the strip boundaries.

9.3. Threshold dependence

As previously discussed, for an ideal microstrip SPC detector, a threshold set close to half the photon energy represents the optimal condition (see Section 5.3). Indeed, in this case, the photon is counted by the channel collecting more than half of the generated charge, when the charge is shared between two strips.

To verify the behaviour of the SPC when the threshold is not properly set, additional spatial scans have been performed by varying the comparator threshold in order to compare the corresponding different PSFs.

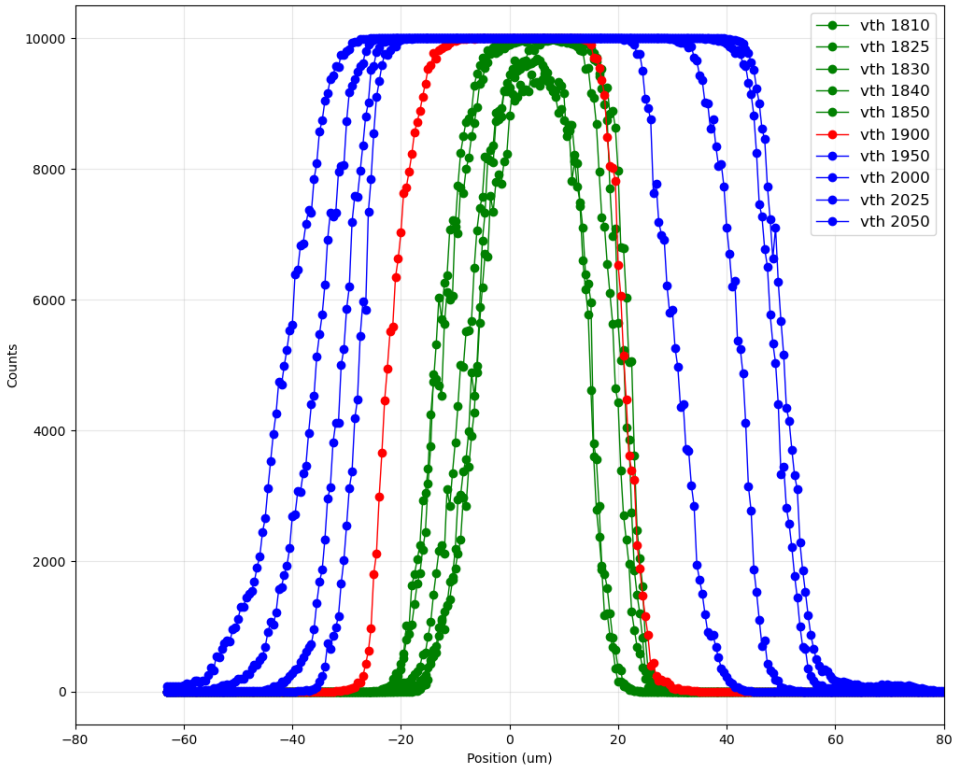


Figure 9.5 Spatial scans for several threshold values[30]

[30] Since the GaAs:Cr detector is operated in negative polarity, increasing the threshold corresponds to moving the threshold closer to the baseline and therefore to a lower effective energy threshold, and vice versa

Figure 9.5 shows indeed, for values around 1900 *DAC*, a PSF with a width close to the physical 50 μm strip pitch. Instead, when the threshold is increased further (blue PSFs), corresponding to lower energy thresholds, the PSF progressively broadens: photons interacting close to the boundary between two strips can generate signals above threshold in both neighbouring channels, increasing the effective strip width (the "multiplicity" effect has been covered in Section 5.3). On the other hand, when the threshold decreases (green ones), corresponding to higher energy thresholds, part of the photons sharing their charge between neighbouring strips does not produce a signal large enough to exceed the latter. These events are then rejected, and the effective width becomes narrower, since it behaves like being collimated at the cost of detection efficiency [4] [8].

9.4. Effects of charge sharing in spatial resolution

In standard SPC, as already addressed in Section 8.1, charge sharing is considered deleterious for the detector response. However, in interpolation mode, it can be exploited to obtain additional information about the photon interaction position.

Indeed, the amount of charge sharing becomes particularly important for the performance of the interpolation mode. For instance, in Bergamaschi et al. [4], the Si sensor investigated had a charge sharing of about $24 \pm 4\%$, with a corresponding region of sharing of about $12 \pm 2\ \mu m$ for a 50 μm strip (if considering the linear charge collection model seen in Section 8.1): the amount of charge sharing mentioned had been deemed as not ideal to achieve an optimized spatial resolution improvement using interpolation.

The situation is, instead, different for the GaAs:Cr investigated in this work, characterized by a charge sharing of the order of 80%, surprisingly independently of the strip pitch (see section 8.1.1): this would contribute to the increase of spatial resolution, thus potentially explaining the 25% increase of first minimum of GaAs:Cr with respect to the one found in Bergamaschi et al. [4] for Silicon. However, spatial resolution would also come down to other factors, like photon energy and rate, sensor thickness and bias which have not been yet investigated [4].

In conclusion, the strong charge sharing of GaAs:Cr, which represents a limitation in standard counting mode, can become an advantage when using the interpolation mode due to the extensive charge spread out that could lead to improving the spatial resolution.

10 Conclusion and future developments

This work focused on the characterization of two prototype chromium compensated GaAs microstrip sensors couplet to the MYTHEN III *Single Photon Counting* detector. Indeed, the measurement showed that MYTHEN III can be successfully operated with GaAs:Cr sensors in negative polarity (electron collection) and that its ASIC provides enough flexibility in order to modify the detector response accordingly.

The detector characterization, by varying the feedback parameters, showed that the preamplifier settings (equivalent resistance and capacitance) have the largest influence on the detector response, in terms of gain and high rate behaviour but also ENC and eventually determining the energy range and the speed. At the same time the shaper parameters (except for its coupling capacitance) show only weaker influence on the mentioned figures of merit, at least within the investigated range of values. Indeed, decreasing the feedback preamplifier capacitance has been found to increase the gain while decreasing the noise and improving the higher rates behaviour, while a trade off has been observed for the feedback preamplifier equivalent resistance, whose increase provides larger gain and lower noise but at the expense of worse high flux behaviour. Therefore, no single configurations have been considered as the paramount ones, as the settings should be adjusted according to the requirements of use. Moreover, the estimated usable energy range confirmed the coverage of the energy region of interest for GaAs:Cr: despite the relatively large ENC, the resulting lower cap remains below the energy region where the Si absorption efficiency becomes irrelevant. At the same time, the electronic dynamic range would extend even beyond the useful range of 500 μm thick GaAs:Cr sensors. Eventually, once ensuring upper energy coverage, settings providing higher gain and lower noise are preferable.

Furthermore, a dedicated trimming procedure has been successfully applied in order to reduce the spread of channel responses, reducing the dispersion down to a few equivalent electrons for all three counters and the two sensors. In addition, an unexpected temperature dependence has been observed, for which over the investigated increasing temperature range the gain increased with decreasing ENC as well, in contrast to the behaviour observed for MYTHEN III equipped with silicon sensors: this behaviour has been justified considering the operation in negative polarity and then a different analogue configuration.

The subsequent investigation on charge transport properties showed a characteristic high charge sharing fraction of approximately 75-80% for the 50 μm sensor, in line

with previous measurements. However, a similar result has been found for the 100 μm which was expected to decrease due to the simple geometrical scaling. The laser scan reproduced the same results by showing approximately trapezoidal charge collection profiles (and then also confirming the described theoretical model) and therefore supporting the explanation that large charge sharing comes down on the sensor itself. Different electrode geometry of the investigated sensors, with respect to ones previously studied, may contribute to this behaviour. In addition to charge sharing, scans obtained with localized laser red (660 nm wavelength) beam showed signals of opposite polarity in neighbouring strips, thus confirming that the so-called "crater effect" might affect the induced signal.

Finally, the spatial response of the detector has been investigated and quantified through the PSF and consequent MTF. Although in standard SPC the large charge sharing may represent a limitation, this becomes an advantage when employing the MYTHEN III inter strip interpolation logic: indeed, for the 50 μm, the central and the side interpolated counters showed narrower spatial responses and consequently first MTF minima at higher spatial frequencies. Therefore, the strong charge sharing may be favourable fort this operating mode.

Overall, the results indicate that GaAs:Cr can extend the potentiality of MYTHEN III detector towards higher photon energies, while keeping the main perks of SPC. However, some aspect could be investigated further, like the unusual temperature dependence which would require dedicated investigation to separate potentially different contributions coming from sensor and the readout electronics. Another relevant topic would be a more detailed study of the large charge sharing observed for the 100 μm strips sensor, investigating the electric and weighting fields, as well as how the electrode geometry affects overall.

Beyond synchrotron applications, high Z SPC detector would be also relevant for medical imaging, such as photon counting CT [47]. In particular, the possibility of selecting multiple thresholds allows the detected photons to be separated into different energy ranges, providing spectral information while rejecting signals and then make possible material discrimination and decomposition in photon counting CT systems.

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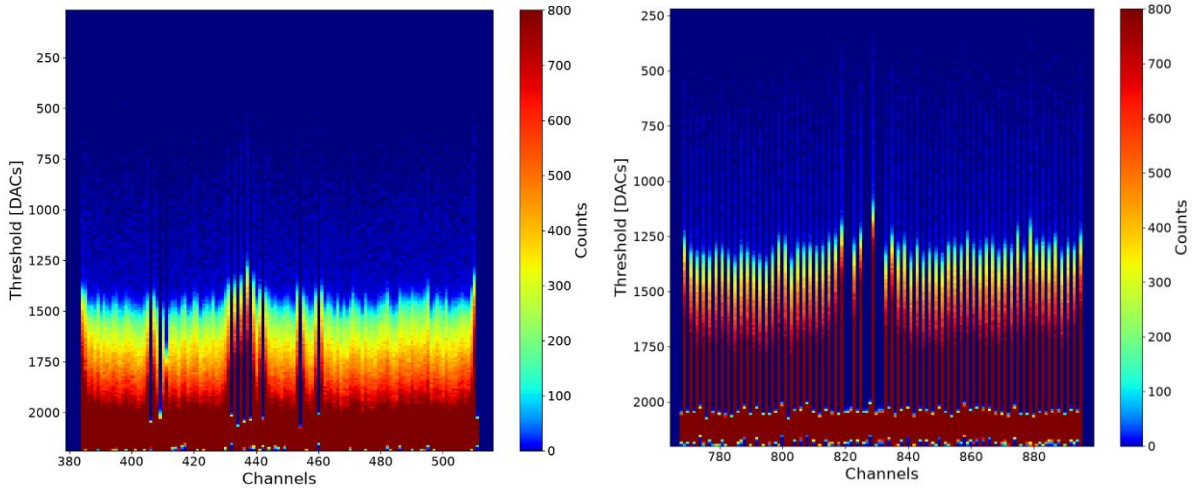
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A Appendix – Threshold Scan Maps of the GaAs:Cr Sensors

The following figures show the threshold scan carried out for both GaAs:Cr sensors. The threshold scan map shows that some channels remain inactive: this has been observed for more *VDC,shaper* values and energies, indicating that it is related to a problematic connection between channel and readout, as suggested in Bergamaschi et al. [5].

In Figure A.1, the colour scale represents the number of counts. Therefore, for each fixed channel, the variation of the counts along the threshold axis corresponds to the threshold scan of that channel.



(a) 50 µm sensor (b) 100 µm sensor

Figure A.1 Overview of threshold scans for the two sensors

Due to malfunctioning wire bonding, not all strips have been used to perform scans, as visible in Figure A.1 where there is no response for some strips and the surroundings sometimes are likely affected and then deemed as not reliable enough. indeed only 95 out of 128 strips belonging to 50 µm have been employed (about 74%), while 55 out of 64 strips for the 100 µm (about 86%).

B Appendix – Semi-analytical calculation of weighting potential

The weighting potential shown in Figure 8.4 has been calculated starting from the results depicted in Heubrandtner et al. [45], which found expression for the weighting field of a stripe electrode geometry. The latter instead is a closed solution of the equation, for a three dimensional case, proposed in Rieglel et al. [25], which found the general solution for the Laplace equation $\nabla^2\varphi_0 = 0$ in case of a pair of plate electrodes, one put at unit potential and the other grounded.

The equation has been solved by writing a proper *Python* script with the correct frame conditions.

The geometry has been approximated as a homogeneous parallel plane system with the readout strip located at $z = 0$, while the opposite at $z = D$, where D is the sensor thickness set at 500 μm. Like in Heubrandtner et al. [45], the strip is assumed to be thin and infinitely extended along the longitudinal direction, to simplify the problem in two directions. Moreover, gaps between the strips are neglected.

Therefore, following the weighing potential definition, the selected strip (whose extension is from $-w/2$ to $w/2$) has been set to unit potential, while the other electrodes are grounded. Eventually, the corresponding weighting potential is given by Heubrandtner et al. [45] as follows:

$$\phi_w(x,z) = \frac{1}{\pi}\left\{\tan^{-1}\left[\cot\left(\frac{\pi z}{2D}\right)\tanh\left(\frac{\pi\left(x+\frac{w}{2}\right)}{2D}\right)\right] - \tan^{-1}\left[\cot\left(\frac{\pi z}{2D}\right)\tanh\left(\frac{\pi\left(x-\frac{w}{2}\right)}{2D}\right)\right]\right\} \quad \text{(B.1)}$$

where D is the sensor thickness, w the strip width, x the lateral position with respect of the centre of the reference strip and z the distance from the readout electrode.

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Variable	Description	Unit
VR,preamp	Feedback preamplifier resistance voltage	1 DAC = 0.61 mV
VR,shaper	Feedback (negative) shaper resistance voltage	1 DAC = 0.61 mV
VDC,shaper	Shaper baseline voltage	1 DAC = 0.61 mV
C,preamp	Feedback preamplifier capacitance	fF
C,shaper	Feedback shaper capacitance	fF
C,AC	AC coupling capacitance	fF

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